Autonomous and Mobile Robotics

Comprehensive Lecture Notes

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Part I

Lecture 1

1 Configuration Space and Workspace

(notes p. 1–2)

1.1 The idea

The central idea is to *define a suitable space in which the robot is a single point.* Once the robot is a point, two core tasks become much simpler to formulate:

- **planning** (find a path for the point from a start to a goal),
- **control** (make the point follow that path).

(Fun fact from the notes: the C-space concept was born to do *collision avoidance.*)

Intuition

Think of a huge, awkwardly shaped sofa that has to be moved through a door. Reasoning about every corner of the sofa at the same time is painful. If instead we find a space where the whole sofa is *one dot*, and the "forbidden" places for the dot are exactly the sofa positions that hit a wall, then moving the sofa becomes moving a dot around obstacles.

1.2 Robots and workspaces

Definition 1.1 (Robot and workspace)**.** A *robot* is a collection of n_b rigid bodies (we assume rigid robots) moving in a *workspace* $\mathcal{W} = \mathbb{R}^2$ or $\mathcal{W} = \mathbb{R}^3$, the physical space in which the robot moves.

- If $\mathcal{W} = \mathbb{R}^2$ the robot is *planar*; if $\mathcal{W} = \mathbb{R}^3$ it is *spatial.*
- If $n_b = 1$ the robot is *single-body*; if $n_b > 1$ it is *multi-body.*

Example 1.2. A car can be modelled as a single body, whereas a car with a trailer is a multi-body system ($n_b = 2$).

1.3 Configuration and generalized coordinates

Definition 1.3 (Posture and configuration)**.** The *posture* of a robot is the position of all its points in $\mathcal{W}$. A *configuration* is a *minimal* set of parameters that defines the posture of the robot in $\mathcal{W}$. The corresponding vector

$$q = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$$

is called a *configuration vector*, and each q_i is a *generalized coordinate.*

The generalized coordinates are of two kinds:

- **Cartesian coordinates**: identify the position of representative points of the robot; they take values in $\mathbb{R}^2$ or $\mathbb{R}^3$.
- **Angular coordinates**: identify the orientation of the bodies of the robot; they take values in SO(2) (planar: one parameter) or SO(3) (spatial: three parameters).

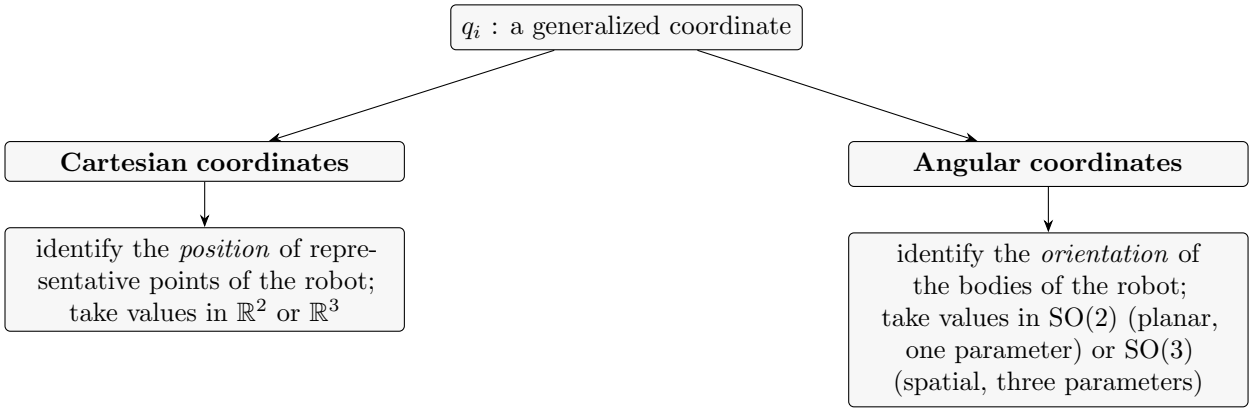


Figure 1: The two kinds of generalized coordinates.

Intuition

"Minimal" is the key word. The posture is the full, redundant description (every point of every body); the configuration is the shortest list of numbers from which the posture can be reconstructed. A rigid rectangle in the plane has infinitely many points, but three numbers (x, y, θ) tell you where all of them are.

1.4 Configuration space

Definition 1.4 (Configuration space)**.** The *configuration space* (C-space) $\mathcal{C}$ of a robot is the set of all possible values of q. Its dimension $n = \dim \mathcal{C}$ is the number of degrees of freedom of the robot.

Intuition

The C-space is the "catalogue of all poses the robot can be in". Each point of $\mathcal{C}$ *is* one pose. A curve in $\mathcal{C}$ is therefore a motion of the robot in the physical space, which is exactly why planning for a robot reduces to planning for a point.

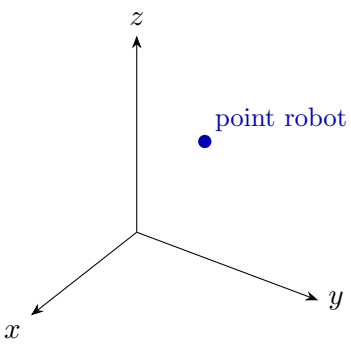
2 Examples of Configuration Spaces

(notes p. 2–3)

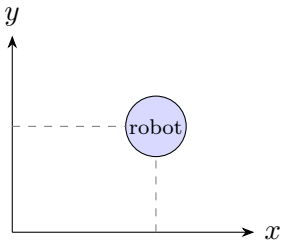
Example 2.1 (Point robot in $\mathcal{W} = \mathbb{R}^3$)**.** $q = (x, y, z)^\top$, so $\mathcal{C} = \mathbb{R}^3 = \mathcal{W}$ and $n = 3$.

Example 2.2 (Disk robot in $\mathcal{W} = \mathbb{R}^2$)**.** Since a disk looks the same under any rotation, its orientation is irrelevant: $q = (x, y)^\top$, $\mathcal{C} = \mathbb{R}^2 = \mathcal{W}$, $n = 2$.

In both examples $\mathcal{C}$ and $\mathcal{W}$ coincide. A case where they differ:



(a) point robot in $\mathcal{W} = \mathbb{R}^3$, $q = (x, y, z)^\top$



(b) disk robot in $\mathcal{W} = \mathbb{R}^2$, $q = (x, y)^\top$

Figure 2: Point robot and disk robot: in both cases $\mathcal{C} = \mathcal{W}$.

Example 2.3 (Disk with a tick)**.** We add a small mark (a "tick") to the disk, so that its orientation now matters. Then

$$q = \begin{pmatrix} x \\ y \\ \varphi \end{pmatrix}, \qquad \mathcal{C} = \mathbb{R}^2 \times \mathrm{SO}(2), \qquad n = 3,$$

where $\times$ is the Cartesian product.

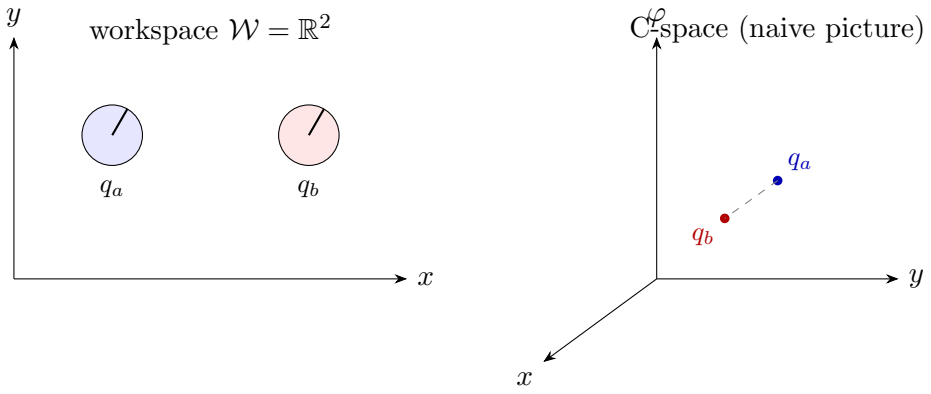


Figure 3: The disk with a tick. Configurations q_a and q_b have the same y and φ but different x. In C-space they differ only along the x-axis. (This picture is *naive*: it is not topologically correct, see Section 4.)

Every change in the configuration space is a path, i.e. a motion, in the physical space.

Example 2.4 (Polygonal robot in $\mathcal{W} = \mathbb{R}^2$)**.** For instance a car moving on the ground. It can rotate about its centre of mass:

$$q = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}, \quad \mathcal{C} = \mathbb{R}^2 \times \mathrm{SO}(2), \quad n = 3.$$

Example 2.5 (Polyhedral robot in $\mathcal{W} = \mathbb{R}^3$)**.** Treat it like the previous case, but add z and two more orientation angles:

$$q = (x, y, z, \alpha, \beta, \gamma)^\top, \qquad \mathcal{C} = \mathbb{R}^3 \times \mathrm{SO}(3), \qquad n = 6.$$

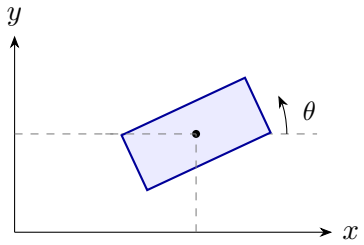


Figure 4: Polygonal robot in the plane: position of the centre of mass (x, y) and orientation θ.

Definition 2.6 (Special Euclidean groups)**.** The spaces $\mathbb{R}^2 \times \mathrm{SO}(2)$ and $\mathbb{R}^3 \times \mathrm{SO}(3)$ are called the *special Euclidean groups*, denoted $\mathrm{SE}(2)$ and $\mathrm{SE}(3)$.

Intuition

$\mathrm{SE}(n)$ is the set of all rigid motions (a rotation followed by a translation) of n-dimensional space. A free rigid body can be moved anywhere and turned any way, so its configuration space is exactly the set of rigid motions: 3 numbers in the plane (2 translations + 1 rotation), 6 in space (3 translations + 3 rotations).

3 Multi-Body Robots: Manipulators

(notes p. 4–5)

3.1 Planar manipulator with n_b links

Consider a planar manipulator with n_b links and n_b (e.g. revolute) joints moving in $\mathcal{W} = \mathbb{R}^2$. Each link is a rigid body with 3 degrees of freedom in the plane, so one might expect $3n_b$ generalized coordinates. This is wrong: the links are *connected*, and every joint introduces constraints.

Proposition 3.1 (Degrees of freedom of a planar serial arm)**.** *A planar arm of n_b links connected to the ground by n_b revolute joints has*

$$3n_b - 2n_b = n_b$$

degrees of freedom. Each revolute joint imposes 2 constraints (the two coordinates of the joint point must coincide on the two bodies it connects).

Hence, to describe the configuration we do not need to remember all the constraints: one angle per joint is enough,

$$q = \begin{pmatrix} q_1 \\ q_2 \\ \vdots \\ q_{n_b} \end{pmatrix}, \quad n = n_b, \qquad \mathcal{C} = \underbrace{\mathrm{SO}(2) \times \cdots \times \mathrm{SO}(2)}_{n_b \text{ times}} = \mathrm{SO}(2)^{n_b}.$$

Intuition

Imagine gluing the links together at the joints. Each glued joint removes two "free sliding" directions. Three links loose in the plane would have 9 degrees of freedom; pin them into a chain anchored at the base and only 3 remain: the three joint angles. The number of angles

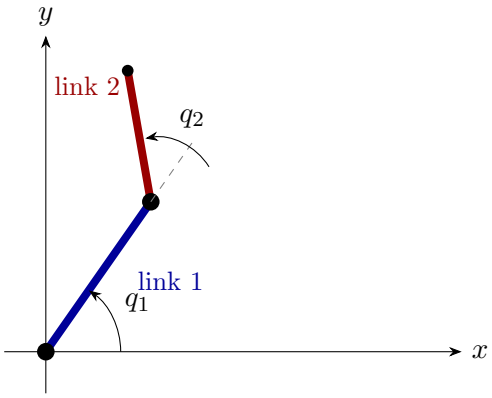


Figure 5: A 2R planar arm: one angle per joint is enough to describe the configuration.

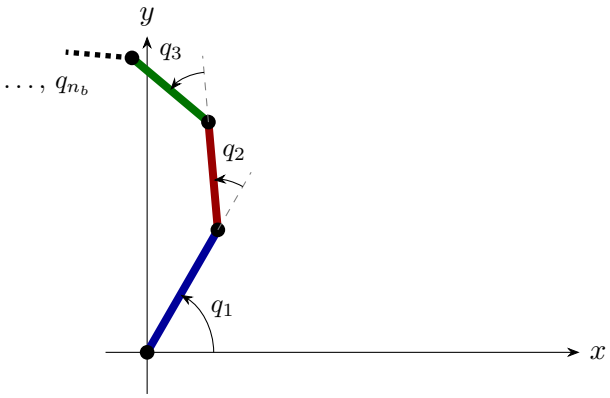


Figure 6: Planar manipulator with n_b links: one joint angle q_i per joint.

= the number of joints.

Proposition 3.2. *For three revolute joints, $\mathcal{C} = \mathrm{SO}(2) \times \mathrm{SO}(2) \times \mathrm{SO}(2)$, which is very different from $\mathrm{SO}(3)$: the two spaces have the same dimension 3 but different topology.*

Intuition

SO(3) describes all orientations of *one* rigid body (roll, pitch, yaw). Three planar joints are three *independent* angles each living on its own circle; they cannot all mix into one 3D rotation. Same number of parameters, different shape: it is the shape (topology), not just the count, that matters later.

3.2 Spatial manipulator with n_b links

Proposition 3.3 (Degrees of freedom of a spatial serial arm)**.** *A spatial arm with n_b links and n_b revolute joints in $\mathcal{W} = \mathbb{R}^3$ has*

$$6n_b - 5n_b = n_b$$

degrees of freedom: each link has 6 degrees of freedom, but a revolute joint imposes 5 constraints (3 positions and 2 orientations are fixed; only rotation about the joint axis remains).

Even though the manipulator is spatial, each joint has a *planar* orientation (a single angle), so

again

$$\mathcal{C} = \mathrm{SO}(2) \times \cdots \times \mathrm{SO}(2) = \mathrm{SO}(2)^{n_b} \qquad (n_b \text{ times}).$$

Intuition

A hinge (revolute joint) always allows exactly one motion: turning about its axis. Whether the world is 2D or 3D, that one motion is a single angle. The ambient dimension changes the *constraint count* (2 vs. 5) but not the end result: one angle per joint.

4 Topology of $\mathcal{C}$

(notes p. 5–6)

4.1 The problem with a flat picture of the 2R robot

Take the 2R planar robot, $\mathcal{C} = \mathrm{SO}(2) \times \mathrm{SO}(2)$, $q = (q_1, q_2)^\top$. Since angles are periodic, q_1 and $q_1 + 2\pi$ give the *same posture*. So if we naively draw $\mathcal{C}$ as the plane $\mathbb{R}^2$ of angle pairs, then two different points give the same posture: the map from this plane to postures is *not injective*, which is bad for us.

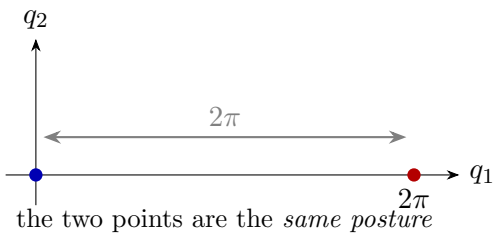


Figure 7: On the plane of angle pairs, $q_1 = 0$ and $q_1 = 2\pi$ (same q_2) correspond to the same posture: the map to postures is many-to-one.

A new attempt: restrict each angle to $[0, 2\pi)$, i.e. draw $\mathcal{C}$ as the square $[0, 2\pi) \times [0, 2\pi)$. Now the correspondence with postures is one-to-one (considering only the two "bad sides" of the square, $q = 0$ and $q = 2\pi$, as the same). But this is still not good enough: two configurations q_a and q_b that are almost the same posture (one just below 2π, the other just above 0) look *far apart* in the square.

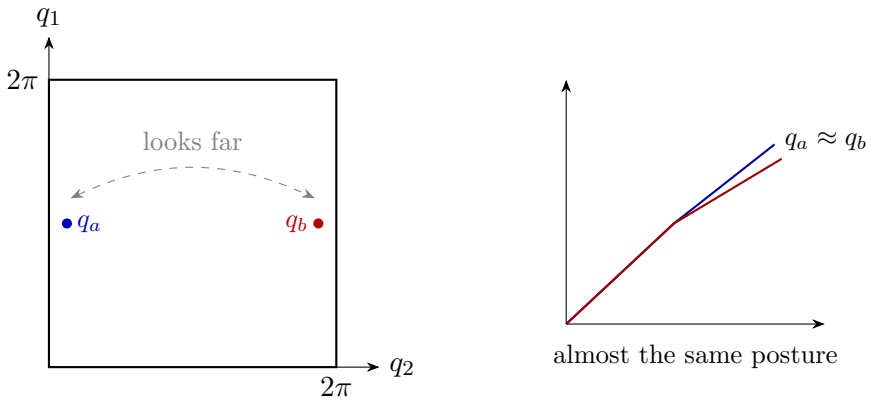


Figure 8: The square picture of $\mathcal{C}$: q_a and q_b are close as postures but far in the flat picture.

So this topology does not describe exactly what is happening.

4.2 Wrapping the space: the torus

The fix is to *wrap* the space: identify the opposite sides of the square (left with right, bottom with top) and embed the resulting 2D object in 3D. We obtain a *torus* (from Latin, "pillow").

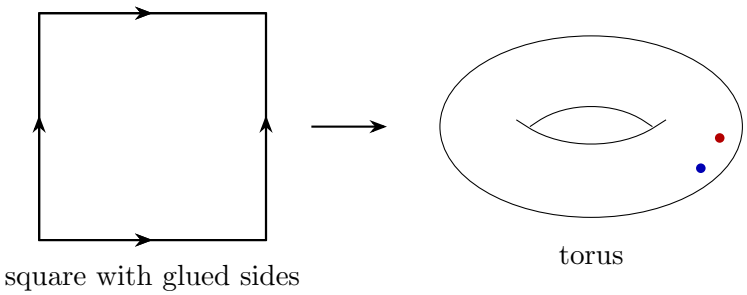


Figure 9: Gluing opposite sides of the square gives a torus. Points that were far apart in the square (across the seam) become neighbours on the torus.

Proposition 4.1. $\mathrm{SO}(2) \cong \mathbb{S}^1$ *(the circle), hence the configuration space of the 2R planar robot is* $\mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{T}^2$*, the torus, and in general* $\mathrm{SO}(2)^{n_b} \cong \mathbb{T}^{n_b}$ *is an* n_b*-dimensional torus.*

Intuition

A joint angle is like a clock hand: after 2π you are back where you started, so one angle lives on a *circle*, not on a line. Two joints give two circles, and a product of two circles is the surface of a doughnut. Walking off the right side of the square and reappearing on the left is just going around the doughnut the short way.

4.3 Manifolds

Definition 4.2 (Manifold)**.** A *manifold* of dimension n is a space in which the neighbourhood of each point "looks like" a neighbourhood of $\mathbb{R}^n$. (For the torus, $\mathbb{R}^2$.)

Proposition 4.3. *Configuration spaces are manifolds (in general* not *Euclidean anymore):* $\mathbb{R}^m$*,* $\mathrm{SO}(2)$*,* $\mathrm{SO}(3)$*,* $\mathrm{SE}(2)$*,* $\mathrm{SE}(3)$ *and their products are all manifolds.*

Intuition

The Earth is the classical example: it is a sphere, yet a small enough patch looks flat, so a city map works. Locally we can use ordinary coordinates and calculus; globally the space can wrap around, and that is what a flat picture cannot show.

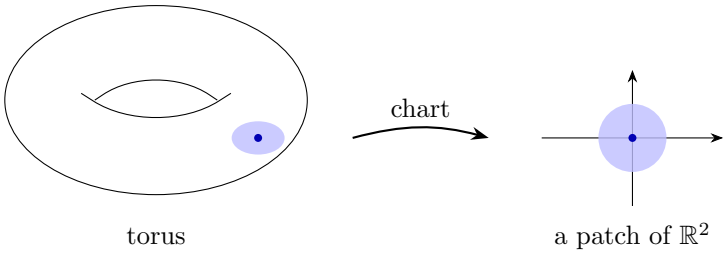


Figure 10: A small neighbourhood of any point of the torus can be flattened (charted) onto a neighbourhood in $\mathbb{R}^2$: this is what "looks like $\mathbb{R}^n$" means.

5 Distances in $\mathcal{C}$

(notes p. 6–8)

5.1 What a distance must satisfy

Definition 5.1 (Distance (metric))**.** A function $d : \mathcal{C} \times \mathcal{C} \to \mathbb{R}$ is a *distance* if for all $q_A, q_B, q_C \in \mathcal{C}$:

(i) $d(q_A, q_B) \geq 0$;
(ii) $d(q_A, q_B) = 0$ if and only if $q_A = q_B$;
(iii) $d(q_A, q_B) = d(q_B, q_A)$ (symmetry);
(iv) $d(q_A, q_B) \leq d(q_A, q_C) + d(q_C, q_B)$ (triangle inequality).

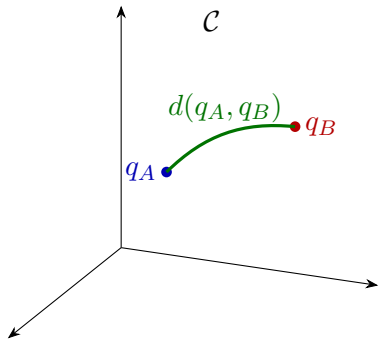
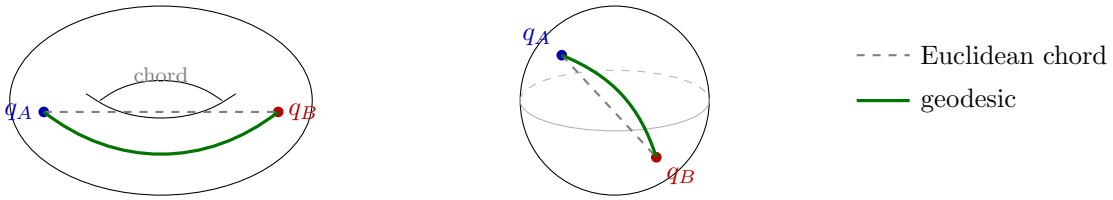


Figure 11: The distance $d(q_A, q_B)$ between two points of the configuration space.

5.2 Distances on manifolds and geodesics

The classic (Euclidean) distance completely ignores the curvature of the space: for two points on a torus, the straight segment between them passes *through the hole*, outside the manifold. So we cannot use the Euclidean distance.



torus: tricky in general

sphere: great circles

Figure 12: The chord between two points leaves the manifold (it crosses the hole of the torus, or the inside of the sphere); the geodesic stays on the surface. On a sphere the geodesic is an arc of a great circle; on a general manifold it is hard to compute.

Definition 5.2 (Geodesic)**.** A *geodesic* is a minimum-length path between two points *on the manifold*. The geodesic distance is the length of that path.

Intuition

For two cities on Earth, the geodesic is the great-circle route an aircraft follows, not the tunnel through the Earth. Geodesics are the "honest" distances on a curved space.

5.3 The displacement metric

Geodesics are easy on a sphere but can be tricky in general (and on a torus). Because of this difficulty, roboticists do not use them. They use instead a *displacement metric*.

Let p denote a point of the robot (e.g. the centre of mass, or the end point of a link), and $p(q)$ its position in $\mathcal{W}$ when the configuration is q.

Definition 5.3 (Displacement metric)**.**

$$d(q_A, q_B) = \max_{p \in \text{robot}} \|p(q_A) - p(q_B)\|.$$

It is the *maximum displacement* of the points of the robot when moving from q_A to q_B.

Proposition 5.4. *The displacement metric satisfies the four axioms of a distance (provided distinct configurations give distinct postures, which is the case for a minimal parametrization).*

Proof. (i),(iii) hold because a norm is non-negative and symmetric, and the maximum of such quantities is too. (ii) If $q_A = q_B$ every point has zero displacement, so $d = 0$; conversely $d = 0$ means every point is at the same place in the two configurations, i.e. the postures coincide, hence $q_A = q_B$. (iv) For every point p,

$$\|p(q_A) - p(q_B)\| \le \|p(q_A) - p(q_C)\| + \|p(q_C) - p(q_B)\| \le d(q_A, q_C) + d(q_C, q_B);$$

taking the maximum over p on the left gives the claim. □

Intuition

Instead of measuring a length *inside* the curved configuration space, we go back to the physical world and ask: "how far does the worst-behaved point of the robot have to travel?" We use ordinary Euclidean distances there, where they are perfectly fine, and the curvature of $\mathcal{C}$ takes care of itself. Nice side effect: this measure is automatically periodic, since q_1 and $q_1 + 2\pi$ give the same positions.

Example 5.5 (1R robot)**.** A single link of length L rotating about the origin; q_A, q_B are two angles differing by $\Delta \in [0, \pi]$. A point at distance $r \le L$ from the joint moves by the chord $2r \sin(\Delta/2)$, largest at the tip, so

$$d(q_A, q_B) = 2L \sin\left(\tfrac{\Delta}{2}\right).$$

We are effectively using the *Euclidean* distance between the two tip positions to measure distance on the manifold.

Example 5.6 (Numbers for the 1R robot, $L = 1$)**.** For $\Delta = \pi/2$: $d = 2\sin(\pi/4) = \sqrt{2} \approx 1.41$. For two angles that differ by $2\pi - 0.1$ (almost a full turn): the wrapped difference is 0.1, so $d = 2\sin(0.05) \approx 0.10$ – correctly tiny, whereas the flat difference of angles (6.18) would call them far apart.

Example 5.7 (2R robot)**.** With two links we take the maximum over the points of both links. The displacement of a point on a link is an affine function of its position along the link, so its norm is convex and the maximum is attained at a link endpoint: the base (which never moves), the elbow joint, or the tip. In Figure 15 (link lengths 1.6 and 1) the tip moves by about 0.64 but the elbow moves by about 1.60, so $d(q_A, q_B) \approx 1.60$: the maximum is *not*

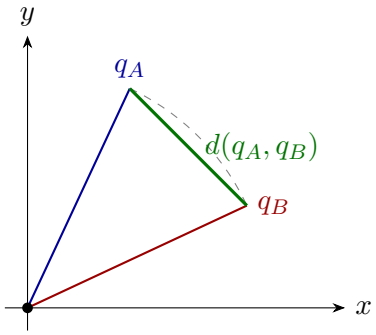


Figure 13: 1R robot: the displacement metric is the Euclidean distance between the tip in the two configurations.

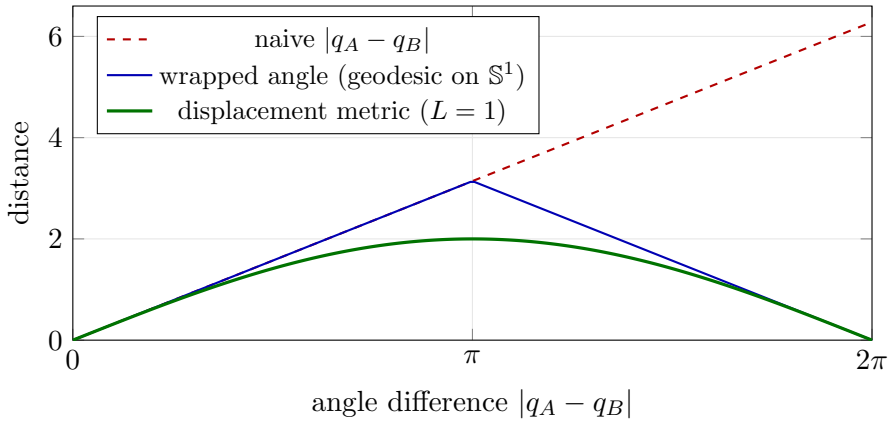


Figure 14: Three ways to measure the distance between two configurations of a 1R robot. The naive difference grows up to 2π; the other two vanish again at 2π because the configuration wraps around.

necessarily at the tip.

5.4 The max over infinitely many points

There is a catch: the definition asks for a maximum over an *infinite* set of points. Intuitively, for polygonal robots we should only look at the *vertices*, but it is not always like this.

Proposition 5.8. *If the robot is a convex polygon (or a union of segments, as a chain of links), the maximum in the displacement metric is attained at a vertex (link endpoint).*

Proof. Under a rigid motion, the position of a body point r is affine in r, so $r \mapsto p(q_A) - p(q_B)$ is affine on the body. The norm of an affine function is convex, and a convex function on a polytope attains its maximum at a vertex. □

Intuition

The point that moves most is the one sticking out the farthest – a corner, or the tip of a link. For a curved body (e.g. a disk with attached parts) there is no finite list of corners to check, so we need an approximation.

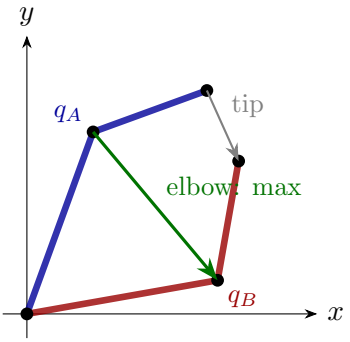


Figure 15: 2R robot in two configurations. The displacement of each characteristic point is drawn; the displacement metric takes the largest one (here the elbow).

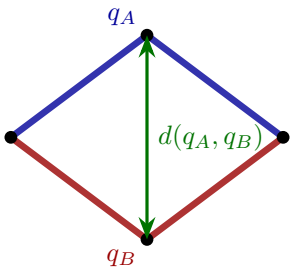


Figure 16: 2R robot with equal links, "elbow up" vs "elbow down", same tip: the tip does not move, and the displacement metric equals the distance between the two elbows.

5.5 A practical alternative: control points

We can use a predetermined number of points, called *control points*, which, if chosen correctly, give a good approximation.

Definition 5.9 (Control-point distance)**.** Fix control points $\{p_1, p_2, \ldots, p_c\}$ on the robot. Define

$$d(q_A, q_B) = \sum_{i=1}^{c} \|p_i(q_A) - p_i(q_B)\|.$$

Algorithm 5.10 (Computing the control-point distance)**.**
1. Choose c representative points $p_1, \ldots, p_c$ on the robot (e.g. centre of mass, link endpoints).
2. For each i, compute the forward kinematics $p_i(q_A)$ and $p_i(q_B)$.
3. Return $\sum_{i=1}^{c} \|p_i(q_A) - p_i(q_B)\|$.

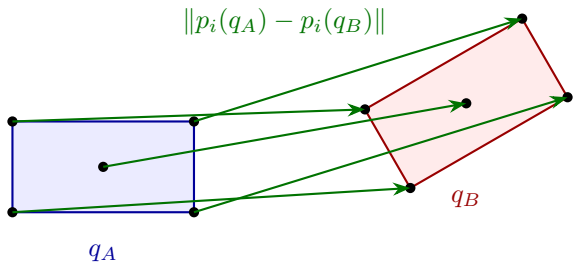


Figure 17: Control points $p_1, \ldots, p_5$ (four corners and the centre) of a rectangular robot. The distance is the sum of the lengths of the displacement arrows.

Proposition 5.11. *The control-point distance is symmetric and satisfies the triangle inequality. It is a genuine distance if the control points are chosen so that $p_i(q_A) = p_i(q_B)$ for all i implies $q_A = q_B$; otherwise it is only a pseudo-distance.*

Proof. Each summand is a norm of a difference, so symmetry holds, and the triangle inequality holds termwise and is preserved by the sum. The condition on control points gives axiom (ii). □

Intuition

Rather than tracking every point, you pin a few "sensors" on the robot (say, the tip and the middle) and add up how far they each move. A moving robot cannot fool three well-placed sensors: if they all end up in the same places, the robot is in the same pose. The cost: an approximation of the true worst-case displacement, but cheap and easy to differentiate.

6 Summary

6.1 Configuration spaces seen in this lecture

Robot	$\mathcal{C}$	$n = \dim \mathcal{C}$	Euclidean?
Point in $\mathbb{R}^3$	$\mathbb{R}^3$	3	yes
Disk in $\mathbb{R}^2$	$\mathbb{R}^2$	2	yes
Disk with tick / polygon in $\mathbb{R}^2$	$\mathrm{SE}(2) = \mathbb{R}^2 \times \mathrm{SO}(2)$	3	no
Polyhedron in $\mathbb{R}^3$	$\mathrm{SE}(3) = \mathbb{R}^3 \times \mathrm{SO}(3)$	6	no
Planar n_b-R arm	$\mathrm{SO}(2)^{n_b} = \mathbb{T}^{n_b}$	n_b	no
Spatial n_b-R arm	$\mathrm{SO}(2)^{n_b} = \mathbb{T}^{n_b}$	n_b	no

6.2 Distances on $\mathcal{C}$

Distance	Definition	Pros / cons
Euclidean in parameters	$\|q_A - q_B\|$	simple, but ignores wrapping (bad on angles)
Geodesic	shortest path on the manifold	exact, but hard to compute in general
Displacement metric	$\max_p \|p(q_A) - p(q_B)\|$	respects topology, physically meaningful; max over infinitely many points
Control points	$\sum_i \|p_i(q_A) - p_i(q_B)\|$	cheap approximation; needs well-chosen points

6.3 Key take-aways

- The configuration space $\mathcal{C}$ is the space of all values of a *minimal* set of parameters describing the robot; in it the robot is a single point, and planning/control become problems about that point.

- Joints reduce degrees of freedom: a planar arm has $3n_b - 2n_b = n_b$ and a spatial arm $6n_b - 5n_b = n_b$ degrees of freedom – one angle per joint.
- Because angles are periodic, $\mathcal{C}$ is generally a *manifold* (e.g. a torus for a 2R arm), not $\mathbb{R}^n$: same dimension does not mean same shape ($\mathrm{SO}(2)^3 \neq \mathrm{SO}(3)$).
- Euclidean distance in parameter space is misleading on a manifold; geodesics are correct but hard to compute.
- Roboticists use the *displacement metric* (max point displacement) or its practical version with a few *control points*.

Part II
Lecture 2

About these notes

These notes have two parts. **Part I** is based on the lecture slides (`WheeledMobileRobots1_Slides.pdf`, slide ranges in parentheses). **Part II** is based on the second half of the lecture, provided as handwritten notes (`AMR02.pdf`, page ranges in parentheses). No companion book chapter was provided for either part, so the formal statements (Definitions, Propositions, Lemmas, Theorems, proofs) formalise the content of the slides/notes with standard results. Whatever is *not* in the source is explicitly tagged *(beyond the slides)* or *(beyond the notes)*. In Part II I corrected a typo in the notes ($\partial h/\partial k$ read as $\partial h/\partial q$) and added the solution of the example left open on page 6. Red dashed or framed boxes marked "Blank" are spots for photographs/videos on the slides that were deliberately not redrawn; all other figures are original TikZ drawings.

Part III
Mechanics of Mobile Robots

7 Outline of this lecture *(slides 1–2)*

The lecture opens the part of the course on *wheeled mobile robots* (WMRs). It covers four topics:

- **ground locomotion**: how a robot moves by contact with the ground;
- **balance**: what it takes for the robot not to fall;
- **wheels**: the three basic wheel types;
- **kinematic structures**: the classical ways of combining wheels on a chassis.

Intuition

Think of this lecture as the *hardware vocabulary* of the course. Later lectures write equations of motion (kinematics, control) for each structure; to do so one first needs to know which wheel is where, which wheels are motorised, and which can steer. Everything here is about that bookkeeping.

8 Ground locomotion *(slide 3)*

Moving on the ground requires *contact* between the robot and the terrain. The slides distinguish two ways of establishing it.

Definition 8.1 (Wheeled mobile robot)**.** A *wheeled mobile robot* (WMR) typically consists of a single rigid body (the *base* or *chassis*) plus a set of wheels that connect it to the ground.

Definition 8.2 (Legged robot)**.** A *legged robot* typically consists of *several* rigid bodies articulated through joints; the contact with the ground is made through the feet.

Some robots achieve ground locomotion with neither wheels nor feet, for instance *snake robots*.

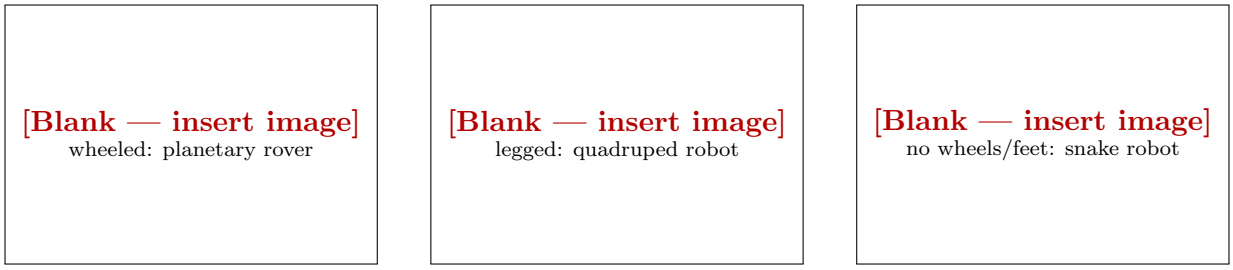


Figure 18: Photographs on slide 3 (not redrawn).

Intuition

The difference is in the *mechanical complexity*: a WMR is one rigid body plus wheels, so its configuration is just the position and orientation of the chassis (plus steering angles); a legged robot is an articulated system, and every joint adds a degree of freedom to control. This is why the course starts with WMRs.

9 Balance (not falling) *(slides 4–5)*

9.1 Static balance and the support polygon

Definition 9.1 (Support polygon)**.** The *support polygon* is the convex hull of the points where the robot touches the ground: the wheel contact points for a WMR, the footprints for a legged robot.

Definition 9.2 (Static balance)**.** A robot is *statically balanced* when the vertical projection of its Center of Mass (CoM) on the ground falls inside the support polygon.

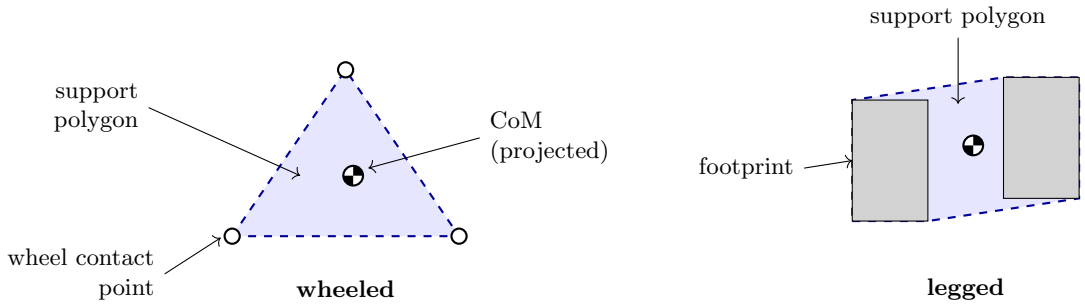


Figure 19: Support polygons (slide 4). Wheeled robot: triangle through three wheel contact points. Legged robot: convex hull of the two footprints. The projected CoM (⊗-like symbol) must lie inside.

For WMRs the slides state a consequence: *one needs 3 wheels*. Here is the formal reason.

Proposition 9.3 (Three wheels are necessary)**.** *A robot whose support polygon has non-empty interior needs at least three ground contact points that are not collinear. Consequently, a WMR that must be statically balanced with a robust margin needs at least 3 wheels.*

Proof. The convex hull of one point is a point; of two points, a segment; of three or more collinear points, still a segment. All of these have empty interior in the plane, so the CoM projection can

lie in them only by an unstable coincidence. Three non-collinear points span a triangle, which has non-empty interior. □

Intuition

A stool with one or two legs falls; a three-legged stool stands. With two wheels the robot can rotate about the line through the two contact points (like a seesaw), unless extra dynamic control is used. This is why the differential-drive robot of slide 9 adds a third, passive wheel (a caster) "for balance".

The support polygon also tells us how the weight is shared among the wheels.

Example 9.4. A differential-drive robot has driving wheels at $p_1 = (0, 0.3)$, $p_2 = (0, -0.3)$ and a caster at $p_3 = (0.4, 0)$ (metres), so $A(p_1, p_2, p_3) = 0.12\,\mathrm{m}^2$. With $c = (0.1, 0.05)$ the algorithm gives $\lambda = (0.458,\ 0.292,\ 0.25)$: all positive, so the robot is balanced and the caster carries 25% of the weight. After loading a payload that moves the CoM to $c = (0.3, 0.1)$, we get $\lambda = (7/24,\ -1/24,\ 3/4)$. Since $\lambda_2 < 0$, wheel 2 lifts and the robot tips over the edge p_1p_3.

Intuition

The weight splits among the wheels like a seesaw: the closer the CoM is to a wheel, the more load that wheel carries. In the example the caster carries a quarter of the weight, so 75% of the weight presses on the driving wheels, which is what gives them traction. If the load moves far from the centre, one wheel is relieved of all weight and the robot tips. With four or more wheels the loads are no longer unique (like a four-legged table that wobbles on uneven ground), which is why real robots with more than three wheels need suspensions.

9.2 Dynamic balance and the Zero Moment Point *(slides 4–5)*

Definition 9.5 (Dynamic balance)**.** In *dynamic balance*, the CoM in the static criterion is replaced by the *Zero Moment Point* (ZMP), the point of the ground about which the horizontal components of the moment of gravity and inertial forces vanish. The robot is dynamically balanced when the ZMP lies inside the support polygon.

Example 9.6. With $z_c = 0.8\,\mathrm{m}$, $\ddot{z}_c = 0$ and $\ddot{x}_c = 1\,\mathrm{m/s}^2$: $x_{\mathrm{zmp}} = x_c - 0.8/9.81 \approx x_c - 8.2\,\mathrm{cm}$. The ZMP lies 8.2 cm behind the CoM projection.

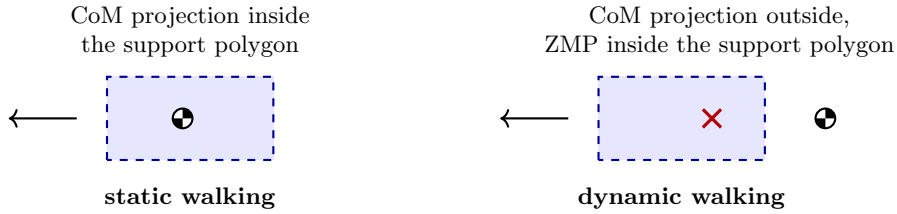


Figure 20: Static vs. dynamic walking (slide 5), shown for a single stance foot, top view. Arrows: walking direction. Red cross: ZMP.

Intuition

Carrying a tray while walking fast: you do not keep the tray's weight right above your feet, you lean and let motion help. Static walking is the cautious version, the CoM never leaves the foot area, so the robot could stop at any instant without falling; it is slow. In dynamic

walking the CoM may be momentarily outside the foot, and the robot stays up because the accelerations move the ZMP back inside. The slide contrasts two humanoids walking one way or the other.

10 Wheels: 3 basic types *(slides 6–8)*

Wheels are classified by how their orientation relates to the chassis and by whether they are motorised (*active*) or free (*passive*).

10.1 Fixed wheel *(slide 6)*

Definition 10.1 (Fixed wheel)**.** A wheel with *fixed orientation* with respect to the chassis. It may be *active* (used for driving) or *passive* (used for balance).

10.2 Orientable (steerable) wheel *(slide 7)*

Definition 10.2 (Orientable wheel)**.** A wheel whose orientation with respect to the chassis is *variable*. It is typically *active* (used for steering).

10.3 Caster wheel *(slide 8)*

Definition 10.3 (Caster wheel)**.** A wheel with variable orientation with respect to the chassis, typically *passive* (used for balance) but possibly active, that *automatically aligns with the direction of motion*.

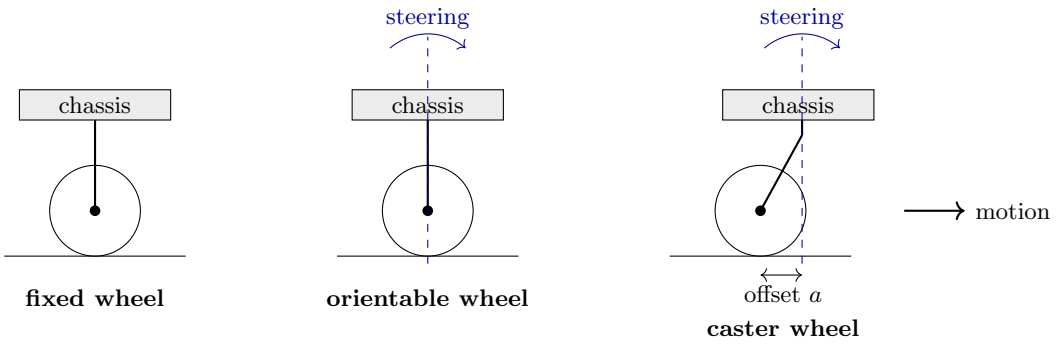


Figure 21: Side views of the three wheel types. The steering axis of the orientable wheel passes through the wheel centre; that of the caster is offset by a distance a, with the wheel trailing behind it.

Wheel	Orientation w.r.t. chassis	Typical role
Fixed	constant	active (driving) or passive (balance)
Orientable	variable	active (steering)
Caster	variable, self-aligning	passive (balance), possibly active

Table 1: The three basic wheel types (slides 6–8).

Intuition

It is the same reason a shopping-cart wheel swings round when you start pushing, or why a trailer follows the car: the wheel is *pulled* by a pivot ahead of it, and a pulled object trails behind. With $v = 0.5\,\mathrm{m/s}$ and $a = 5\,\mathrm{cm}$ the time constant is $a/v = 0.1\,\mathrm{s}$: the wheel aligns in about a tenth of a second. The *offset* is essential: with $a = 0$ the wheel would be an ordinary orientable wheel, which stays wherever it is steered. Fixed wheels give grip and balance, orientable wheels give steering, casters give free support.

11 WMR kinematic structures *(slides 9–15)*

The wheel types are combined on a chassis in a few classical ways. The slides present seven structures.

11.1 Differential-drive mobile robot *(slide 9)*

Definition 11.1 (Differential-drive WMR)**.** A WMR with **2 fixed wheels** on a common axle, **independently actuated**, plus **1 caster** added for balance.

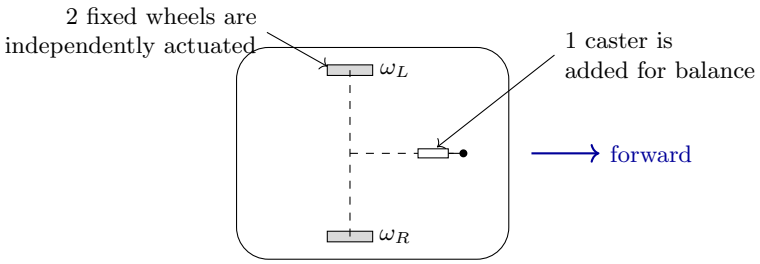


Figure 22: Differential-drive robot (slide 9), top view.

Intuition

It is a tank (or a wheelchair): equal speeds drive straight, different speeds curve towards the slower wheel, opposite speeds spin on the spot. With $r = 0.1\,\mathrm{m}$, $d = 0.4\,\mathrm{m}$: $\omega_R = \omega_L = 5\,\mathrm{rad/s}$ gives $v = 0.5\,\mathrm{m/s}$, $\omega = 0$; $\omega_R = -\omega_L = 5\,\mathrm{rad/s}$ gives $v = 0$, $\omega = 2.5\,\mathrm{rad/s}$. The slides show an indoor mobile base and a robot vacuum cleaner as examples. The caster only keeps the robot from tipping, as the balance section requires.

11.2 Synchro-drive mobile robot *(slide 10)*

Definition 11.2 (Synchro-drive WMR)**.** A WMR with **3 orientable wheels** that are **simultaneously actuated**. The orientation of the chassis remains constant.

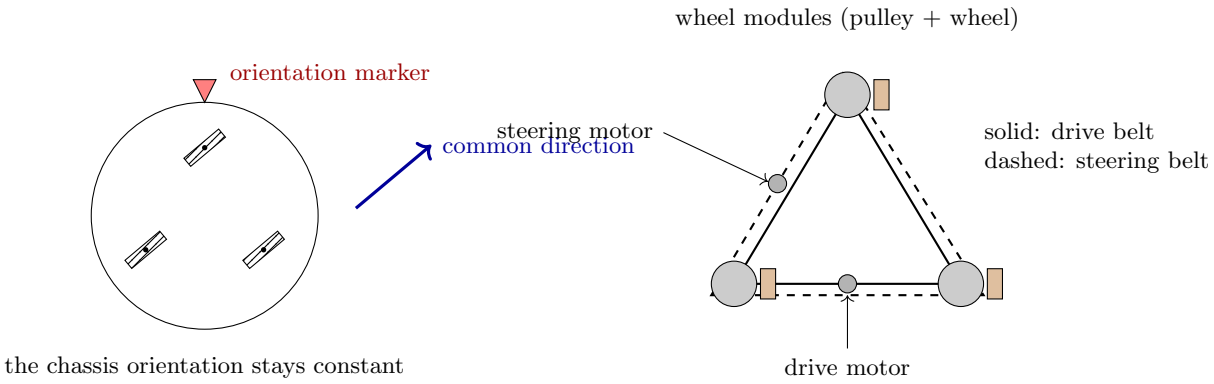


Figure 23: Synchro-drive (slide 10). Left: three orientable wheels, always parallel. Right: simplified drive train redrawn from the slide: one drive motor and one steering motor act on all the wheels through belts.

Intuition

The three wheels are like three people pushing a table in exactly the same direction at the same speed: the table translates and never turns. The robot changes its direction of travel by turning all wheels together, but its body keeps pointing the same way; that is why the sensors or the "head" of a synchro-drive robot would need their own rotation mechanism (the slide shows a laptop riding on top). The price of this simplicity is a drive train with belts, but only two motors for three wheels.

11.3 Tricycle and car-like robots *(slide 11)*

Definition 11.3 (Tricycle)**. 2 fixed wheels** on a common axle and **1 orientable wheel**.

Definition 11.4 (Car-like)**. 2 fixed wheels** on a common axle and **2 orientable wheels** on a common axle.

Both may be front-wheel drive or rear-wheel drive.

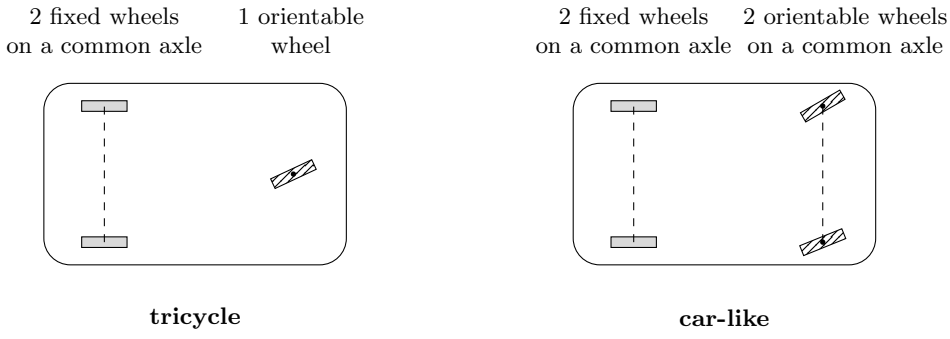


Figure 24: Tricycle (left) and car-like robot (right), top views (slide 11). The tilted wheels are steered.

Intuition

When a car turns, each wheel must roll along a circle with the same centre, and the inner circle is smaller; the inner wheel therefore has to turn more than the outer one. Numbers:

$\ell = 2.5\,\text{m}$, $w = 1.5\,\text{m}$, $\delta_i = 30°$: $\cot\delta_o = 1.732 + 0.6 = 2.332$, so $\delta_o \approx 23.2°$. A tricycle, with only one steered wheel, does not have this issue. Front-wheel drive means the steered wheel(s) are also driven; rear-wheel drive means the fixed axle is driven, which requires the device of the next subsection when the two rear wheels share a motor.

11.4 The differential *(slide 12)*

Definition 11.5 (Differential)**.** A mechanical device needed whenever two *driving* wheels are mounted on a *common axle* (driven by the same motor), which allows the two wheels to move at *different speeds*.

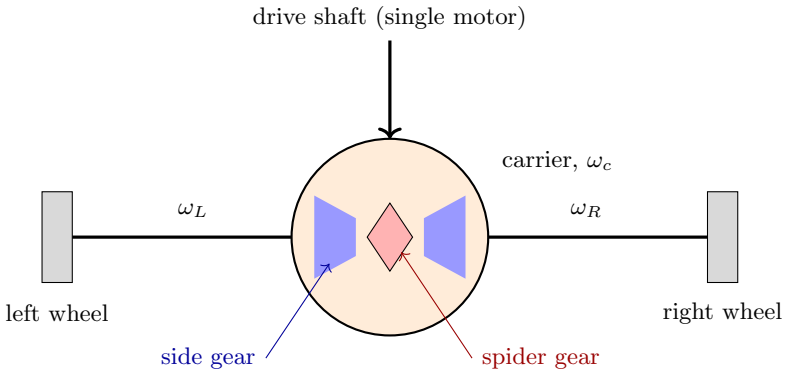


Figure 25: Schematic of a differential (idealised). The slide shows an animated 3D model.

Intuition

Why is it needed? When a car turns, the outer wheel travels along a longer arc than the inner one, so the two must rotate at different speeds; if rigidly tied to one shaft they would drag and slip. The differential lets the wheels share the motor while spinning differently. Numbers: wheels at 4 and 6 rad/s $\Rightarrow$ the carrier turns at 5 rad/s. A known side effect of equal torque split: if one wheel is on ice, it can only transmit a tiny torque, and so can the other.

11.5 Firetruck *(slide 13)*

Definition 11.6 (Firetruck)**.** A **tractor** (car-like) coupled to a **trailer** having **2 orientable wheels on a common axle**.

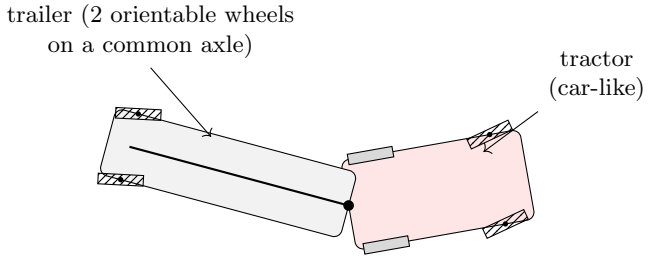


Figure 26: Firetruck (slide 13), top view; the dot is the hitch.

Intuition

A ladder firetruck is very long. A second person (the "tiller operator", as the slide's overlay says) steers the *rear* axle so that the trailer does not cut the corner and can pass around cars and curbs. In kinematic terms, it is a car-like robot pulling a trailer with its own steering.

11.6 Omnidirectional robots with actuated caster wheels *(slide 14)*

Definition 11.7 (Omnidirectional WMR)**.** A WMR that can instantaneously move in any direction of the plane, *without first re-orienting the chassis*, and rotate independently of translation.)

The structure on the slide uses **active caster wheels**, i.e. caster wheels that are both steered and driven.

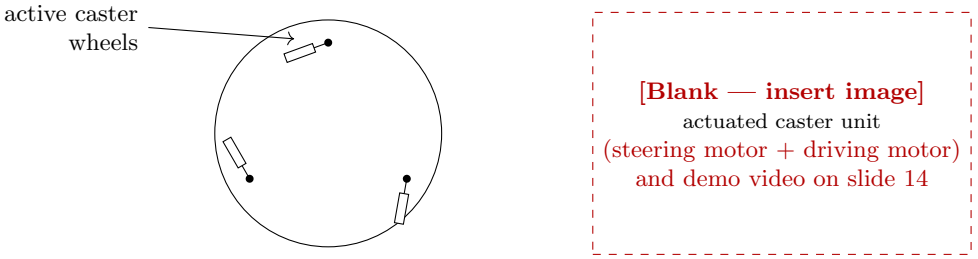


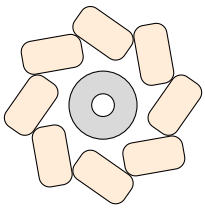
Figure 27: Omnidirectional robot with three active casters (slide 14, left drawing) and a spot for the slide's photo/video.

Intuition

Each wheel can point wherever it likes and be driven at the right speed, so it can follow the velocity that its own mount point must have. If all wheels point the same way, the robot translates in that direction; if their directions differ, the robot rotates. The effect is like a shopping cart that can instantly go sideways.So something that a car can't do, however the cost is mechanical: two motors per wheel. So more expensive, more complex and also more difficult to control than a differential-drive robot, but it can move in any direction without turning first.

11.7 Mecanum (Swedish) wheels *(slide 15)*

Definition 11.8 (Mecanum wheel)**.** A wheel with a ring of free-spinning rollers mounted around its rim, with the roller axes oblique to the wheel axle. Omnidirectional mobile robots can be built using Mecanum (Swedish) wheels.



[Blank — insert image]
Mecanum robot (slide 15, top photo)

[Blank — insert image]
Mecanum wheelchair (slide 15, bottom photo)

Figure 28: A Mecanum wheel seen along its axle (slide 15, left drawing) and spots for the two photos.

Intuition

The rollers turn the "wheel" into one that can roll forward *and* slide sideways freely: the ground force of each wheel acts along the axis of its roller instead of along the wheel plane. Choosing the speeds of several such wheels, one can obtain any combination of forward, sideways and rotational motion of the chassis, with no steering mechanism at all, a purely fixed-axle structure that is nonetheless omnidirectional.The problem is that the rollers have to be really close to the ground to transmit force, so the robot has to be very low to the ground. The slide shows a small robot and a wheelchair as examples.

Part IV

Wheels: Constraints and Mobility

12 Wheels (constraints & mobility) *(page 1)*

Wheels do not merely support the robot: they *introduce constraints* on its motion. We can study them in four steps:

- **geometric constraints** (GC), which limit the configurations q the robot can take;
- **kinematic constraints** (KC), which limit the velocities $\dot{q}$ it can have;
- the **relationships** between GC and KC;
- the resulting **mobility** of the robot,

and then apply everything to the example of **rolling wheels**.

Intuition

Compare a bead on a wire with an ice skater. The bead can never leave the wire: a restriction on *where* it can be. The skater's blade cannot slide sideways at any instant: a restriction on *how* she can move, yet she can still reach every point of the rink. Both are "constraints" but of two different natures, and this part of the lecture is about telling them apart. Wheels turn out to be of the skater type.

13 Geometric constraints (GC) *(pages 1–2)*

13.1 Constraints on the configuration *(page 1)*

Definition 13.1 (Configuration space)**.** The configuration of the robot is a vector $q \in \mathcal{C}$ of generalized coordinates, where the configuration space $\mathcal{C}$ is (locally) $\simeq \mathbb{R}^n$. The number n is the number of coordinates.

Constraints are classified along two axes: *equality* (bilateral) vs. *inequality* (unilateral), and *time-variant* vs. *time-invariant*.

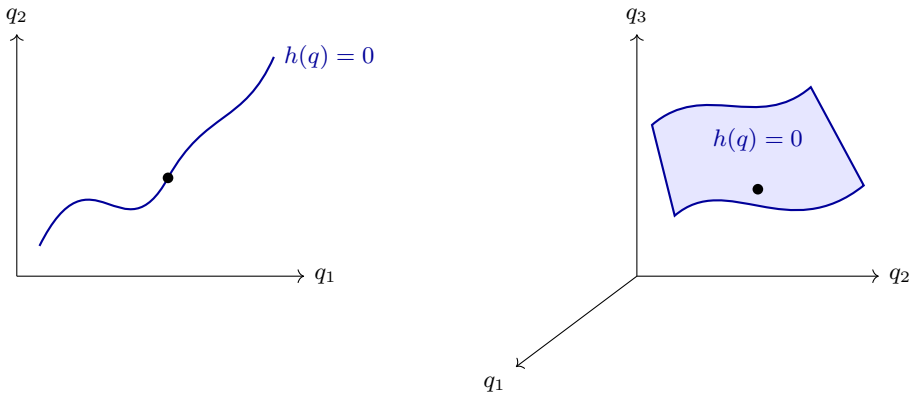
	time-invariant	time-variant
equality (bilateral)	$h(q) = 0$	$h(q,t) = 0$
inequality (unilateral)	$h(q) \geq 0$	$h(q,t) \geq 0$

Table 2: Classification of constraints on q (page 1). Only the first column of the first row is studied here.

Definition 13.2 (Geometric constraint)**.** Among the equality, time-invariant constraints, a *geometric constraint* (GC) has the form

$$h_i(q) = 0, \qquad i = 1, \ldots, k,$$

and *depends only on* q.



$\mathcal{C} \simeq \mathbb{R}^2$, $k = 1$
the robot can only stay on the curve

$\mathcal{C} \simeq \mathbb{R}^3$, $k = 1$
the robot can only stay on the surface

Figure 29: Geometric constraints (page 1): the dot is the robot, which must remain on the curve (left) or surface (right) $h(q) = 0$.

Intuition

A bead threaded on a wire has $n = 2$ coordinates in the plane but one constraint ($k = 1$): it lives on a curve. A point constrained to a surface in space ($n = 3$, $k = 1$) lives on a surface. In general, k constraints leave an $(n - k)$-dimensional world for the robot to live in.

13.2 Mobility limitation and reduced coordinates *(page 2)*

Proposition 13.3 (Global mobility limitation)**.** *If the constraints $h_1(q) = \cdots = h_k(q) = 0$ are independent, i.e. $\partial h/\partial q$ has rank k, the set $\{q : h(q) = 0\}$ is (locally) an $(n-k)$-dimensional subset of $\mathcal{C}$, and the robot configuration must remain in it for all times.*

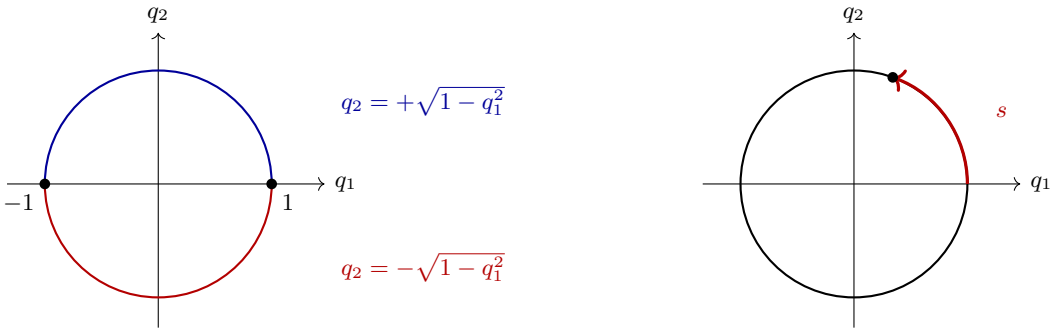
Proof. This is the implicit function theorem (regular value theorem): around any solution, k coordinates can be expressed as smooth functions of the remaining $n-k$. □

We call this a *global* mobility limitation: wherever the robot is, it is confined to a lower-dimensional subset.

Intuition

Are we then "wasting" the k remaining coordinates? Yes: they are determined by the other $n-k$. The number $n-k$ is the *number of degrees of freedom* (DOF), and one may try to describe the robot with a "reduced" configuration vector of $n-k$ coordinates: solve the GC for k coordinates as functions of the independent ones and keep only the latter.

Example 13.4 (The circle)**.** Take $q = (q_1, q_2) \in \mathbb{R}^2$ with $q_1^2 + q_2^2 = 1$. Solving for q_2 gives $q_2 = \pm\sqrt{1-q_1^2}$. This is only a *local* resolution: it needs a choice of branch, and at $q_1 = \pm 1$ the derivative $\partial h/\partial q_2 = 2q_2$ vanishes, so the square root breaks down (a *singularity*). So *we don't do that.* We use different coordinates instead, here the arc length $s \in [0, 2\pi)$, with $q = (\cos s, \sin s)$.



local resolution: two branches, singular at $q_1 = \pm 1$

arc-length coordinate $s \in [0, 2\pi)$

Figure 30: The circle $q_1^2 + q_2^2 = 1$ (page 2): solving the GC for one coordinate versus using a single new coordinate s.

Intuition

It is the difference between describing a point on the Earth's equator by "the latitude as a square root of something" and simply by its *longitude*: the longitude (our arc length s) is one honest coordinate that wraps around, with no branches and no singular points. In general one picks new coordinates that automatically satisfy the GC, instead of solving the constraint equations.

14 Kinematic constraints (KC) *(page 3)*

Definition 14.1 (Kinematic constraint)**.** A *kinematic constraint* (KC) has the form

$$a_i(q, \dot{q}) = 0, \qquad i = 1, \dots, k,$$

where $\dot{q}$ are the *generalized velocities* (time derivatives). It depends on q *and* $\dot{q}$; at any q it is a constraint on $\dot{q}$.

Definition 14.2 (Pfaffian form)**.** A KC is in *Pfaffian form* when $\dot{q}$ appears *linearly*:

$$a_i^T(q)\,\dot{q} = 0, \quad i = 1, \dots, k \qquad \Longleftrightarrow \qquad A(q)\,\dot{q} = 0, \quad A(q) = \begin{bmatrix} a_1^T(q) \\ \vdots \\ a_k^T(q) \end{bmatrix}.$$

KC are typically found in this form.

Proposition 14.3 (Admissible velocities)**.** *At each configuration q the admissible generalized velocities are exactly those in the null space $\mathcal{N}(A(q))$. If $A(q)$ has rank k, this is a linear subspace of dimension $n - k$.*

Proof. $A(q)\dot{q} = 0$ is the definition of the null space, whose dimension is $n - \operatorname{rank} A(q) = n - k$ by rank–nullity. □

Example 14.4. For $n = 3$ and the single constraint $a^T = (2, -1, 0)$, i.e. $2\dot{q}_1 - \dot{q}_2 = 0$, the admissible velocities are $\dot{q} = \alpha(1, 2, 0)^T + \beta(0, 0, 1)^T$: a plane ($n - k = 2$ dimensions) in velocity space.

Intuition

The KC does not forbid any *place*; it forbids some *ways of moving* at each place: at every q the allowed velocities form a plane (or subspace), and the plane can change with q. This is a **local** mobility limitation. The number of instantaneous degrees of freedom is still $n - k$, but *we cannot "reduce" the configuration vector*: all n coordinates are still needed (whether this is really so is the subject of the next section).

15 Relationships between GC and KC *(pages 4–6)*

15.1 Any GC implies a KC *(pages 4–5)*

Proposition 15.1 (GC ⇒ KC)**.** *Every GC $h(q) = 0$ implies the Pfaffian KC*

$$\dot{h}(q) = \frac{\partial h}{\partial q}\,\dot{q} = \left(\nabla_q h\right)^T \dot{q} = 0,$$

i.e. $a^T(q) = \partial h / \partial q$. Hence admissible velocities belong to the tangent plane of the surface $h(q) = 0$ at every q.

Proof. For the constraint to hold as time goes by, it must hold continuously: $h(q(t)) \equiv 0$, so

$\dot{h} = 0$. Since q depends on time, the chain rule gives $\frac{\partial h}{\partial q}\dot{q} = 0$, where $\left(\frac{\partial h}{\partial q_1}, \dots, \frac{\partial h}{\partial q_n}\right) = (\nabla_q h)^T$ is the gradient. This is a KC in Pfaffian form. □

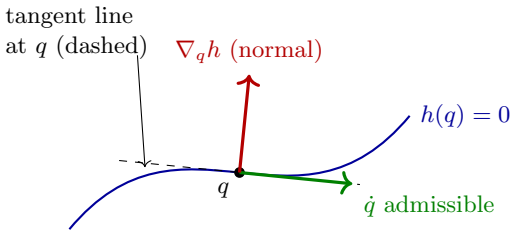


Figure 31: The gradient $\nabla_q h$ is normal to the surface $h(q) = 0$; admissible velocities are tangent to it, so $(\nabla_q h)^T \dot{q} = 0$ (page 4).

Intuition

Imagine hiking on a hillside: your velocity must be tangent to the slope, never pointing into the ground or up into the air. The gradient is the arrow sticking out of the surface (the normal), and "$a^T \dot{q} = 0$" simply says that the velocity is *orthogonal to the normal*: a dot product between orthogonal vectors is zero. So every geometric constraint secretly hides a velocity constraint.

15.2 The converse: integrable vs. non-integrable *(pages 5–6)*

Can we say the contrary, i.e. that a KC $a^T(q)\dot{q} = 0$ implies a GC $h(q) = 0$? In general no, sometimes yes: it depends on whether the KC can be *integrated*.

Definition 15.2 (Holonomic / nonholonomic)**.** A Pfaffian KC $a^T(q)\dot{q} = 0$ is *integrable* (*holonomic*) if there exist a function $h(q)$ and a function $\gamma(q) \neq 0$ such that

$$\frac{\partial h}{\partial q} = \gamma(q)\, a^T(q),$$

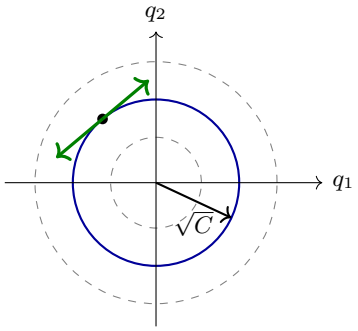
so that the KC is equivalent to $\dot{h} = 0$, i.e. to the GC $h(q) = \text{const}$. Otherwise it is *non-integrable* (*nonholonomic*), which is the general case.

Example 15.3 (Integrable: circles)**.** In $\mathbb{R}^2$, $q = (q_1, q_2)^T$, take $q_1\dot{q}_1 + q_2\dot{q}_2 = 0$, i.e. $a^T(q) = (q_1,\ q_2)$. Consider $q_1^2 + q_2^2 = C$ and differentiate: $2q_1\dot{q}_1 + 2q_2\dot{q}_2 = 0$; dividing by 2 gives the KC. So it is *integrable*, with $h = q_1^2 + q_2^2$ ($\gamma = 2$) and C an integration constant.

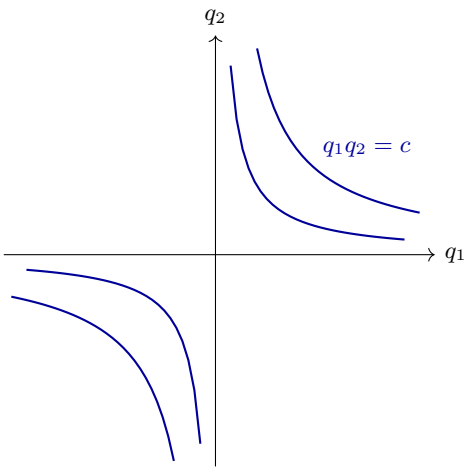
Proposition 15.4 (Local limitation ⇒ global limitation)**.** *If a KC is integrable, every admissible trajectory stays on a level set of h: $h(q(t)) = h(q(0))$ for all t.*

Proof. $\frac{d}{dt}h(q(t)) = \frac{\partial h}{\partial q}\dot{q} = \gamma(q)\, a^T(q)\dot{q} = 0$. □

Example 15.5 (Left open by the professor: $q_2\dot{q}_1 + q_1\dot{q}_2 = 0$ *(solution by me, not in the notes)*)**.** Here $a^T = (q_2,\ q_1)$ and $h(q) = q_1 q_2$ satisfies $\partial h/\partial q = (q_2, q_1) = a^T$ (with $\gamma = 1$): the KC is integrable and the robot is confined to a hyperbola $q_1 q_2 = c$.



$q_1^2 + q_2^2 = C$



$q_1 q_2 = c$

Figure 32: Left: the KC $q_1\dot{q}_1 + q_2\dot{q}_2 = 0$ is tangent to circles: starting at a point we cannot leave the circle (page 6). Right: the KC $q_2\dot{q}_1 + q_1\dot{q}_2 = 0$ integrates to hyperbolas.

Intuition

At each point the admissible velocity is tangent to a curve; chaining these tiny tangent steps, you simply trace the curve out: you can never hop onto another circle. This is why we can say: *the local limitation (of the KC) implies a global limitation (of the GC)*. The function γ is just a rescaling: multiplying the constraint by a non-zero factor does not change which velocities are admissible. How does one know if a constraint is integrable without guessing h? There are mathematical tools, such as **Frobenius' theorem**.

Theorem 15.6 (Frobenius, single constraint in $\mathbb{R}^3$ *(stated without proof)*)**.** *Let $a(q) \neq 0$, $q \in \mathbb{R}^3$. The constraint $a^T(q)\dot{q} = 0$ is integrable if and only if $a \cdot (\nabla \times a) = 0$.*

16 Example: rolling wheels *(pages 7–9)*

16.1 The rolling disk and the pure rolling constraint *(page 7)*

If the KC is **not integrable** (nonholonomic), the mobility limitation is *purely local*. The key example is the rolling disk.

Definition 16.1 (Rolling disk)**.** A disk (wheel) that stays upright while rolling on the plane. Its configuration is $q = (x, y, \theta)^T$, where (x, y) is the contact point and θ the orientation of the wheel plane (heading).

Lemma 16.2 (Pure rolling constraint)**.** *Pure rolling without lateral slip means that the instantaneous velocity of the contact point has no component along the unit normal $n = (\sin\theta, -\cos\theta)^T$ to the wheel plane:*

$$\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0, \quad \textit{i.e.} \quad \underbrace{(\sin\theta,\ -\cos\theta)}_{n^T} \begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = 0.$$

In the full configuration space this is the Pfaffian KC

$$a^T(q)\,\dot{q} = (\sin\theta,\ -\cos\theta,\ 0)\begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{pmatrix} = 0.$$

Proof. A velocity $(\dot{x},\dot{y})$ orthogonal to n is parallel to the heading $(\cos\theta,\sin\theta)$; the formula is the orthogonality condition $n^T(\dot{x},\dot{y})^T = 0$, with a zero entry for $\dot{\theta}$ since the constraint does not involve it. □

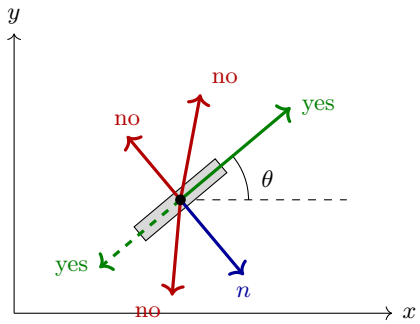


Figure 33: Top view of the rolling disk (page 7); the dot is the contact point (x,y). Velocities of the contact point with a component along the normal n are forbidden ("no"); only motion along the wheel plane is allowed ("yes").

Intuition

It is the ice-skater's blade: the contact point can glide forwards or backwards along the blade, but can never slide sideways, so the velocity is always lined up with θ. Since the condition concerns *instantaneous* velocity, it is a *local* mobility constraint. In fact the null space of $a^T(q)$ is spanned by $g_1 = (\cos\theta,\sin\theta,0)^T$ (roll) and $g_2 = (0,0,1)^T$ (spin): $\dot{q} = g_1 v + g_2\omega$, with $n-k = 3-1 = 2$ admissible directions at each configuration.

16.2 Is pure rolling integrable? *(page 8)*

Proposition 16.3 (Pure rolling is non-integrable)**.** *The pure rolling constraint is non-integrable.*

We decide to avoid the use of Frobenius' theorem and prove it *constructively*, by proposing a maneuver that moves the disk from anywhere to everywhere: the *universal maneuver*.

Algorithm 16.4 (Universal maneuver)**.** Input: start $q_s = (x_s,y_s,\theta_s)$ and goal $q_g = (x_g,y_g,\theta_g)$, with $(x_g,y_g) \neq (x_s,y_s)$. Let $\varphi = \operatorname{atan2}(y_g - y_s,\ x_g - x_s)$.

1. **Pure rotation** about the vertical axis, $\dot{x} = \dot{y} = 0$, from θ_s to φ (the KC is satisfied).
2. **Pure rolling**, $\dot{q} = v(\cos\varphi,\sin\varphi,0)^T$, until (x_g,y_g) is reached (the KC is satisfied because the contact point velocity is always aligned with $\theta = \varphi$).
3. **Pure rotation** from φ to θ_g (the KC is satisfied).

(If $(x_g,y_g) = (x_s,y_s)$, only a rotation is needed.)

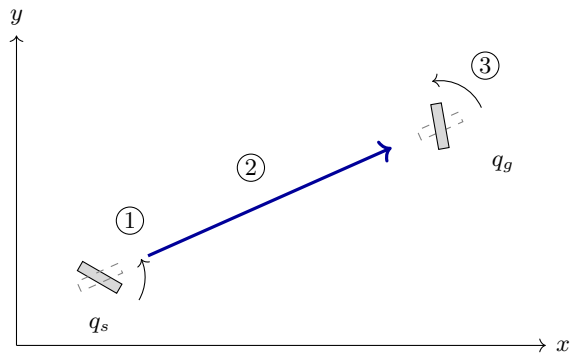


Figure 34: The universal maneuver (page 8): ① rotate in place towards the goal, ② roll straight to (x_g, y_g), ③ rotate to the final orientation. Dashed: intermediate configurations.

Proposition 16.5 (The maneuver is universal)**.** *For any $q_s, q_g \in \mathbb{R}^2 \times S^1$, Algorithm 16.5 produces a path that satisfies the pure rolling constraint at every instant and connects q_s to q_g.*

Proof. During phases 1 and 3, $\dot{x} = \dot{y} = 0$ so $a^T \dot{q} = 0 \cdot \sin\theta - 0 \cdot \cos\theta + 0 \cdot \dot{\theta} = 0$. During phase 2, $\theta = \varphi$ is constant and $\dot{q} = v(\cos\varphi, \sin\varphi, 0)$, so $a^T \dot{q} = v(\sin\varphi\cos\varphi - \cos\varphi\sin\varphi) = 0$. The three phases end respectively at heading φ, at position (x_g, y_g) and at heading θ_g. □

Corollary 16.6 (Non-integrability)**.** *The pure rolling constraint is non-holonomic.*

Proof. If it were integrable, by Proposition 15.4 every admissible trajectory would stay on a level set of a function h with $\partial h/\partial q = \gamma a^T \neq 0$, so h would not be constant; but by the universal maneuver 16.5 admissible paths connect any two configurations, so h would be constant on $\mathbb{R}^2 \times S^1$. Contradiction. □

Example 16.7. Let $q_s = (0, 0, 0)$ and $q_g = (1, 1, \pi/2)$ (metres, radians). Then $\varphi = \pi/4$: rotate by $\pi/4$ in place, roll $\sqrt{2} \approx 1.41\,\text{m}$ along the diagonal, then rotate by another $\pi/4$ to reach $\theta_g = \pi/2$.

Intuition

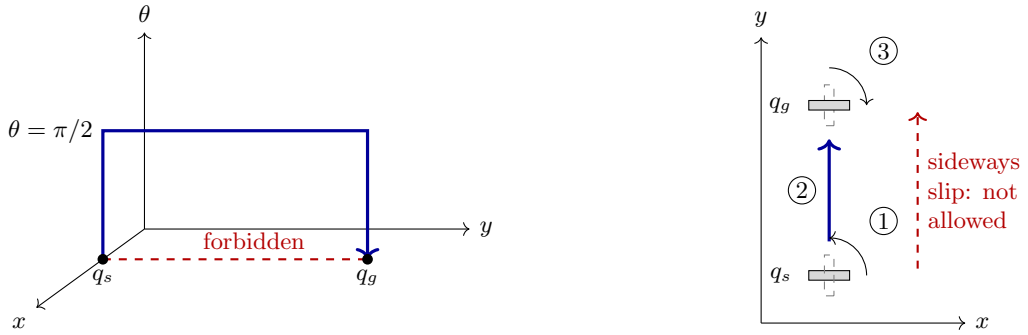
It is the recipe of "turn towards the goal, walk straight, turn to the final direction". Each of the three motions is allowed by the KC, and together they reach *any* configuration, even though at every instant only 2 of the 3 directions of q-space are admissible. A constraint that could be integrated would trap the robot on a curve or surface, whereas here nothing traps it.

16.3 No global mobility limitation *(page 9)*

Hence there is **no global mobility limitation**: the KC *must be non-holonomic*. Here we illustrate this with a "naive representation" of $\mathcal{C}$ and with parallel parking.

Example 16.8 (Parallel parking of a rolling disk)**.** Let $q_s = (0, 0, 0)$ and $q_g = (0, 2, 0)$: the goal is 2 m to the side, with the same orientation. The direct motion $\dot{y} \neq 0$ with $\theta = 0$ is not admissible, since $a^T \dot{q} = -\dot{y} \neq 0$: parallel parking *cannot be done instantaneously*. It can be

done with three admissible moves: (1) rotate to $\theta = \pi/2$; (2) roll 2 m along y (at $\theta = \pi/2$ the constraint reads $\dot{x} = 0$, so y is free); (3) rotate back to $\theta = 0$.



(a) path in $\mathcal{C}$

(b) top view

Figure 35: Parallel parking (page 9). (a) In $\mathcal{C} = \mathbb{R}^2 \times S^1$ the direct path at $\theta = 0$ is forbidden; the admissible path climbs along the θ axis, moves along y, and comes back down. (b) The same maneuver seen from above.

Intuition

Think of the three coordinates (x, y, θ) as a 3D world in which the disk may move only along two of the three directions at each point; yet by combining the motions, like a sailor tacking against the wind, every point becomes reachable. So the kinematic constraint *reduces how you can move* (local), but *not where you can get to* (global). Compare the two kinds of constraint: an integrable KC is a GC in disguise (global limitation, the dimension of the reachable set drops to $n - k$); a non-integrable KC (rolling) keeps all n coordinates alive and limits only the instantaneous velocities.

17 Summary

17.1 Part I: kinematic structures of WMRs

Structure	Wheels	Key feature	Slide examples
Differential-drive	2 fixed, independently actuated, + 1 caster (balance)	Steering through the speed difference of the two wheels	indoor base, robot vacuum cleaner
Synchro-drive	3 orientable, simultaneously actuated	Wheels always parallel; chassis orientation constant	cylindrical robot with laptop
Tricycle	2 fixed on a common axle + 1 orientable	Front- or rear-wheel drive	—
Car-like	2 fixed + 2 orientable, each pair on a common axle	Front- or rear-wheel drive; differential for a driven common axle	—
Firetruck	car-like tractor + trailer with 2 orientable wheels on a common axle	Trailer steered separately by the tiller operator	ladder truck
Omni, active casters	active caster wheels (steered and driven)	Any direction without re-orienting; 2 motors per wheel	actuated caster unit
Omni, Mecanum	wheels with oblique free rollers	Omnidirectional with fixed wheel axles	small robot, wheelchair

Table 3: Part I: comparison of the WMR kinematic structures (slides 9–15).

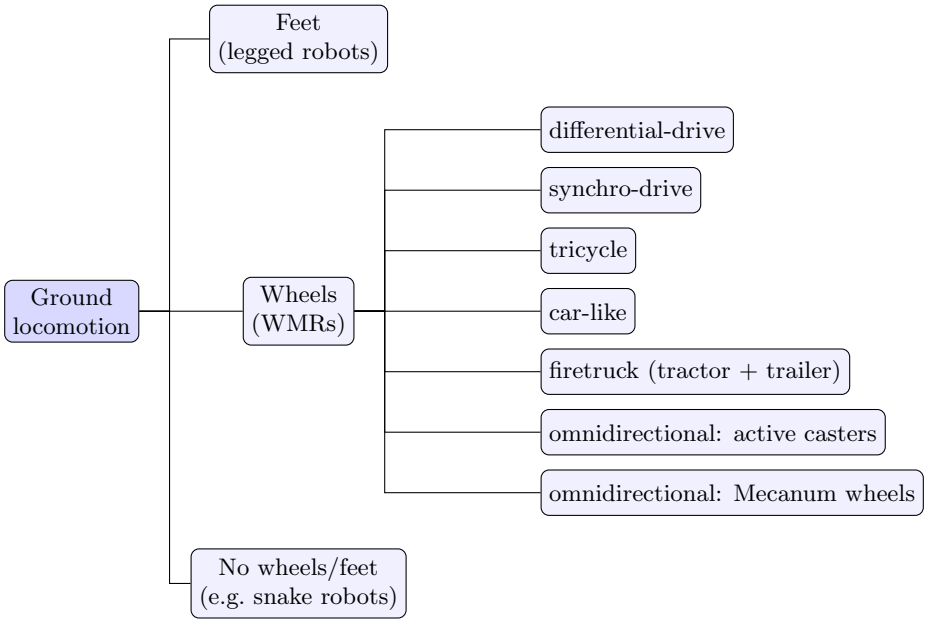


Figure 36: Part I map: from ground locomotion to the WMR kinematic structures.

17.2 Part II: constraints and mobility

Constraint	Form	Mobility limitation	Example
Geometric (GC)	$h(q) = 0$, depends only on q	*Global*: q confined to an $(n-k)$-dimensional set; reduced coordinates possible	circle $q_1^2 + q_2^2 = 1$, arc length s
Kinematic, integrable (holonomic)	$a^T(q)\dot{q} = 0$ with $\partial h/\partial q = \gamma\, a^T$	Equivalent to a GC: local limitation implies a global one	$q_1\dot{q}_1 + q_2\dot{q}_2 = 0$: circles
Kinematic, non-integrable (nonholonomic)	$a^T(q)\dot{q} = 0$, no such h	*Purely local*: $\dot{q} \in \mathcal{N}(A(q))$; every configuration is reachable; q cannot be reduced	rolling disk $\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0$

Table 4: Part II: types of constraints and their effect on mobility (pages 1–9).

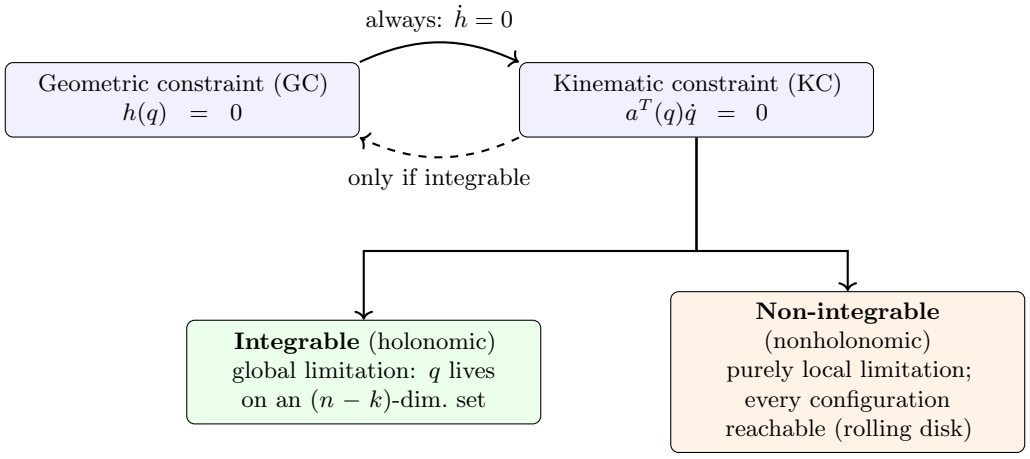


Figure 37: Part II map: relationships between geometric and kinematic constraints.

17.3 Key take-aways

- Ground locomotion needs contact (wheels or feet). Static balance requires the CoM projection inside the support polygon, hence at least 3 wheels; dynamic balance replaces the CoM with the ZMP.
- Wheels come in three types (fixed, orientable, caster) which combine into the classical structures: differential-drive, synchro-drive, tricycle, car-like, tractor–trailer, and omnidirectional robots (active casters or Mecanum wheels); a differential is needed when two driven wheels share an axle.
- Wheels introduce constraints on q. A geometric constraint $h(q) = 0$ confines the robot globally to an $(n-k)$-dimensional set; a kinematic constraint $a^T(q)\dot{q} = 0$ restricts the velocities locally to the null space of $A(q)$.
- Every GC implies a KC (the velocity must be tangent to the surface, $\dot{h} = 0$); the converse holds only for integrable (holonomic) KC, for which the local limitation implies a global one.
- Pure rolling, $\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0$, is non-integrable (Frobenius, or the universal "turn, roll, turn" maneuver): there is no global mobility limitation, all of (x, y, θ) is reachable, but not instantaneously, as in parallel parking.

Part V

Lecture 3

18 Kinematic constraints and admissible motions

(page 1)

Let the robot have a configuration vector $q \in \mathcal{C} \simeq \mathbb{R}^n$, where n is the number of generalized coordinates. The wheels impose *kinematic constraints* (KCs) that are linear in the generalized velocities.

Definition 18.1 (Kinematic constraints)**.** A set of k kinematic constraints is written in Pfaffian form

$$A(q) = 0, \qquad A(q) \in \mathbb{R}^{k \times n},$$

where k is the number of constraints and n the number of generalized coordinates.

Definition 18.2 (Admissible motions)**.** A generalized velocity is *admissible* at q if $\in (A(q))$, the null space of $A(q)$. If the constraints are independent, $\dim(A(q)) = n - k =: m$.

Intuition

The constraints do not forbid places, they forbid *instantaneous directions of motion.* At each configuration q the robot may move only along the directions in the null space. Think of a skate on ice: from any position it may glide along its blade, but not sideways. The set of allowed directions changes with the configuration, which is why A depends on q.

Pick a basis $\{g_1(q), \dots, g_m(q)\}$ of $(A(q))$. Each $g_i : \mathcal{C} \to \mathbb{R}^n$ is a *vector field*, and every admissible velocity is a combination

$$= \sum_{i=1}^{m} g_i(q)\, u_i = \begin{pmatrix} g_1(q) & \cdots & g_m(q) \end{pmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_m \end{pmatrix},$$

where the u_i are scalar coefficients, i.e. the *admissible velocities* (also called pseudo-velocities).

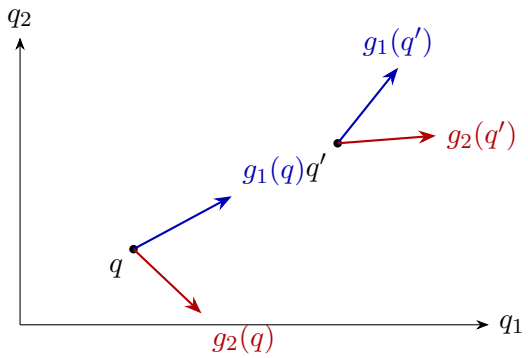


Figure 38: The basis vectors $g_i(q)$ are vector fields: at each configuration they give (generally different) admissible directions.

19 Explicit representation: the kinematic model

(pages 2–3)

Definition 19.1 (Kinematic model)**.** The *kinematic model* (KM) of a robot with m independent admissible directions is

$$= \sum_{i=1}^{m} g_i(q)\, u_i = G(q)\, u, \qquad G(q) \in \mathbb{R}^{n\times m} \text{ (a tall matrix)}, \; u \in \mathbb{R}^m.$$

For a chosen input u, at each q it returns the instantaneous motion the robot executes.

The KM is a *dynamical system* with

- state $q \in \mathcal{C} \simeq \mathbb{R}^n$ and input $u \in \mathbb{R}^m$;
- *nonlinear* dependence on the state q (linear in the input u);
- *driftless*: $u = 0 \Rightarrow = 0$, i.e. with no input the robot stays where it is.

Intuition

"Driftless" means there is no momentum in a kinematic model: stop commanding and the robot stops. (Momentum and forces appear only in the *dynamic* model, which is a different level of description.) "Tall matrix" simply says that we have more coordinates (n) than inputs (m): fewer commands than degrees of freedom of the configuration, which is exactly what the constraints cause.

Proposition 19.2 (The KM is not unique)**.** *Let $= G(q)u$ be a kinematic model and $T(q) \in \mathbb{R}^{m\times m}$ any smooth invertible matrix. Then $= G(q)T(q)\,u'$ is another kinematic model of the same robot, and the two describe the same set of admissible velocities.*

Proof. The columns of $G' = GT$ are linear combinations of the columns of G, hence lie in $(A(q))$; they are independent because T is invertible, so they are m independent vectors in an m-dimensional space, i.e. a basis. Conversely any admissible $= Gu = GT(T^{-1}u)$ is generated by $u' = T^{-1}u$. □

Intuition

The basis $\{g_1, \dots, g_m\}$ can be chosen in infinitely many ways (rotate, rescale, mix the vector fields), so *all these models are equivalent*. Some choices are nevertheless better because the inputs u_i get a physical *interpretation* (e.g. "forward speed" and "turning speed"). We will see this with the unicycle.

19.1 The unconstrained robot

(page 3)

Example 19.3 (KM of an unconstrained robot)**.** If there are no kinematic constraints ($k = 0$) and no geometric constraints, then $m = n$, A is absent and a natural basis of $\mathbb{R}^n$ is the canonical one:

$$= \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} u_1 + \cdots + \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} u_n = u.$$

This is the case of, e.g., a free-flying robot. Every generalized velocity is directly commanded: the robot is *fully actuated*.

Block schemes. The unconstrained robot is a bank of n simple integrators; a constrained robot places the matrix $G(q)$ between the pseudo-velocities u and the integrators, with q fed back to G.

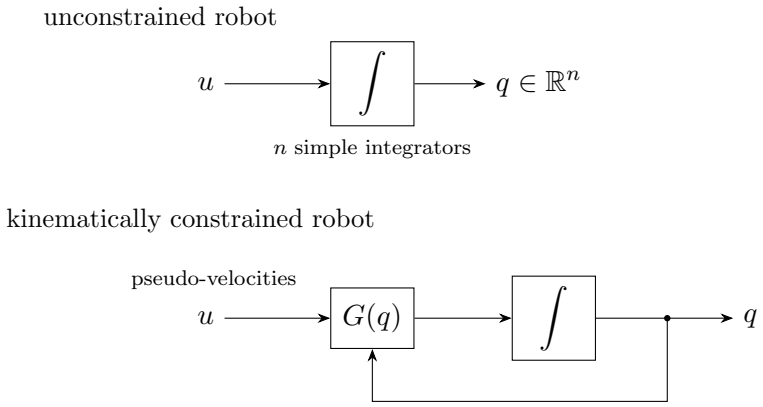


Figure 39: Block schemes of the kinematic model.

20 Controllability of kinematic models

(page 4)

Definition 20.1 (Controllable KM)**.** A kinematic model is *controllable* if for any initial configuration q_i and any final configuration q_f there exist a time T and an input $u : [0, T] \to \mathbb{R}^m$ such that the solution of $= G(q)u$ starting at q_i reaches q_f at time T.

Controllability can be established in two ways: *constructively*, by looking for a *universal maneuver* that connects any pair of configurations, or by applying a *test for controllability* (a rank condition, see Remark 22.3).

Link with the constraints. The notes relate controllability to the integrability of the kinematic constraints:

- the KM is controllable $\iff$ the KCs are *nonholonomic* (not integrable);
- the KM is not controllable $\iff$ the KCs are *holonomic*, i.e. integrable (partially or completely).

Intuition

If a constraint is integrable, it can be rewritten as a restriction on positions, $h(q) = \text{const}$ (for example $y = 0$): the robot is trapped on a lower-dimensional surface and can never reach the other configurations, so the KM cannot be controllable. If the constraints are not integrable they leave the robot able to reach every configuration by sequencing admissible motions, as a car does when parallel parking: it cannot move sideways instantaneously, but it can reach any pose.

21 The unicycle

(pages 4–6)

Definition 21.1 (Unicycle)**.** The unicycle is a vehicle with a single *orientable* wheel. Its configuration is $q = (x, y, \theta)^\top$, where (x, y) is the contact point and θ the orientation of the wheel's *sagittal axis* with respect to the x-axis. The configuration space is $\mathcal{C} = \mathbb{R}^2 \times \mathrm{SO}(2)$, locally $\simeq \mathbb{R}^3$ (hence $n = 3$).

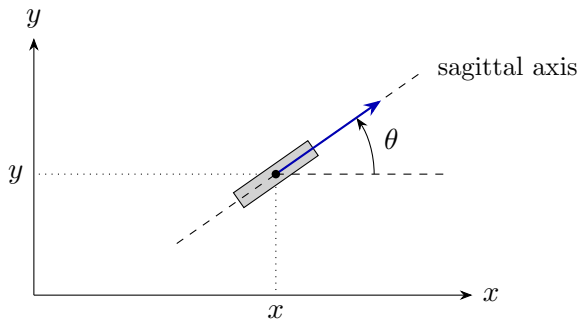


Figure 40: The unicycle seen from above: contact point (x, y), sagittal axis at angle θ.

21.1 The pure rolling constraint

(page 5)

The wheel rolls without slipping, so the contact point cannot move along the wheel axis (orthogonal to the sagittal axis):

$$\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0.$$

Equivalently, $(\dot{x}, \dot{y})$ must be *aligned with the sagittal axis*, i.e. it has no component along the direction orthogonal to it. In Pfaffian form

$$\underbrace{\begin{pmatrix}\sin\theta & -\cos\theta & 0\end{pmatrix}}_{A(q)} \begin{pmatrix}\dot{x}\\ \dot{y}\\ \dot{\theta}\end{pmatrix} = 0, \qquad k = 1,\ n = 3,\ m = n - k = 2.$$

Intuition

Worked numbers. At $\theta = 0$: $A = (0, -1, 0)$, so the constraint reads $\dot{y} = 0$: the wheel may translate along x and spin in place, never sideways. At $\theta = \pi/2$: $A = (1, 0, 0)$, so $\dot{x} = 0$: now it may move along y only. The forbidden direction rotates together with the wheel.

Proposition 21.2 (Basis of the admissible velocities)**.** *A basis of* $\big((\sin\theta \quad -\cos\theta \quad 0)\big)$*, a 2-dimensional linear space, is*

$$g_1(q) = \begin{pmatrix}\cos\theta\\ \sin\theta\\ 0\end{pmatrix}, \qquad g_2(q) = \begin{pmatrix}0\\ 0\\ 1\end{pmatrix}.$$

Proof. $Ag_1 = \sin\theta\cos\theta - \cos\theta\sin\theta = 0$ and $Ag_2 = 0$. The two vectors are linearly independent for every θ, and $\dim(A) = 3 - 1 = 2$. □

Kinematic model of the unicycle.

$$= \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u_2 \quad \Longleftrightarrow \quad \begin{cases} \dot{x} = \cos\theta\, u_1 \\ \dot{y} = \sin\theta\, u_1 \\ \dot{\theta} = u_2. \end{cases}$$

21.2 Interpretation of the inputs

(pages 5–6)

From the model, $\dot{x}^2 + \dot{y}^2 = \cos^2\theta\, u_1^2 + \sin^2\theta\, u_1^2 = u_1^2$, hence

$$u_1 = \pm\sqrt{\dot{x}^2 + \dot{y}^2} \quad \text{(equivalently } u_1 = \cos\theta\,\dot{x} + \sin\theta\,\dot{y}\text{)},$$

the magnitude of the velocity of the contact point, with sign + for forward and − for backward motion. We therefore rename

$$u_1 = v \text{ (driving velocity)}, \qquad u_2 = \dot{\theta} = \omega \text{ (steering velocity)}.$$

The driving velocity is related to the angular velocity of the wheel about its own axis by $v = r\,\omega_{\text{roll}}$ (r = wheel radius, side view in Fig. 41); ω is the angular velocity of the wheel about the vertical axis.

Definition 21.3 (Unicycle kinematic model)**.**

$$= \begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix} v + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \omega, \qquad \text{i.e.} \quad \begin{cases} \dot{x} = v\cos\theta \\ \dot{y} = v\sin\theta \\ \dot{\theta} = \omega. \end{cases}$$

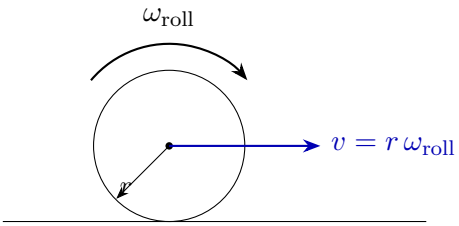


Figure 41: Side view of the wheel: the driving velocity is $v = r\,\omega_{\text{roll}}$.

Intuition

The basis $\{g_1, g_2\}$ was chosen so that the inputs are exactly what a driver feels: v is "how fast I roll along where I am pointing" and ω is "how fast I turn". Any other basis (Proposition 19.2) would give an equivalent but less readable model. Constant inputs give circles: with $v = 1$ m/s and $\omega = 0.5$ rad/s the contact point moves on a circle of radius $v/\omega = 2$ m (see the figure in the Summary).

22 Controllability of the unicycle

(page 7)

Proposition 22.1. *The unicycle KM is controllable: it can be steered from any $q_i = (x_i, y_i, \theta_i)$ to any $q_f = (x_f, y_f, \theta_f)$.*

The notes answer "yes" by applying a universal maneuver, the same idea used earlier for the disk, in three steps.

Algorithm 22.2 (Universal maneuver for the unicycle)**.** Let $\varphi = \operatorname{atan2}(y_f - y_i,\ x_f - x_i)$ and $d = \sqrt{(x_f - x_i)^2 + (y_f - y_i)^2}$.

1. $v = 0$, $\omega =$ constant, until $\theta = \varphi$ (rotate in place to point at the goal).
2. $v =$ constant, $\omega = 0$, until $(x, y) = (x_f, y_f)$ (drive straight, distance d).
3. $v = 0$, $\omega =$ constant, until $\theta = \theta_f$ (rotate in place to the final orientation).

Proof of Proposition 22.1. During phases 1 and 3, $v = 0$ gives $\dot{x} = \dot{y} = 0$ and $\dot{\theta} = \omega$, so the position is frozen and θ changes at will; phase 1 takes θ from θ_i to φ in time $T_1 = (\varphi - \theta_i)/\omega$. During phase 2, $\omega = 0$ keeps $\theta = \varphi$ constant, so $(\dot{x}, \dot{y}) = v(\cos\varphi, \sin\varphi)$ is a straight line, reaching (x_f, y_f) after $T_2 = d/v$. Phase 3 takes θ from φ to θ_f. (If $d = 0$ phase 2 is skipped.) □

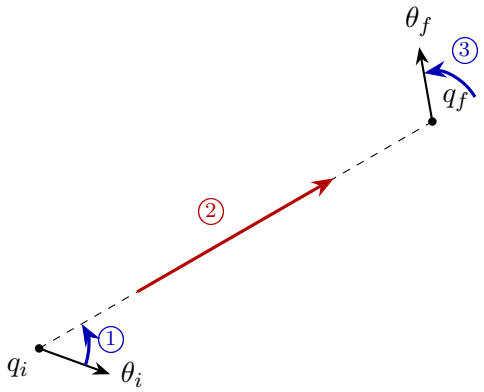


Figure 42: Universal maneuver: rotate in place, drive straight, rotate in place.

Intuition

It is a "point, drive, align" strategy. Worked numbers: from $q_i = (0, 0, 0)$ to $q_f = (3, 4, \pi/2)$ we have $\varphi = \arctan(4/3) \approx 0.927$ rad (53.1°) and $d = 5$. With $|\omega| = 1$ rad/s and $v = 1$ m/s: rotate for 0.927 s, drive for 5 s, rotate for $\pi/2 - 0.927 \approx 0.644$ s (36.9°). The two rotations "cost" nothing in position, which is possible only because the model lets us set $v = 0$ and turn freely.

Intuition

[Consistency with Section 3] Despite the constraint $\dot{y} = 0$ at $\theta = 0$ (no sideways motion), the unicycle can reach any pose, because the pure rolling constraint is *not integrable*: this is exactly the controllable/nonholonomic case.

Lemma 22.3 (Rank test, supplementary)**.** *Let* $[g_1, g_2] = \frac{\partial g_2}{\partial q} g_1 - \frac{\partial g_1}{\partial q} g_2$ *be the Lie bracket. For the unicycle* $[g_1, g_2] = (\sin\theta, -\cos\theta, 0)^\top$*, and* $\{g_1, g_2, [g_1, g_2]\} = 3 = n$ *for every* q*, so the controllability rank condition (Chow) holds.*

Proof. $\partial g_2/\partial q = 0$ and $\partial g_1/\partial q\, g_2 = (-\sin\theta, \cos\theta, 0)^\top$, hence $[g_1, g_2] = (\sin\theta, -\cos\theta, 0)^\top$, which is orthogonal to g_1 and independent of g_2. □

(This is the "test for controllability" mentioned on page 4; the explicit computation is not in the handwritten notes.)

Intuition

$[g_1, g_2]$ points *sideways*, exactly the forbidden direction: the wiggle "drive, turn, back up, turn back" produces a net sideways displacement, the same effect as parallel parking.

23 Mechanically equivalent vehicles

(pages 7–8)

Several real vehicles have the unicycle KM as soon as the right inputs are chosen.

23.1 Differential drive

(page 7)

Two parallel wheels (left and right) with a common axis; the reference point is the *midpoint between the two contact points*. Its KM is the unicycle one,

$$\begin{cases} \dot{x} = \cos\theta\, v \\ \dot{y} = \sin\theta\, v \\ \dot{\theta} = \omega \end{cases} \qquad v = r\,\frac{\omega_R + \omega_L}{2}, \qquad \omega = r\,\frac{\omega_R - \omega_L}{d},$$

where ω_R, ω_L are the roll velocities of the right and left wheel, r the wheel radius and d the distance between the wheels (the denominator is cut off in the handwritten page).

Proposition 23.1. *For a differential drive, the physical inputs (ω_R, ω_L) are related to the unicycle inputs (v, ω) by an invertible linear map.*

Proof. The contact points move with speeds $v_R = r\omega_R$ and $v_L = r\omega_L$. The midpoint has velocity $v = (v_R + v_L)/2$ and the body rotates with $\omega = (v_R - v_L)/d$. The matrix $\frac{r}{2}\begin{pmatrix} 1 & 1 \\ 2/d & -2/d \end{pmatrix}$ has determinant $-2r^2/d \neq 0$. □

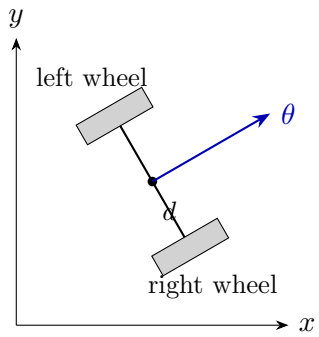


Figure 43: Differential drive seen from above; the reference point is the midpoint of the axle.

Intuition

Worked numbers: $r = 0.1$ m, $d = 0.4$ m, $\omega_R = 10$ rad/s, $\omega_L = 6$ rad/s. Then $v = 0.1 \cdot 16/2 = 0.8$ m/s and $\omega = 0.1 \cdot 4/0.4 = 1$ rad/s, a turn of radius $v/\omega = 0.8$ m. Equal wheel speeds give $\omega = 0$ (straight line); opposite speeds give $v = 0$ (rotation in place, the first and third phase of the universal maneuver).

23.2 Synchro-drive

(page 8)

A synchro-drive has three (or more) wheels that are steered and driven in a synchronized way, so that all wheels always point in the same direction θ. The notes state that the KM *is* the unicycle one and that *no input transformation is needed*:

$$\begin{cases} \dot{x} = \cos\theta\, v \\ \dot{y} = \sin\theta\, v \\ \dot{\theta} = \omega, \end{cases}$$

with v the common driving velocity and ω the common steering velocity.

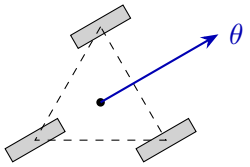


Figure 44: Synchro-drive: three wheels steered together, all with orientation θ.

Intuition

Because every wheel is always aligned with the others, the vehicle behaves as one big unicycle: the *body orientation* that the model calls θ is the common steering angle. Unlike the differential drive, the physical inputs (driving and steering velocity of the synchronized wheels) already coincide with (v, ω), so the model works as is.

24 Summary

System	n	k	m	Inputs u	Model
Unconstrained robot	n	0	n	$u \in \mathbb{R}^n$ (fully actuated)	$= u$ (canonical basis)
Unicycle	3	1	2	v (driving), ω (steering)	$= g_1 v + g_2 \omega$
Differential drive	3	1	2	$\omega_R, \omega_L \Rightarrow v = r\frac{\omega_R+\omega_L}{2}$, $\omega = r\frac{\omega_R-\omega_L}{d}$	unicycle KM
Synchro-drive	3	1	2	v, ω directly (no transformation)	unicycle KM

Table 5: Kinematic models covered. All are driftless and nonlinear in q; the three wheeled vehicles are controllable.

Key take-aways

- Wheels impose Pfaffian constraints $A(q) = 0$; admissible velocities form $(A(q))$, of dimension $m = n - k$.
- The kinematic model $= G(q)u$ is a nonlinear, driftless system; G is not unique, and bases are chosen for interpretability.

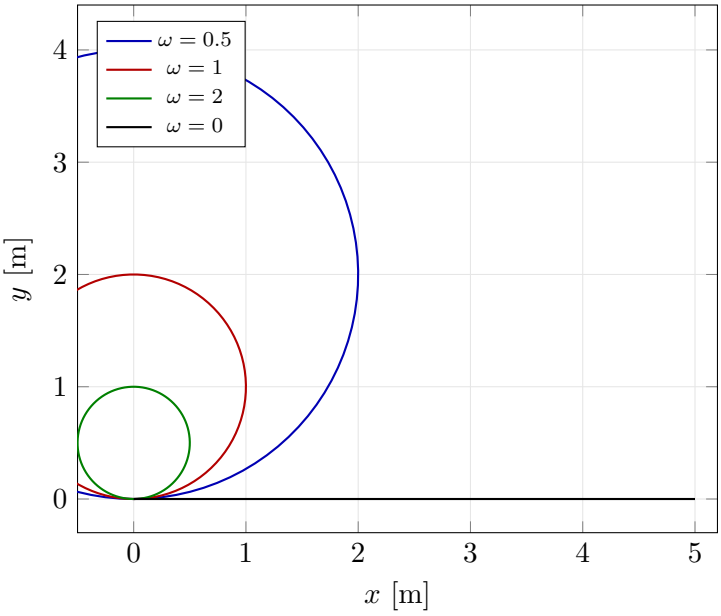


Figure 45: Unicycle with $v = 1$ m/s, $q_i = (0, 0, 0)$ and constant ω: circles of radius v/ω (straight line for $\omega = 0$).

- The unicycle $(n = 3, k = 1, m = 2)$ has $\dot{x} = v\cos\theta,\ \dot{y} = v\sin\theta,\ \dot{\theta} = \omega$; $v = r\omega_{\text{roll}}$ is the driving velocity and ω the steering velocity.
- Controllability of a KM is tied to nonholonomy of the constraints; the unicycle is controllable via the *rotate–drive–rotate* maneuver.
- Differential drive and synchro-drive share the unicycle KM (the former after the linear input change $v = r\frac{\omega_R+\omega_L}{2}$, $\omega = r\frac{\omega_R-\omega_L}{d}$).

Part VI

Lecture 4

Conventions. The driving and steering inputs are written u_1, u_2 (for the unicycle also v, ω). The Lie bracket is defined as $g_1 g_2 = \frac{\partial g_2}{\partial q} g_1 - \frac{\partial g_1}{\partial q} g_2$. $g_1 g_2 = \frac{\partial g_2}{\partial q} g_1 - \frac{\partial g_1}{\partial q} g_2$.

25 Recap: kinematic models and the unicycle

Definition 25.1 (Kinematic model)**.** Let a wheeled robot with configuration $q \in \mathbb{R}^n$ be subject to k independent Pfaffian (kinematic) constraints

$$A(q)\,\dot{q} = 0, \qquad A(q) \in \mathbb{R}^{k \times n}.$$

Admissible velocities are exactly $\dot{q} \in (A(q))$, a subspace of dimension $m = n - k$. Choosing a basis $g_1(q), \ldots, g_m(q)$ of the null space gives the *kinematic model*

$$\dot{q} = \sum_{i=1}^{m} g_i(q)\, u_i.$$

Example 25.2 (Unicycle)**.** With $q = (x, y, \theta)$ the single pure-rolling constraint is $\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0$, and

$$\dot{q} = \underbrace{\begin{pmatrix} \cos\theta \\ \sin\theta \\ 0 \end{pmatrix}}_{\text{drive VF } g_1} v + \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\text{steer VF } g_2} \omega.$$

Intuition

The two vector fields are the two "elementary motions" of the robot: g_1 rolls it along its heading without changing the heading, g_2 spins it in place. Every admissible velocity is a mix of the two. Example: at $\theta = \pi/4$ with $v = 1$, $\omega = 0$ the robot moves with $\dot{q} = (0.71, 0.71, 0)^T$, a straight diagonal line. We have $n = 3$ coordinates but only $m = 2$ inputs: the missing direction (sideways) is exactly what is forbidden by the constraint.

26 The bicycle model

26.1 Configuration and constraints

The bicycle has a fixed rear wheel and a steerable front wheel. Let (x, y) be the rear-wheel contact point, θ the orientation of the body, ϕ the steering angle, ℓ the wheelbase, and (x_f, y_f) the front-wheel contact point:

$$q = \begin{pmatrix} x \\ y \\ \theta \\ \phi \end{pmatrix}, \qquad \mathcal{C} = \mathbb{R}^2 \times SO(2) \times SO(2), \qquad n = 4.$$

Both wheels roll without lateral slipping:

$$\sin\theta\,\dot{x} - \cos\theta\,\dot{y} = 0 \qquad \text{(pure rolling, rear wheel)} \qquad \text{(R)}$$
$$\sin(\theta+\phi)\,\dot{x}_f - \cos(\theta+\phi)\,\dot{y}_f = 0 \qquad \text{(pure rolling, front wheel)} \qquad \text{(F)}$$

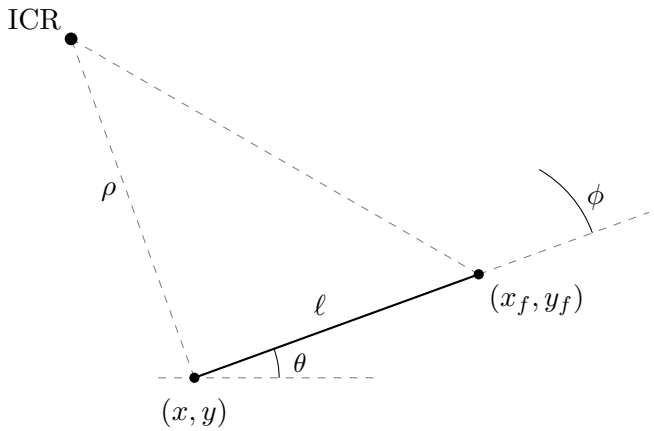


Figure 46: Bicycle geometry. The axes of the two wheels (perpendicular to the rolling directions) meet at the instantaneous centre of rotation (ICR); the whole robot rotates instantaneously about it, with turning radius $\rho = \ell/\tan\phi$.

Intuition

At every instant a rigid body that moves in the plane is rotating about a single point, the ICR. A rolling wheel can only move perpendicular to its axle, so the ICR must lie on the line through the axle. With two wheels, the ICR is the intersection of the two axle lines (Fig. 46) and *all* points of the robot contribute to this one rotation. Numbers: for $\ell = 1$ and $\phi = 30^\circ$ the radius is $\rho = 1/\tan 30^\circ \approx 1.73$; as $\phi \to 0$ the axle lines become parallel and the ICR escapes to infinity (straight motion).

26.2 Writing the front constraint with rear-wheel coordinates

Lemma 26.1 (Front-wheel constraint)**.** *Constraint (F) is equivalent to*

$$\sin(\theta+\phi)\,\dot{x} - \cos(\theta+\phi)\,\dot{y} - \ell\cos\phi\,\dot{\theta} = 0.$$

Proof. The front contact point is $x_f = x + \ell\cos\theta$, $y_f = y + \ell\sin\theta$, hence

$$\dot{x}_f = \dot{x} - \ell\sin\theta\,\dot{\theta}, \qquad \dot{y}_f = \dot{y} + \ell\cos\theta\,\dot{\theta}.$$

Substituting in (F):

$$\sin(\theta+\phi)\dot{x} - \cos(\theta+\phi)\dot{y} - \ell\dot{\theta}\big[\sin(\theta+\phi)\sin\theta + \cos(\theta+\phi)\cos\theta\big] = 0,$$

and the bracket equals $\cos\big((\theta+\phi)-\theta\big) = \cos\phi$. □

Proposition 26.2 (Constraint matrix)**.** *With $k = 2$ constraints, $A(q)\dot{q} = 0$ where*

$$A(q) = \begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 \\ \sin(\theta+\phi) & -\cos(\theta+\phi) & -\ell\cos\phi & 0 \end{pmatrix}.$$

Hence $\dim(A(q)) = n - k = 2$ *and two linearly independent vector fields are needed.*

26.3 Rear-wheel drive (RWD) kinematic model

Proposition 26.3 (RWD bicycle)**.** *A basis of* $(A(q))$ *is*

$$g_1(q) = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{\ell}\tan\phi \\ 0 \end{pmatrix}, \qquad g_2(q) = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

so that

$$\dot{q} = \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \frac{1}{\ell}\tan\phi \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2, \qquad u_1 = \pm\sqrt{\dot{x}^2 + \dot{y}^2}, \quad u_2 = \dot{\phi}.$$

Proof. $g_2 \in (A)$ trivially, because the last column of A is zero. For g_1 take $(\cos\theta, \sin\theta, \alpha, 0)^T$; the first row gives $\sin\theta\cos\theta - \cos\theta\sin\theta = 0$ for any α, and the second gives

$$\sin(\theta+\phi)\cos\theta - \cos(\theta+\phi)\sin\theta - \ell\cos\phi\,\alpha = \sin\phi - \ell\cos\phi\,\alpha = 0 \implies \alpha = \tfrac{1}{\ell}\tan\phi.$$

Finally $\dot{x}^2 + \dot{y}^2 = (\cos^2\theta + \sin^2\theta)u_1^2 = u_1^2$, so u_1 is the driving speed of the rear wheel, and $u_2 = \dot{\phi}$ is the steering velocity. □

Intuition

Think of the rear wheel as the "engine" and the front wheel as just a rudder. The rear point moves along the heading with speed u_1 (first two components of g_1), while the heading changes at the rate $\dot{\theta} = \frac{u_1}{\ell}\tan\phi = u_1/\rho$: exactly the angular rate of a point travelling at speed u_1 on a circle of radius $\rho = \ell/\tan\phi$. The steering vector field g_2 only turns the handlebar. Numbers: $\ell = 1$, $\phi = 30°$, $u_1 = 1$ gives $\dot{\theta} \approx 0.58$ rad/s.

Validity and singularity. The model is good *if the bicycle is rear-wheel drive.* At $\phi = \pi/2$ the vector field g_1 "blows up" ($\tan\phi \to \infty$): a singularity. Physically, with the front wheel perpendicular to the body, a nonzero rear speed $v \neq 0$ would require the front wheel to *slip* sideways, which the pure-rolling hypothesis excludes.

27 The front-wheel drive case

For a front-wheel-drive (FWD) vehicle the natural control is the speed of the front wheel, $u_1 = \pm\sqrt{\dot{x}_f^2 + \dot{y}_f^2}$.

Proposition 27.1 (FWD bicycle). *Rescaling $g_1 \mapsto g_1 \cos\phi$ and keeping g_2,*

$$\dot{q} = \begin{pmatrix} \cos\theta\cos\phi \\ \sin\theta\cos\phi \\ \frac{1}{\ell}\sin\phi \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2, \qquad \dot{x}_f^2 + \dot{y}_f^2 = u_1^2, \quad u_2 = \dot{\phi}.$$

Proof. From $\dot{x}_f = \dot{x} - \ell\sin\theta\,\dot{\theta}$ and $\dot{y}_f = \dot{y} + \ell\cos\theta\,\dot{\theta}$ we get

$$\dot{x}_f = u_1(\cos\theta\cos\phi - \sin\theta\sin\phi) = u_1\cos(\theta+\phi), \qquad \dot{y}_f = u_1\sin(\theta+\phi),$$

so $\dot{x}_f^2 + \dot{y}_f^2 = u_1^2$. Since $\cos\phi\, g_1 \in (A)$ and $g_2 \in (A)$, this is still a basis of the null space wherever $\cos\phi \neq 0$, and the rescaled field is also valid (smooth, nonzero) at $\phi = \pi/2$. □

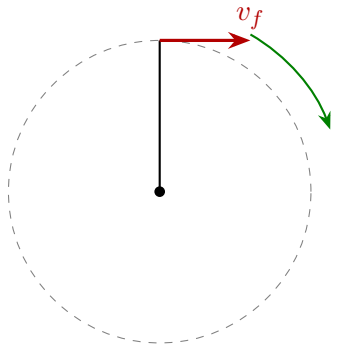


Figure 47: FWD bicycle at $\phi = \pi/2$: the front wheel pushes the vehicle sideways and it pivots about the rear-wheel contact point.

Intuition

If you *drive* the front wheel you can steer it all the way to $\phi = \pi/2$ and still move: the front wheel rolls in its own direction, perpendicular to the body, and the body simply rotates about the rear contact point (Fig. 47) with $\dot{\theta} = u_1/\ell$. So you can move in circle around the rear point *without hitting any singularity*. In the RWD model the same configuration would require infinite $\dot{\theta}$ per unit of rear speed.

28 Controllability

28.1 Is there a universal manoeuvre?

Algorithm 28.1 (Universal manoeuvre for the bicycle). To go from q_i to any q_f:

(1) use the ICR: with the wheel steered, drive along a circle until the heading points towards the goal;

(2) steer the wheel back to straight;

(3) go straight along a line, then finish with another circular arc to match the final heading and steering.

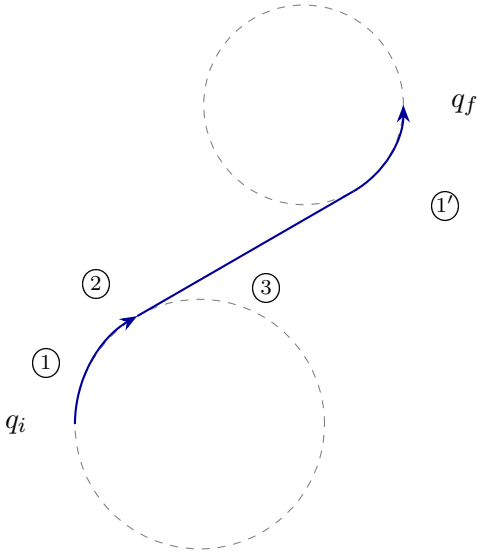


Figure 48: Schematic universal manoeuvre (arc around an ICR, straight line, arc). The numbers refer to the steps of the algorithm; $1'$ is the final arc that matches the target heading.

Intuition

It is the same idea as parking: any pose can be reached by combining only two primitives, rotating about an ICR and driving straight. Hence, although the bicycle has only two inputs for four configuration variables, *it can reach every configuration.* This is possible precisely because the velocity constraints are *non-integrable* (nonholonomic): if they were integrable, the robot would be confined to a lower-dimensional surface of $\mathcal{C}$ and could not reach everything.

28.2 Nonlinear controllability

We study driftless nonlinear systems

$$\dot{x} = \sum_{i=1}^{m} g_i(x) u_i, \qquad x \in \mathbb{R}^n,\ u \in \mathbb{R}^m,$$

("driftless": no term independent of u; for us x is the configuration q).

Definition 28.2 (Controllability)**.** The system is *controllable* if, for any pair (x_i, x_f), there exist $T > 0$ and an input $u(t)$, $t \in [0, T]$, such that the trajectory starting at $x(0) = x_i$ satisfies $x(T) = x_f$.

29 The Lie bracket

Definition 29.1 (Lie bracket)**.** Let g_1, g_2 be two vector fields on $\mathbb{R}^n$. The vector field

$$g_1 g_2(x) = \frac{\partial g_2}{\partial x}\, g_1(x) - \frac{\partial g_1}{\partial x}\, g_2(x)$$

(where $\partial g_i / \partial x$ is the Jacobian) is the *Lie bracket* (LB) of g_1, g_2.

Immediate properties: $g_1 g_2 = -g_2 g_1$ (hence $gg = 0$), and $g_1 g_2 = 0$ for constant fields.

Example 29.2 (A bracket in $\mathbb{R}^3$)**.** Let $g_1 = (\cos x_3, \sin x_3, 0)^T$ and $g_2 = (0, 0, x_1)^T$. Then

$$g_1 g_2(x) = \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{\partial g_2/\partial x} \begin{pmatrix} \cos x_3 \\ \sin x_3 \\ 0 \end{pmatrix} - \underbrace{\begin{pmatrix} 0 & 0 & -\sin x_3 \\ 0 & 0 & \cos x_3 \\ 0 & 0 & 0 \end{pmatrix}}_{\partial g_1/\partial x} \begin{pmatrix} 0 \\ 0 \\ x_1 \end{pmatrix}$$
$$= \begin{pmatrix} 0 \\ 0 \\ \cos x_3 \end{pmatrix} - \begin{pmatrix} -x_1 \sin x_3 \\ x_1 \cos x_3 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 \sin x_3 \\ -x_1 \cos x_3 \\ \cos x_3 \end{pmatrix}.$$

This is a *new* vector field which, in this case, is linearly independent of g_1, g_2 (the determinant of $[g_1 \; g_2 \; g_1 g_2]$ is x_1^2, nonzero for $x_1 \neq 0$).

Intuition

The bracket measures *how much the flows of two vector fields fail to commute*: moving first along g_1 then g_2 ends in a different place than g_2 then g_1. The difference of the two endpoints is (to second order) the bracket. If the bracket points in a direction that g_1, g_2 cannot generate directly, then "wiggling" between the two fields gives access to new directions of motion, which is the key to controllability.

30 Interpretation of the Lie bracket

Consider the 2-input system $\dot{x} = g_1(x)u_1 + g_2(x)u_2$ and the piecewise-constant input of duration 4ε.

Algorithm 30.1 (Lie-bracket manoeuvre)**.**

$$(u_1, u_2) = \begin{cases} (1, 0) & t \in [0, \varepsilon) \\ (0, 1) & t \in [\varepsilon, 2\varepsilon) \\ (-1, 0) & t \in [2\varepsilon, 3\varepsilon) \\ (0, -1) & t \in [3\varepsilon, 4\varepsilon). \end{cases}$$

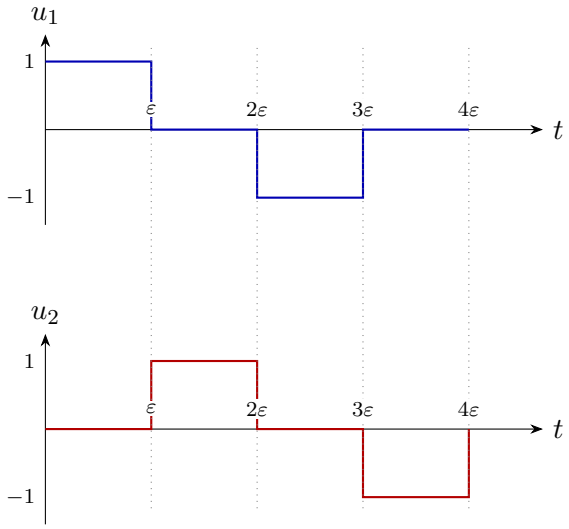


Figure 49: Control sequence of Algorithm 30.1: forward g_1, forward g_2, backward g_1, backward g_2.

Proposition 30.2 (Displacement of the LB manoeuvre)**.** *Starting from $x(0) = x_0$, the state after the sequence of Algorithm 30.1 is*

$$x(4\varepsilon) = x(0) + \varepsilon^2\, g_1 g_2(x_0) + O(\varepsilon^3),$$

where the remainder goes to zero as ε^3 or faster when $\varepsilon \to 0$.

Proof. Write $a = g_1(x_0)$, $b = g_2(x_0)$, $A = \partial g_1/\partial x(x_0)$, $B = \partial g_2/\partial x(x_0)$. The flow of a field g for time t is $x + t\,g(x) + \frac{t^2}{2}\frac{\partial g}{\partial x} g(x) + O(t^3)$. Applying the four phases in turn, up to terms $O(\varepsilon^2)$:

$$\begin{aligned} x_1 &= x_0 + \varepsilon a + \tfrac{\varepsilon^2}{2} A a, \\ x_2 &= x_1 + \varepsilon g_2(x_1) + \tfrac{\varepsilon^2}{2} B b = x_0 + \varepsilon(a+b) + \varepsilon^2\big(\tfrac{1}{2}Aa + Ba + \tfrac{1}{2}Bb\big), \\ x_3 &= x_2 - \varepsilon g_1(x_2) + \tfrac{\varepsilon^2}{2} A a = x_0 + \varepsilon b + \varepsilon^2\big(Ba + \tfrac{1}{2}Bb - Ab\big), \end{aligned}$$

using $g_1(x_2) = a + \varepsilon A(a+b) + O(\varepsilon^2)$. Finally $g_2(x_3) = b + \varepsilon B b + O(\varepsilon^2)$ and

$$x_4 = x_3 - \varepsilon g_2(x_3) + \tfrac{\varepsilon^2}{2} B b = x_0 + \varepsilon^2(Ba - Ab) + O(\varepsilon^3) = x_0 + \varepsilon^2 g_1 g_2(x_0) + O(\varepsilon^3). \qquad \square$$

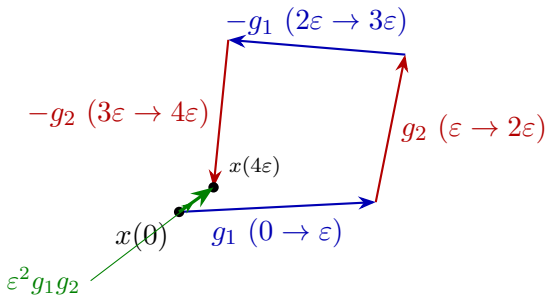


Figure 50: The four legs of the manoeuvre almost close a loop; the gap is the displacement $\varepsilon^2 g_1 g_2(x_0)$ (schematic, exaggerated).

Intuition

Going forward-sideways-backward-sideways "almost" returns to the start, like the parking manoeuvre. The residual gap is second order in the time, ε^2, and is the Lie bracket. Unicycle check ($\theta = 0$): go forward ε, rotate by $+\varepsilon$, go backward ε along the rotated heading, rotate back. The net motion in y is $-\varepsilon \sin \varepsilon \approx -\varepsilon^2$, i.e. a sideways shift, exactly $\varepsilon^2 g_1 g_2 = \varepsilon^2(\sin\theta, -\cos\theta, 0) = \varepsilon^2(0, -1, 0)$, even though the unicycle cannot move sideways directly. Numbers: with $\varepsilon = 0.1$ the path costs $4\varepsilon = 0.4$ time units but moves only $\varepsilon^2 = 0.01$ sideways: bracket motions are cheap in inputs but slow.

31 Distributions, accessibility and Chow's theorem

Definition 31.1 (Input distribution)**.** For $\dot{x} = \sum_{i=1}^{m} g_i(x)u_i$ the *input distribution* $= \{g_1(x), \ldots, g_m(x)\}$ associates to each x the subspace $\{g_1(x), \ldots, g_m(x)\}$ (all directions in which the inputs can push directly).

Definition 31.2 (Accessibility distribution)**.** The *accessibility distribution* is the distribution generated by the input fields and all their iterated brackets:

$$= \Big\{ g_1, \ldots, g_m,\ \ldots, g_i g_j,\ \ldots,\ g_i g_j g_k,\ \ldots \Big\},$$

with all 1st-order brackets, all 2nd-order brackets, and so on.

Theorem 31.3 (Chow)**.** *Consider the driftless system* $\dot{x} = \sum_i g_i(x)u_i$ *on* $\mathbb{R}^n$ *and assume* $\dim(x)$ *is constant for all* x*. Then the system is controllable if and only if*

$$\dim = n.$$

This result is known as *Chow's theorem*, the *Lie algebra rank condition* or the *accessibility rank condition*. In practice, for low-order brackets one computes

$$\Big(g_1 \ \ g_m \ \cdots \ g_i g_j \ \cdots \ g_i g_j g_k \ \cdots \Big) = n,$$

adding brackets of increasing order until the rank reaches n (or no new independent direction appears).

Intuition

The inputs give you m directions for free. Each bracket manoeuvre buys you one more direction (Prop. 30.2), brackets of brackets buy yet more. If after collecting all of them the directions span the whole state space, there is no direction in which you are stuck, so you can steer anywhere. Analogy: you can only drive a car forward/backward and turn, yet by combining these ($\to$ parallel parking) you effectively obtain a "slide sideways" control.

32 Application to the unicycle

Example 32.1 (Controllability of the unicycle)**.** With $q = (x, y, \theta)^T$, $g_1 = (\cos\theta, \sin\theta, 0)^T$

(drive) and $g_2 = (0,0,1)^T$ (steer):

$$g_1 g_2(q) = \underbrace{0}_{\partial g_2/\partial q = 0} \cdot g_1 - \begin{pmatrix} 0 & 0 & -\sin\theta \\ 0 & 0 & \cos\theta \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \sin\theta \\ -\cos\theta \\ 0 \end{pmatrix} =: g_3(q),$$

and

$$\begin{pmatrix} \cos\theta & 0 & \sin\theta \\ \sin\theta & 0 & -\cos\theta \\ 0 & 1 & 0 \end{pmatrix} = 3 = n.$$

(The determinant is 1 for all θ.)

Corollary 32.2. *The unicycle is controllable by Theorem 31.3.*

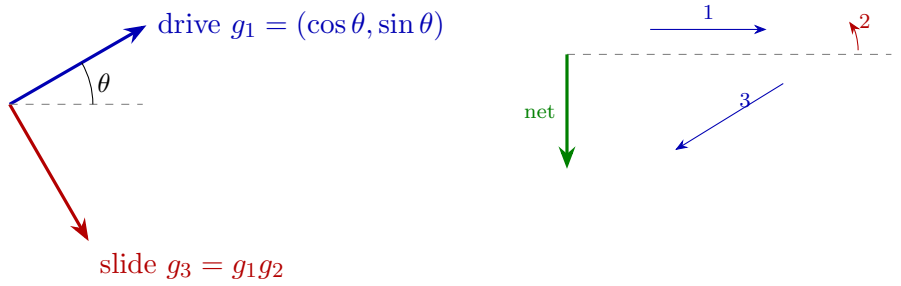


Figure 51: Left: drive and slide vectors of the unicycle. Right: LB manoeuvre (1 forward, 2 rotate, 3 backward, then rotate back): the net displacement is sideways (angles exaggerated).

Intuition

The new vector field $g_3 = g_1 g_2$ points perpendicular to the heading: it is the *slide* that the wheel forbids. Together g_1 (drive), g_2 (steer) and g_3 (slide) span all of $\mathbb{R}^3$ at every configuration, so the unicycle can reach every (x, y, θ).

33 Application to the bicycle

Take the RWD bicycle ($n = 4$)

$$\dot{q} = \begin{pmatrix} \cos\theta \\ \sin\theta \\ \tan\phi/\ell \\ 0 \end{pmatrix} v + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \omega.$$

The same computation applies to the FWD model. Since $n = 4$ and only two inputs are available, we need the brackets up to the second order.

Proposition 33.1 (Brackets of the bicycle)**.** *With g_1, g_2 as above,*

$$g_1 g_2 = -\frac{\partial g_1}{\partial \phi} = \begin{pmatrix} 0 \\ 0 \\ -\dfrac{1}{\ell \cos^2 \phi} \\ 0 \end{pmatrix},$$

$$g_1 g_1 g_2 = \begin{pmatrix} -\dfrac{\sin\theta}{\ell \cos^2 \phi} \\ \dfrac{\cos\theta}{\ell \cos^2 \phi} \\ 0 \\ 0 \end{pmatrix}.$$

Proof. Only g_1 depends on ϕ, and $g_2 = \partial/\partial\phi$ is constant, so $g_1 g_2 = -\frac{\partial g_1}{\partial q} g_2 = -\frac{\partial g_1}{\partial \phi} = -(0, 0, \frac{1}{\ell \cos^2 \phi}, 0)^T$. Let $w = g_1 g_2 = (0, 0, -\sigma, 0)^T$ with $\sigma = 1/(\ell \cos^2 \phi)$. Then $\frac{\partial w}{\partial q} g_1 = 0$ (since w depends only on ϕ and g_1 has no ϕ-component) and $\frac{\partial g_1}{\partial q} w = -\sigma \frac{\partial g_1}{\partial \theta} = -\sigma(-\sin\theta, \cos\theta, 0, 0)^T$. Hence $g_1 w = -\frac{\partial g_1}{\partial q} w = \sigma(-\sin\theta, \cos\theta, 0, 0)^T$. □

With the bracket convention above, the first-order bracket has a negative sign; the second-order bracket is then exactly as stated. The sign does not affect any rank computation.

Theorem 33.2. *The bicycle is controllable: for $\phi \neq \pm\pi/2$,*

$$\big(g_1,\ g_2,\ g_1 g_2,\ g_1 g_1 g_2\big) = 4 = n.$$

Proof. Order the coordinates (x, y, θ, ϕ). The matrix has g_2 as the only vector with a ϕ-component, $w = g_1 g_2$ with a θ-component $-\sigma \neq 0$ that also appears in g_1, and the last vector spans the (x, y)-plane orthogonally to the heading. Direct expansion (along the ϕ-row, then the column of w) gives $\det = \sigma^2 = \frac{1}{\ell^2 \cos^4 \phi} \neq 0$. □

The determinant of the matrix with the columns in the order above is σ^2, nonzero wherever the model is defined, which is all that matters. For the FWD model, one checks that the same rank-4 test holds, also at $\phi = \pi/2$, where the second-order bracket is $\frac{1}{\ell}(-\sin\theta, \cos\theta, 0, 0)^T$.

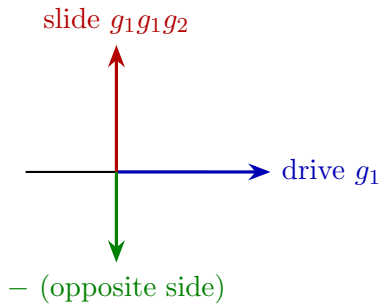


Figure 52: Bicycle at $\theta = 0$: the second-order bracket $g_1 g_1 g_2 \propto (-\sin\theta, \cos\theta, 0, 0)$ is the *sliding* direction, perpendicular to the heading.

Intuition

The picture is the same as for the unicycle but one level deeper: *steering* the wheel changes the *turning rate* (first bracket, $\partial_\phi \dot{\theta}$ direction: a pure rotation of the body), and driving while rotating then produces a net *sideways slide* (second-order bracket, with magnitude growing as $1/\cos^2\phi$). This is the car's parallel-parking motion. Four independent directions (drive, steer, rotate, slide) cover the four-dimensional state space.

34 Summary

34.1 Comparison of the models

	Unicycle	**Bicycle RWD**	**Bicycle FWD**
State q (n)	(x, y, θ) (3)	(x, y, θ, ϕ) (4)	(x, y, θ, ϕ) (4)
Constraints k / inputs m	1 / 2	2 / 2	2 / 2
Drive field g_1	$(\cos\theta, \sin\theta, 0)$	$(\cos\theta, \sin\theta, \frac{\tan\phi}{\ell}, 0)$	$(\cos\theta\cos\phi, \sin\theta\cos\phi, \frac{\sin\phi}{\ell}, 0)$
Steer field g_2	$(0, 0, 1)$	$(0, 0, 0, 1)$	$(0, 0, 0, 1)$
Drive input u_1	body speed v	rear-wheel speed	front-wheel speed
Singularity	none	$\phi = \pm\pi/2$	none
Brackets needed	$g_1 g_2$	up to $g_1 g_1 g_2$	same
Rank / controllable	$3 = n$, yes	$4 = n$, yes	$4 = n$, yes

Table 6: Kinematic models and their controllability.

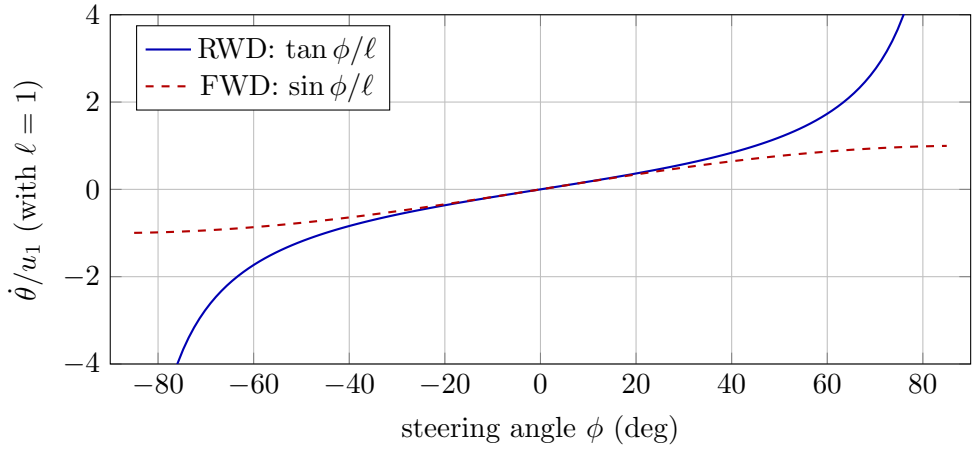


Figure 53: Rotation rate per unit of drive speed: it blows up at $\phi = \pm\pi/2$ for the rear-wheel-drive model but stays bounded for the front-wheel-drive one.

34.2 Key take-aways

- The bicycle has $n = 4$ coordinates and 2 independent pure-rolling constraints: the null space has dimension 2, giving the drive and steer vector fields. RWD uses $g_1 = (\cos\theta, \sin\theta, \tan\phi/\ell, 0)$ and is singular at $\phi = \pi/2$; FWD uses $g_1 \cos\phi$ and has no singularity.

- Controllability of driftless nonlinear systems is about *directions*: the Lie bracket $g_1 g_2 = \frac{\partial g_2}{\partial x} g_1 - \frac{\partial g_1}{\partial x} g_2$ gives new directions, and the four-leg manoeuvre displaces the state by $\varepsilon^2 g_1 g_2 + O(\varepsilon^3)$.
- Chow's theorem (Lie algebra / accessibility rank condition): the system is controllable iff the accessibility distribution generated by g_i and all iterated brackets has dimension n.
- Unicycle: $g_1 g_2 = (\sin\theta, -\cos\theta, 0)$ is the forbidden sliding direction, so rank 3 and it is controllable. Bicycle: $g_1 g_2$ and $g_1 g_1 g_2$ supply the missing rotation and slide, rank 4, hence controllable.
- Controllability with fewer inputs than states is equivalent to the kinematic constraints being non-integrable, i.e. nonholonomic.

Part VII

Lecture 5

35 Vehicles Kinematically Equivalent to the Bicycle

Several vehicles that look different share the same kinematic model as the bicycle: the *tricycle* (front-wheel or rear-wheel drive) and the *car-like* robot. In all of them the pair of front (steering) wheels can be replaced by a single *virtual* wheel placed at the midpoint between the two wheels, and the configuration is

$$= \begin{pmatrix} x \\ y \\ \theta \\ \phi \end{pmatrix},$$

where (x, y) is the position of the midpoint of the rear axle, θ the orientation of the body and ϕ the steering angle. A *differential-drive* robot has instead the same model as the unicycle, with the reference point centred on the axis of the driving wheels.

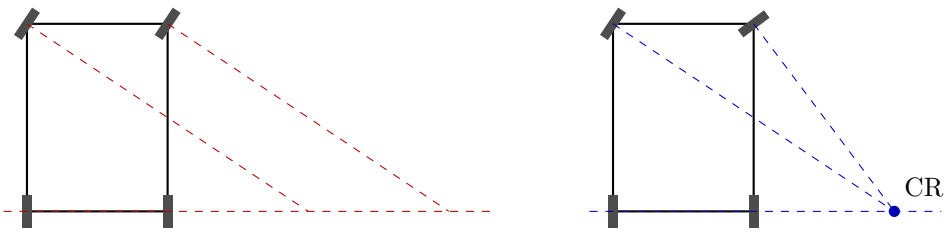
Intuition

Replacing two wheels by one is legitimate because of the *instantaneous centre of rotation*: at any instant a rigid body moves as if rotating about a single point (the CR), and pure rolling forces the motion of every wheel to be perpendicular to the line joining the wheel to that point. Whatever happens on the left and the right is therefore described by one "average" wheel on the vehicle's axis of symmetry.

35.1 Ackermann steering

Consider a car-like robot whose two front wheels are steered. Each wheel, rolling without slipping, cannot move along its axle, so its *zero-motion line* is the line containing the axle. For the vehicle to move, all zero-motion lines must meet at one common point, the CR.

If the two front wheels were steered by the *same* angle, they would be parallel: the two zero-motion lines would also be parallel, no common point could exist together with the rear-axle line, and the car would necessarily *slip*. The cure is the **Ackermann steering geometry**: the external front wheel is steered slightly *less* than the internal one, so that the three axle lines meet in a single CR lying on the rear axle line. In practice this is obtained mechanically with a four-bar linkage (the steering trapezoid).



(a) parallel wheels: no common point (b) Ackermann: single CR

Figure 54: Zero-motion lines of the wheels (dashed) of a car turning right. With parallel front wheels (a) they never meet in one point; with Ackermann steering (b) the outer (left) wheel is turned less than the inner one and everything converges in one CR.

Intuition

Think of the car going around a bend: the inner front wheel travels on a smaller circle than the outer one, so it must be turned *more*. If both were turned equally, one of them would be fighting the other, dragging sideways on the road, and that is exactly the slipping that the kinematic model forbids.

36 Chained Forms

The kinematic models of many wheeled robots can be brought into a (rather) *canonical* form, the chained form.

Definition 36.1 ($(2,n)$ chained form)**.** A driftless system with two inputs and n state variables is in $(2,n)$ *chained form* if (the first number is the number of inputs, the second the number of coordinates)

$$\begin{aligned}\dot{z}_1 &= w_1,\\ \dot{z}_2 &= w_2,\\ \dot{z}_3 &= z_2\, w_1,\\ &\vdots\\ \dot{z}_n &= z_{n-1}\, w_1,\end{aligned}$$

or, in vector form,

$$\dot{} = \begin{pmatrix}1\\0\\z_2\\\vdots\\z_{n-1}\end{pmatrix} w_1 + \begin{pmatrix}0\\1\\0\\\vdots\\0\end{pmatrix} w_2 = \gamma_1() \, w_1 + \gamma_2 \, w_2.$$

The structure is a *cascade of integrators*: w_1 is integrated to give z_1; w_2 is integrated to give z_2; and every following variable is obtained by multiplying the previous one by w_1 and integrating.

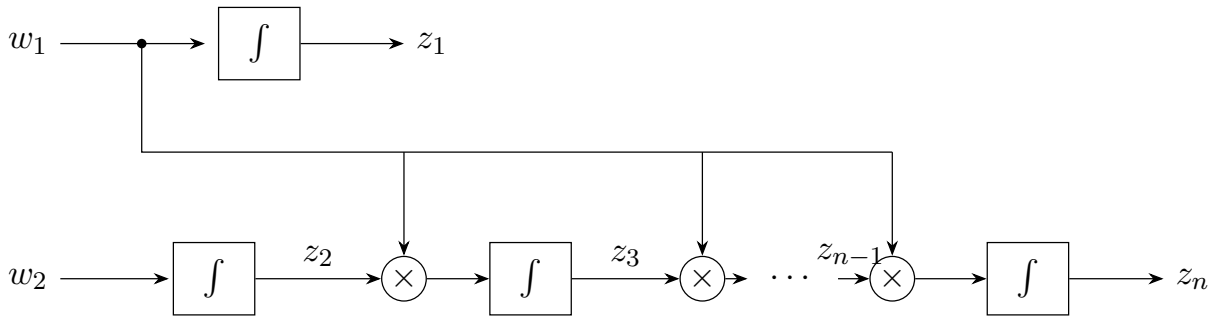


Figure 55: Block diagram of a $(2,n)$ chained form: the signal w_1 is distributed to all the multipliers.

36.1 Controllability

Proposition 36.2. *Every $(2, n)$ chained form is controllable: the accessibility distribution has dimension $\dim \Delta_A = n$ at every .*

Proof. With $\gamma_2 = e_2$ constant, the Lie bracket is $[\gamma_1, \gamma_2] = \frac{\partial \gamma_2}{\partial} \gamma_1 - \frac{\partial \gamma_1}{\partial} \gamma_2 = -\frac{\partial \gamma_1}{\partial} e_2 = -e_3$, because the only non-zero derivative of γ_1 with respect to z_2 is that of its third entry. Repeating the same argument,

$$\overset{k}{\gamma_1} \gamma_2 = (-1)^k e_{k+2}, \qquad k = 0, \dots, n-2.$$

Hence $\gamma_1, \gamma_2, \overset{}{\gamma_1} \gamma_2, \dots, \overset{n-2}{\gamma_1} \gamma_2$ are n vectors that span $\mathbb{R}^n$ (from $e_2, \dots, e_n$ one gets e_1 out of γ_1), i.e. $\dim \Delta_A = n$. By Chow's theorem the system is controllable. □

Intuition

The two inputs directly move only z_1 and z_2. But w_1 "feeds" z_3 through z_2, z_3 feeds z_4, and so on: the effect of w_2 propagates *down the chain* through the interplay with w_1. Each extra Lie bracket (an infinitesimal "wiggle": move along γ_1, then γ_2, then back) opens up exactly one new direction, and there are just enough of them to reach all n coordinates.

36.2 Why chained forms are useful

Proposition 36.3 (Linearity under constant input)**.** *If $w_1(t) \equiv c$ is constant, the chained form becomes a* linear *time-invariant system in $(z_2, \dots, z_n)$ with input w_2 (and $z_1 = ct + z_1(0)$). If w_1 is piecewise constant, the system is piecewise linear.*

Proof. For $w_1 = c$ the equations read $\dot{z}_1 = c$, $\dot{z}_2 = w_2$, $\dot{z}_i = c\, z_{i-1}$, $i = 3, \dots, n$: all right-hand sides are linear in $(z_2, \dots, z_n, w_2)$. □

Intuition

The nonlinearity of the chained form is entirely due to the *product* of a state with the input w_1. Freeze w_1 and the product becomes a plain gain. This is very handy in digital control, where inputs are held constant between sampling instants: within each sampling interval the system is linear, and one can use all of linear-systems theory.

Proposition 36.4 (Transformability, fact)**.** *All controllable driftless systems with $n \leq 4$ and 2 inputs can be put in chained form (under standard regularity assumptions). This means there exist*

- *a* coordinate transformation $= \alpha()$ *from the original coordinates to the chained-form coordinates, and*
- *an* input transformation $= \beta()$ *, with $\beta()$ invertible (so that are the original, linearly independent, inputs).*

Intuition

The chained form is a change of "language": the robot is the same, but written in coordinates and with inputs that make its structure transparent. The transformations are built so that the system after the change is exactly a cascade of integrators.

37 Chained Form of the Unicycle

The unicycle kinematics

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \omega$$

is a $(2,3)$ system: it should be mapped to $\dot{z}_1 = w_1$, $\dot{z}_2 = w_2$, $\dot{z}_3 = z_2 w_1$.

Proposition 37.1. *The coordinate and input transformations*

$$\begin{aligned} z_1 &= \theta, & w_1 &= \omega, \\ z_2 &= x\cos\theta + y\sin\theta, & w_2 &= v - z_3\,\omega, \\ z_3 &= x\sin\theta - y\cos\theta, \end{aligned}$$

bring the unicycle into $(2,3)$ *chained form.*

Proof. Direct check. First $\dot{z}_1 = \dot{\theta} = \omega = w_1$. Then

$$\begin{aligned} \dot{z}_2 &= \dot{x}\cos\theta - x\sin\theta\,\dot{\theta} + \dot{y}\sin\theta + y\cos\theta\,\dot{\theta} \\ &= v(\cos^2\theta + \sin^2\theta) - (x\sin\theta - y\cos\theta)\dot{\theta} = v - z_3\,\omega = w_2, \\ \dot{z}_3 &= \dot{x}\sin\theta + x\cos\theta\,\dot{\theta} - \dot{y}\cos\theta + y\sin\theta\,\dot{\theta} \\ &= v(\cos\theta\sin\theta - \sin\theta\cos\theta) + (x\cos\theta + y\sin\theta)\dot{\theta} = z_2\,\omega = z_2\,w_1. \end{aligned}$$

The map is invertible: $x = z_2\cos z_1 + z_3\sin z_1$, $y = z_2\sin z_1 - z_3\cos z_1$. □

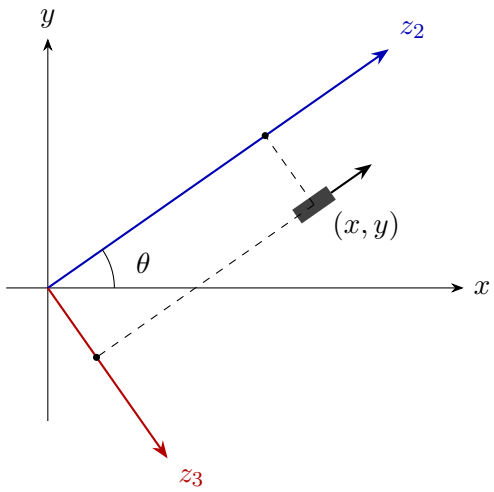


Figure 56: Meaning of the chained coordinates of the unicycle: z_2 and z_3 are the components of the position (x, y) along the axis z_2 (the direction of heading θ) and along the perpendicular axis z_3 obtained by rotating it by $-90°$; $z_1 = \theta$.

Intuition

The new coordinates are simply the robot's position *expressed in a frame rotated by the heading angle*. Rotating the robot ($w_1 = \omega$) changes the frame, and thereby changes z_2 and z_3: the position in the rotating frame changes even if the robot does not translate, which is the origin of the product term $z_2\,w_1$. The input $w_2 = v - z_3\omega$ is the forward speed corrected by this frame-rotation effect.

38 Chained Form of the Bicycle

The bicycle (equivalently the car-like robot) has $n = 4$ coordinates and is therefore a $(2, 4)$ system:

$$\dot{z}_1 = w_1, \quad \dot{z}_2 = w_2, \quad \dot{z}_3 = z_2 w_1, \quad \dot{z}_4 = z_3 w_1.$$

Proposition 38.1. *For the rear-wheel drive car with model $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = \frac{v}{\ell}\tan\phi$, $\dot{\phi} = \omega$, in the region $|\theta| < \pi/2$ the transformation*

$$z_1 = x, \quad z_2 = \frac{\tan\phi}{\ell\cos^3\theta}, \quad z_3 = \tan\theta, \quad z_4 = y, \qquad w_1 = v\cos\theta, \quad w_2 = \dot{z}_2$$

gives the $(2, 4)$ chained form.

Proof. $\dot{z}_1 = v\cos\theta = w_1$ and $\dot{z}_2 = w_2$ by definition (explicitly, $w_2 = \dfrac{\omega}{\ell\cos^3\theta\cos^2\phi} + \dfrac{3\sin\theta\tan^2\phi}{\ell^2\cos^4\theta}\, v$, invertible with respect to ω). Next

$$\dot{z}_3 = \frac{\dot{\theta}}{\cos^2\theta} = \frac{v\tan\phi}{\ell\cos^2\theta} = v\cos\theta\cdot\frac{\tan\phi}{\ell\cos^3\theta} = z_2 w_1, \qquad \dot{z}_4 = v\sin\theta = v\cos\theta\tan\theta = z_3 w_1. \quad \square$$

Intuition

Here z_1 and z_4 are the Cartesian coordinates, z_3 is (the tangent of) the heading, and z_2 is (a scaled version of) the curvature of the path. The chain reads: speed along x drives the slope of the path; the slope drives y; and the steering rate changes the curvature. It is the familiar picture of "curvature → heading → position" written as a cascade.

39 Adding Trailers

Consider adding one (or more) **trailers** to the unicycle or to the bicycle. The new vehicle is *still transformable to chained form*, as long as each trailer is *hinged at the midpoint between the two wheels* of the preceding body (the axle midpoint of the car or of the previous trailer).

Intuition

Hitching at the axle midpoint keeps the geometry "rolling-friendly": each trailer wheel is again the point of a body that can only move along its own heading direction, exactly like the rear wheel of the car. The new body therefore adds one new coordinate (its angle θ_t) and one new link in the chain, rather than breaking the structure. A fully worked example (a tricycle with a trailer, with its constraints and kinematic model) is given in Section 40.

40 Class Test 2009 – Problem 1: Tricycle with a Trailer

Example 40.1 (Problem 1)**.** Consider the mobile robot obtained by attaching a trailer to a rear-wheel drive tricycle. The trailer is a rigid body with an axle carrying two fixed wheels, and is connected to the midpoint of the tricycle rear axle through a revolute joint. Both two-wheel axles can be assimilated to a single wheel located at the midpoint of the axle.

1. Find a set of generalized coordinates for the robot, and clearly indicate them on the drawing.

2. Write the Pfaffian kinematic constraints to which the robot is subject.
3. Derive a kinematic model of the system.

40.1 Generalized coordinates

A convenient choice is

$$= (x \;\; y \;\; \theta \;\; \phi \;\; \theta_t)^T,$$

i.e. the generalized coordinates of the tricycle plus one additional coordinate for the trailer. Here (x, y) is the midpoint P of the rear axle, θ the heading of the tricycle, ϕ the steering angle of the front wheel, and θ_t the orientation of the trailer; P_f is the centre of the front wheel (at distance ℓ from P) and P_t the midpoint of the trailer axle (at distance ℓ_t from P). The robot has three wheels: the tricycle front wheel, the tricycle rear wheel and the trailer wheel.

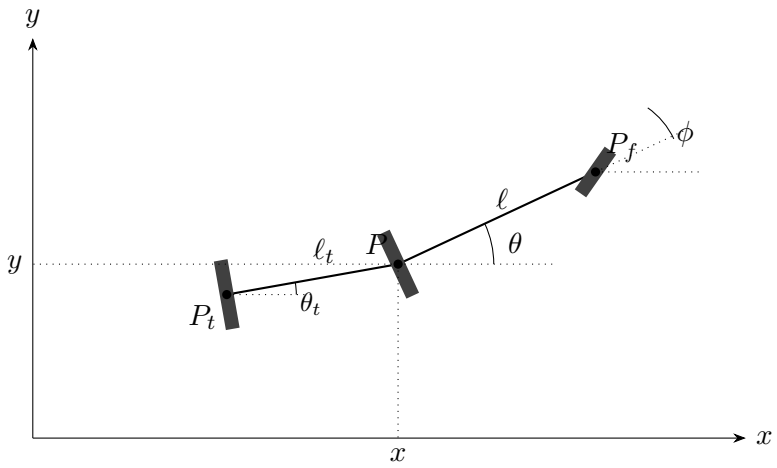


Figure 57: Tricycle with trailer. Generalized coordinates $= (x, y, \theta, \phi, \theta_t)$.

40.2 Pfaffian kinematic constraints

There are three kinematic constraints (one pure-rolling condition per wheel):

$$\begin{aligned} \dot{x}_f \sin(\theta + \phi) - \dot{y}_f \cos(\theta + \phi) &= 0, \\ \dot{x} \sin\theta - \dot{y} \cos\theta &= 0, \\ \dot{x}_t \sin\theta_t - \dot{y}_t \cos\theta_t &= 0, \end{aligned}$$

where (x_f, y_f) and (x_t, y_t) are the Cartesian coordinates of P_f and P_t:

$$x_f = x + \ell\cos\theta, \quad y_f = y + \ell\sin\theta, \qquad x_t = x - \ell_t\cos\theta_t, \quad y_t = y - \ell_t\sin\theta_t.$$

Differentiating and substituting (for instance $\dot{x}_f = \dot{x} - \ell\dot{\theta}\sin\theta$, $\dot{y}_f = \dot{y} + \ell\dot{\theta}\cos\theta$, and the identities $\sin\theta\sin(\theta+\phi) + \cos\theta\cos(\theta+\phi) = \cos\phi$ and $\sin^2\theta_t + \cos^2\theta_t = 1$) one obtains

$$\begin{aligned} \dot{x}\sin(\theta + \phi) - \dot{y}\cos(\theta + \phi) - \dot{\theta}\,\ell\cos\phi &= 0, \\ \dot{x}\sin\theta - \dot{y}\cos\theta &= 0, \\ \dot{x}\sin\theta_t - \dot{y}\cos\theta_t + \ell_t\dot{\theta}_t &= 0, \end{aligned}$$

or, in Pfaffian form $A^T(\dot{}) = 0$,

$$\begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 & 0 \\ \sin(\theta+\phi) & -\cos(\theta+\phi) & -\ell\cos\phi & 0 & 0 \\ \sin\theta_t & -\cos\theta_t & 0 & 0 & \ell_t \end{pmatrix} \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \\ \dot{\theta}_t \end{pmatrix} = 0.$$

Intuition

Each wheel contributes one equation of the form "velocity of the wheel centre $\perp$ axle direction" (no lateral slipping). The first two rows (rear and front wheel of the tricycle) are the constraints of the bicycle; the third row is new and says that the trailer wheel is, again, forced to move along the direction of the trailer.

40.3 Kinematic model

Note that the submatrix made of the first two rows and the first four columns of A^T coincides with the constraint matrix of the bicycle. A basis of the null space of A^T can then be written by "completing" the basis found for the bicycle (in particular the rear-wheel drive version, since the tricycle has rear-wheel drive) with a suitable fifth entry:

$$G(\,) = \begin{pmatrix} g_1 & g_2 \end{pmatrix} = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \tan\phi/\ell & 0 \\ 0 & 1 \\ -\sin(\theta_t-\theta)/\ell_t & 0 \end{pmatrix}.$$

The kinematic control system is

$$\dot{} = g_1(\,)\,v + g_2(\,)\,\omega,$$

where v and ω are respectively the driving velocity and the steering velocity of the tricycle.

Check. The third constraint applied to g_1 gives $\cos\theta\sin\theta_t - \sin\theta\cos\theta_t + \ell_t \cdot \frac{-\sin(\theta_t-\theta)}{\ell_t} = \sin(\theta_t - \theta) - \sin(\theta_t-\theta) = 0$, and the other two rows vanish exactly as in the bicycle.

Intuition

The driving input v moves the tricycle exactly as in the bicycle and, in addition, *pulls* the trailer: $\dot{\theta}_t$ is proportional to $\sin(\theta-\theta_t)$, so the trailer tends to align behind the tricycle when it moves forward (the further it is from alignment, the faster it rotates, and it stops rotating when aligned). The steering input ω only changes ϕ and leaves the others untouched.

A different choice of coordinates. The model depends on the chosen ; a different choice leads to an equivalent model via a change of coordinates. For example with $' = (x\ y\ \theta\ \phi\ \delta)^T$, $\delta = \theta - \theta_t$, one has $\dot{'} = g_1'(')v + g_2'(')\omega$ with

$$G'(') = \begin{pmatrix} \cos\theta & 0 \\ \sin\theta & 0 \\ \tan\phi/\ell & 0 \\ 0 & 1 \\ \tan\phi/\ell - \sin\delta/\ell_t & 0 \end{pmatrix},$$

since $\dot{\delta} = \dot{\theta} - \dot{\theta}_t = v\big(\tan\phi/\ell - \sin\delta/\ell_t\big)$.

41 Worked Exercise: A Rolling Wheel on the Plane

Setting. A unicycle (a disk of radius R) moves on a plane. Extend the configuration vector by including the *wheel (rolling) angle* ϕ. We want to: (1) define the configuration space; (2) write the constraints; (3) show controllability; (4) describe a sequential manoeuvre.

41.1 Configuration space and kinematic model

The configuration is

$$= \begin{pmatrix} x \\ y \\ \theta \\ \phi \end{pmatrix} \in \mathbb{R}^2 \times SO(2) \times SO(2), \qquad \dim = 4,$$

(θ is the heading, ϕ the angle of rotation of the wheel about its horizontal axis). The unicycle equations $\dot{x} = v\cos\theta$, $\dot{y} = v\sin\theta$, $\dot{\theta} = \omega$ are completed by $\dot{\phi} = \omega_\phi$, where ω_ϕ is the angular velocity of the wheel about its horizontal axis. Since the wheel rolls, $v = R\,\omega_\phi$, so

$$\begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ 0 \\ 1 \end{pmatrix} \omega_\phi + \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \omega.$$

41.2 Constraints

Two pure-rolling conditions act on the wheel:

$$\dot{x}\sin\theta - \dot{y}\cos\theta = 0 \qquad \text{(rolling laterally: no side slip, along the axle)},$$
$$v = R\,\omega_\phi \qquad \text{(rolling along the sagittal axis)}.$$

Kinematically, the second one reads $\pm\sqrt{\dot{x}^2 + \dot{y}^2} - R\dot{\phi} = 0$, which is *not* Pfaffian (it is not linear in $\dot{}$).

How to solve this. Use $v = R\dot{\phi}$ and $v(\cos^2\theta + \sin^2\theta) = \dot{x}\cos\theta + \dot{y}\sin\theta$ to obtain

$$\dot{x}\cos\theta + \dot{y}\sin\theta - R\dot{\phi} = 0,$$

which *is* Pfaffian. The constraint matrix is

$$A^T() = \begin{pmatrix} \sin\theta & -\cos\theta & 0 & 0 \\ \cos\theta & \sin\theta & 0 & -R \end{pmatrix},$$

the first row being the lateral rolling condition and the second the rolling condition along the sagittal axis. The null space of $A^T()$ gives the same kinematic model as before, $\dot{} = g_1\omega_\phi + g_2\omega$ with

$$g_1 = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ 0 \\ 1 \end{pmatrix}, \qquad g_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Intuition

The "physical" rolling condition is a statement about *speeds*: the distance covered on the ground equals the radius times the angle turned. A speed contains a square root of the velocity components, which is not linear. The trick is to project the velocity on the heading

direction, $\dot{x}\cos\theta + \dot{y}\sin\theta$, which is the signed forward speed, and which *is* linear in $\dot{}$. This keeps the sign (forward/backward) and brings the problem back to the Pfaffian world.

41.3 Controllability

Proposition 41.1. *The rolling wheel on the plane is controllable (the accessibility distribution has dimension 4 everywhere).*

Proof. Compute the Lie brackets with $[g_i, g_j] = \frac{\partial g_j}{\partial} g_i - \frac{\partial g_i}{\partial} g_j$. Since g_2 is constant,

$$g_3 := [g_1, g_2] = -\frac{\partial g_1}{\partial} g_2 = -\begin{pmatrix} 0 & 0 & -R\sin\theta & 0 \\ 0 & 0 & R\cos\theta & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} R\sin\theta \\ -R\cos\theta \\ 0 \\ 0 \end{pmatrix}.$$

This is linearly independent of g_1, g_2, so already 3 directions are available. Further brackets:

$$[g_1, g_3] = 0, \qquad [g_2, g_3] = \frac{\partial g_3}{\partial \theta} = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ 0 \\ 0 \end{pmatrix} =: g_4.$$

Finally, the matrix $\begin{pmatrix} g_1 & g_2 & g_3 & g_4 \end{pmatrix}$ has the structure (reordering rows and columns, a block triangular matrix) in which only the 2×2 blocks on the diagonal matter. Its determinant is

$$\det \begin{pmatrix} R\cos\theta & 0 & R\sin\theta & R\cos\theta \\ R\sin\theta & 0 & -R\cos\theta & R\sin\theta \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix} = -R^2 \neq 0.$$

So $\{g_1, g_2, g_3, g_4\} = 4$ and by Chow's theorem the system is controllable. □

Intuition

Four coordinates, two inputs. Rolling forward and turning generate the sideways motion g_3 (the parking manoeuvre: roll, turn, roll back, turn back); and turning the wheel's heading while a sideways displacement is present generates a further direction g_4. Crucially, ϕ can only change when the wheel *moves* (rolling), which is why all four directions need the brackets and not only the two inputs.

41.4 A sequential manoeuvre

Algorithm 41.2 (Reaching a desired $(x_g, y_g, \theta_g, \phi_g)$)**.** Let the initial configuration be $(x_s, y_s, \theta_s, \phi_s)$.

1. Steps 1–3 are as for the unicycle: drive the position and heading to the goal values, $x \to x_g$, $y \to y_g$, $\theta \to \theta_g$. At the end of this phase the wheel angle ϕ has some value which, in general, differs from ϕ_g.
2. Perform a *circular motion* that brings ϕ to ϕ_g and leaves the robot in the same position and orientation. The wheel rolls along a circle of radius r once around, so the rolled distance $2\pi r$ equals the arc length corresponding to the required change of the wheel

angle:

$$2\pi r = R\,(\phi_g - \phi_s),$$

where r is the radius of the circle and R is the wheel radius (ϕ_s denotes the wheel angle at the start of this last phase).

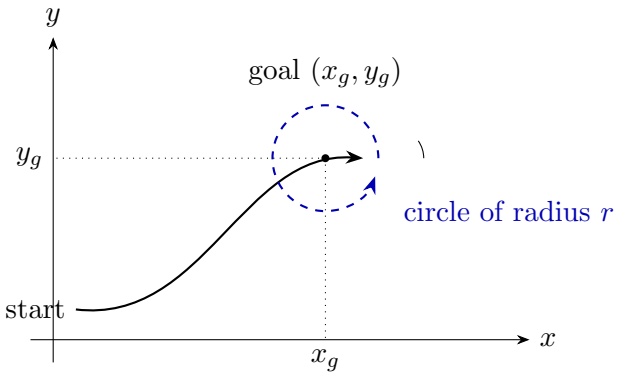


Figure 58: Sequential manoeuvre: first the unicycle-like motion to the goal position and heading (solid), then a circular motion at the goal (dashed) that only changes ϕ.

Intuition

The wheel angle ϕ is only changed by rolling, but rolling also moves the robot. The trick is to roll *in a closed loop*: after one full circle the robot is back at the same position and (modulo 2π) the same heading, while the wheel has turned by an amount proportional to the length of the circle. Choosing r gives exactly the missing wheel rotation. Systematic ways to do this will be seen in the next lecture, when we discuss planning.

42 Summary

System	n	Constraints	Controllable	Chained form
Unicycle / differential drive	3	1	yes	$(2,3)$ via $z_1 = \theta$, $z_2 = x\cos\theta + y\sin\theta$, $z_3 = x\sin\theta - y\cos\theta$
Bicycle, tricycle, car-like (rear-wheel drive)	4	2	yes	$(2,4)$ via $z_1 = x$, $z_2 = \frac{\tan\phi}{\ell\cos^3\theta}$, $z_3 = \tan\theta$, $z_4 = y$ $(\lvert\theta\rvert < \pi/2)$
Tricycle with trailer (hinged at the axle midpoint)	5	3	yes	still transformable to chained form
Rolling wheel on the plane (with ϕ)	4	2 (after linearization of rolling)	yes (brackets g_3, g_4)	reached with a sequential manoeuvre

Key take-aways.

- Bicycle, tricycle and car-like robots share the same kinematic model; an Ackermann steering geometry (outer front wheel turned less than the inner one) guarantees a single CR and therefore pure rolling.
- The $(2,n)$ chained form $\dot z_1 = w_1$, $\dot z_2 = w_2$, $\dot z_i = z_{i-1}w_1$ is a cascade of integrators that is always controllable (via the Lie brackets $\overset{k}{\gamma_1}\gamma_2 \propto e_{k+2}$).

- With w_1 constant (or piecewise constant, as with digital control) the chained form is linear (piecewise linear), which makes control design easy.
- Unicycle and bicycle are put in chained form through a coordinate transformation = $\alpha()$ and an input transformation = $\beta()$; adding trailers hinged at the axle midpoint preserves transformability.
- Constraints that are not Pfaffian at first sight (rolling of the wheel with ϕ) can be made Pfaffian by projecting the velocity on the heading direction; controllability then follows from the Lie algebra rank condition and non-trivial configurations are reached by sequential manoeuvres.

Part VIII

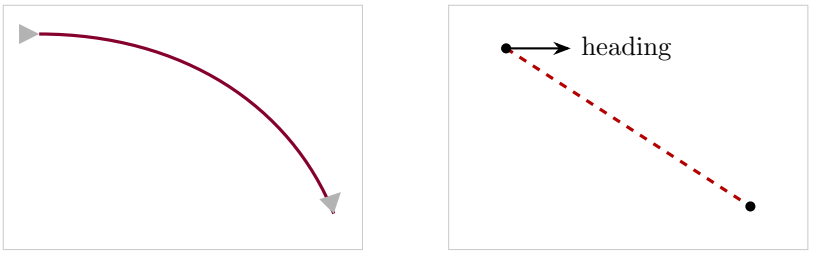
Lecture 6

Roadmap. The lecture is about computing motions for a wheeled mobile robot (WMR) that *respect its nonholonomic constraints*, in free space (no obstacles). The plan is: separate space from time; characterize which *paths* are admissible (geometric model); exploit *differential flatness* to generate admissible paths by pure interpolation; apply this to the unicycle via Cartesian polynomials, via the chained form, and via parametrized inputs; look at numerical parking maneuvers; and finally see how a timing law turns a path into a trajectory.

43 Planning admissible motions *(slides 3)*

Definition 43.1 (Motion planning without obstacles)**.** Given an initial configuration and a final configuration , find a motion (t) that transfers the WMR from to and is *admissible*, i.e., satisfies the nonholonomic constraints $A()\dot{} = 0$ at every time instant.

Meeting the boundary conditions is not enough: a curve that joins to but demands, say, a sideways slip of the wheels is useless. Admissibility is what makes this problem different from plain curve interpolation.



admissible path tangent ≠ heading: not admissible

Figure 59: A unicycle-like robot starting along the horizontal direction can follow a smooth curve that bends toward (left), but not the straight segment (right), whose tangent disagrees with the heading at .

Intuition

Think of parking a car: you cannot simply draw the straight line from where you are to the target spot, because the car cannot move sideways. The planner must hand the robot a curve it is *capable* of tracking. Obstacles are deliberately left out here; the whole difficulty already lives in the nonholonomic constraints.

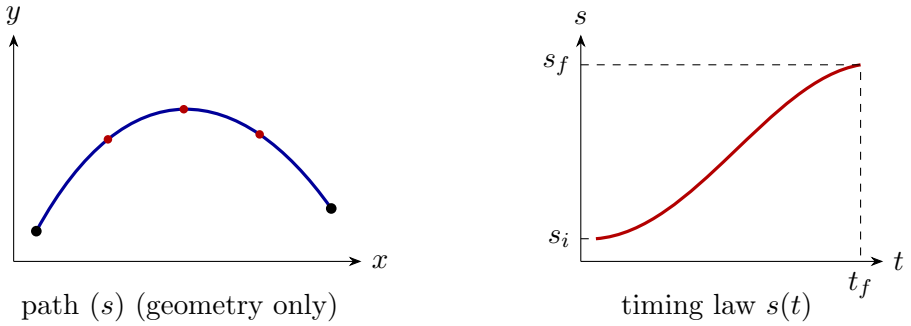
44 Space-time separation *(slide 4)*

Definition 44.1 (Path, timing law, trajectory)**.** A configuration-space *trajectory* (t), $t \in [t_i, t_f]$, with $(t_i) =$ and $(t_f) =$, can be split into:

- a geometric *path* (s), where s is a path parameter running from $s_i = s(t_i)$ to $s_f = s(t_f)$;
- a *timing law* $s = s(t)$, strictly monotonic (increasing).

One possible choice of s is the arc length.

- **Path planning:** plan an admissible path from to .
- **Trajectory planning:** plan an admissible trajectory from to . It can be *direct*, or *indirect*: first plan a path, then add a timing law.



path (s) (geometry only) timing law $s(t)$

Figure 60: Space-time separation: the same path (left) can be run with many timing laws (right). The dots on the path are equally spaced in s, hence equally spaced in time only when $s(t)$ is linear.

Intuition

A path is the *railway track*, the timing law is the *timetable*. The same track can be traveled quickly or slowly, but the track itself decides whether the train can follow it at all. This is why a robot that cannot track a given curve cannot be rescued by changing speed.

In symbols. A trajectory (t), $t \in [t_i, t_f]$, with $(t_i) =$ and $(t_f) =$, is rewritten as the composition $(t) \to (s(t))$. The first factor, (s) with $s \in [s_i, s_f]$, is the *path*: it carries only the *geometry* of the motion. The second factor, $s(t)$ with $t \in [t_i, t_f]$, is the *timing law*: it says *when* each point of the path is reached.

45 Admissible paths and geometric model *(slides 5–8)*

45.1 Geometric path admissibility *(slides 5–6)*

Intuitively, the local mobility limitation caused by the nonholonomic constraints also shows up on *paths*: if a motion is not admissible, speeding up or slowing down will not make it admissible. To see this, start from the space-time separation:

$$\dot{} = \frac{}{t} = \frac{}{s}\dot{s} =' \dot{s}, \tag{1}$$

where the prime denotes differentiation with respect to s. The vector $'$ is directed as the *tangent* to the path in configuration space, and $\dot{s}$ only changes its modulus.

Example 45.1 (Tangent vector and arc length)**.** Let $\mathcal{C} = \mathbb{R}^2$, with coordinates $(q_1, q_2) = (x, y)$, and consider the path

$$x(s) = \sqrt{1 - s^2}, \qquad y(s) = s, \qquad s \in [0, 1],$$

a quarter of the unit circle going from $= (1, 0)$ to $= (0, 1)$. The tangent vector is

$$'(s) = \begin{pmatrix} x'(s) \\ y'(s) \end{pmatrix} = \begin{pmatrix} -s/\sqrt{1 - s^2} \\ 1 \end{pmatrix}, \qquad \|'(s)\| = \frac{1}{\sqrt{1 - s^2}} \neq 1,$$

so this parameter s is *not* the arc length (for which one would have $\|\gamma'\| = 1$, e.g., $x = \cos\sigma$, $y = \sin\sigma$).

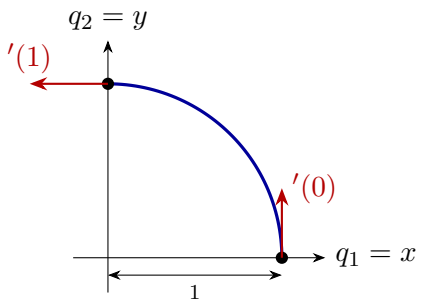


Figure 61: The path of Example 45.1: the tangent direction is the same for any parametrization, only the modulus of γ' depends on s.

Intuition

Reparametrizing a path (e.g., from s to arc length σ) leaves the curve unchanged; it only rescales the tangent vector γ'. This is why, in the geometric model, only the *direction* of γ' matters for admissibility, and why in all the planners below we are free to use a convenient s rather than the arc length.

Proposition 45.2 (Geometric path admissibility)**.** *A path $\gamma(s)$ is the support of an admissible trajectory (for any timing law with $\dot{s} > 0$) if and only if*

$$A(\gamma(s))\,\gamma'(s) = 0 \qquad \text{at all points of the path.} \tag{2}$$

Proof. Using (1), the constraints become $A(\gamma(s))\,\dot{\gamma}(s) = A(\gamma(s))\,\gamma'(s)\,\dot{s} = 0$. Since $\dot{s}(t) > 0$ for all t we may divide by $\dot{s}$, which gives $A(\gamma(s))\,\gamma'(s) = 0$; conversely, if $A(\gamma(s))\,\gamma'(s) = 0$ holds along the path, then $A(\gamma(s))\,\dot{\gamma}(s) = 0$ for any $\dot{s}$. □

Definition 45.3 (Geometric model and geometric inputs)**.** Let the columns of $G(q)$ span $\ker(A(q))$. All geometrically admissible paths are solutions of the *geometric model*

$$\gamma'(s) = G(\gamma(s))\,\omega(s), \tag{3}$$

where ω are the *geometric inputs*. They are related to the velocity inputs of the kinematic model by

$$\boldsymbol{u}(t) = \omega(s(t))\,\dot{s}(t). \tag{4}$$

Once the geometric inputs $\omega(s)$ are chosen, the path is determined; selecting a timing law $s = s(t)$ then identifies a particular trajectory along that path.

Kinematic versus geometric model. The geometric model is built exactly in parallel to the kinematic model (KM). Letting $\boldsymbol{g}_1, \dots, \boldsymbol{g}_m$ be the columns of $G(q)$:

$$\text{(KM)} \quad A(q)\,\dot{q} = 0 \;\Rightarrow\; \dot{q} \in \ker(A(q)) \;\Rightarrow\; \dot{q} = \sum_{i=1}^{m} \boldsymbol{g}_i(q)\,u_i, \tag{5}$$

$$\text{(GM)} \quad A(q)\,\gamma'(s) = 0 \;\Rightarrow\; \gamma'(s) \in \ker(A(q)) \;\Rightarrow\; \gamma'(s) = \sum_{i=1}^{m} \boldsymbol{g}_i(q)\,\tilde{u}_i. \tag{6}$$

Multiplying the geometric model by $\dot{s}$ and using $\dot{\cdot} = {}' \dot{s}$,

$$\dot{\cdot} = {}' \ \dot{s} = \sum_{i=1}^{m} \boldsymbol{g}_i() \, \tilde{u}_i \, \dot{s},$$

and comparing with the kinematic model (the $\boldsymbol{g}_i$ are linearly independent) gives $u_i = \tilde{u}_i \, \dot{s}$. The u_i are the *pseudovelocities* (or *velocity inputs*); the $\tilde{u}_i$ are the *geometric inputs*. In short: the kinematic model is a *trajectory generator*, the geometric model is a *path generator*.

Intuition

The condition $A()' = 0$ is the kinematic model with time "factored out". The nonholonomic constraints say which *directions* of motion are allowed; speed has no say in direction. So the geometric model is just the kinematic model run with a different clock: s instead of t, and instead of $\boldsymbol{u}$.

45.2 The unicycle example *(slides 7–8)*

For the unicycle the pure-rolling constraint is $\dot{x} \sin\theta - \dot{y} \cos\theta = 0$, i.e., $(\sin\theta \quad -\cos\theta \quad 0)\dot{} = 0$. Replacing $\dot{\cdot}$ by ${}'\dot{s}$ and dividing by $\dot{s} > 0$ gives $(\sin\theta \quad -\cos\theta \quad 0)' = 0$, the *path admissibility* condition, which reads

$$x' \sin\theta - y' \cos\theta = 0, \tag{7}$$

i.e., the Cartesian path tangent must align with the sagittal axis of the robot. The robot may travel along the tangent in either sense: forward motion if the heading points along the tangent, backward motion if it points opposite.

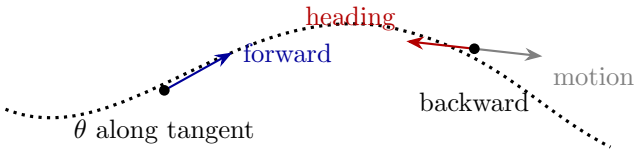


Figure 62: Along a Cartesian path the unicycle heading is tangent to the path; it points the same way as the motion (forward) or the opposite way (backward).

A Cartesian path with a *discontinuous tangent* is not admissible, unless the unicycle is allowed to stop at the discontinuity points to realign with the new tangent.

The geometric model of the unicycle is

$$\begin{aligned} x' &= \cos\theta, \\ y' &= \sin\theta, \\ \theta' &=, \end{aligned} \tag{8}$$

with geometric inputs related to the velocity inputs by $v(t) = (s)\,\dot{s}(t)$ and $\omega(t) = (s)\,\dot{s}(t)$.

Example 45.4 (A Cartesian path with a corner)**.** Consider a Cartesian path made of a horizontal segment followed by a vertical one. Is it admissible for a unicycle? Two paths in the configuration space (x, y, θ) have this same *footprint* in the (x, y) plane:

(1) θ stays at 0 (or jumps at the corner). On the horizontal segment $y' = 0$ and $x' \sin 0 - y' \cos 0 = 0$ holds, but on the vertical segment $x' = 0$, $y' \neq 0$ and the condition fails; a jump of θ would make the path discontinuous in $\mathcal{C}$. **Not admissible.**

(2) $\theta = 0$ on the horizontal segment; at the corner θ goes from 0 to $\pi/2$ while $x' = y' = 0$ (rotation on the spot); then $\theta = \pi/2$ on the vertical segment. The condition holds at

every point, so the path **is admissible**, but only if the unicycle *stops* at the corner.

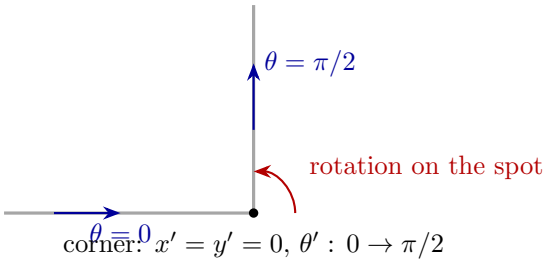


Figure 63: Path (2) of Example 45.4: the footprint has a corner, but the robot realigns with the new tangent by rotating at the corner.

Intuition

A corner in the Cartesian path is a place where the robot would have to change heading instantly. A unicycle can do that only by stopping and rotating on the spot, i.e., by "cheating" the pure path picture.

Planning formulation. Path planning for WMRs is generally formulated and solved with the *geometric* model. Trajectory planning can be solved *directly* with the kinematic model, or *indirectly* by first planning a geometric path and then choosing a timing law.

46 Flat outputs *(slides 9–13)*

46.1 Differential flatness *(slides 9–10)*

Definition 46.1 (Differential flatness)**.** The system $\dot{\boldsymbol{x}} = f(\boldsymbol{x}) + G(\boldsymbol{x})\boldsymbol{u}$ is *differentially flat* if there exists a set of outputs $\boldsymbol{y}$, called *flat outputs*, such that the state and the control inputs can be expressed *algebraically* as functions of $\boldsymbol{y}$ and its time derivatives:

$$\boldsymbol{x} = \boldsymbol{x}\big(\boldsymbol{y}, \dot{\boldsymbol{y}}, \ddot{\boldsymbol{y}}, \ldots, \boldsymbol{y}^{(r)}\big), \tag{9}$$

$$\boldsymbol{u} = \boldsymbol{u}\big(\boldsymbol{y}, \dot{\boldsymbol{y}}, \ddot{\boldsymbol{y}}, \ldots, \boldsymbol{y}^{(r)}\big). \tag{10}$$

The definition applies equally to *geometric* models if s replaces t and derivatives with respect to t are replaced by derivatives with respect to s. If the path (trajectory) of $\boldsymbol{y}$ is known, the path (trajectory) of $\boldsymbol{x}$ and the history of $\boldsymbol{u}$ in s (or t) can be *reconstructed*.

Intuition

Flat outputs are "free handles": pick any sufficiently smooth curve for $\boldsymbol{y}$, and everything else follows by differentiation, with no differential equation to integrate. For a unicycle, picking a smooth planar curve $(x(s), y(s))$ is enough: the heading is the direction of the curve, and the inputs are its speed and curvature.

Proposition 46.2 (Flatness of the unicycle and the bicycle)**.** *The unicycle and the bicycle are differentially flat with x and y as flat outputs. For the geometric model of the unicycle, a Cartesian path $(x(s), y(s))$ determines the path of the whole state $(s) = (x(s), y(s), \theta(s))$ and*

the geometric inputs:

$$\theta(s) = \arctan\frac{y'(s)}{x'(s)} = \big(y'(s), x'(s)\big) + k\pi, \tag{11}$$

$$(s) = \pm\sqrt{(x'(s))^2 + (y'(s))^2}, \tag{12}$$

$$(s) = \frac{y''(s)x'(s) - x''(s)y'(s)}{(x'(s))^2 + (y'(s))^2}. \tag{13}$$

For forward (backward) motion take $k = 1$ ($k = 0$) and the + (−) sign. For the bicycle, let L be the wheelbase and delta the front-wheel steering angle; its kinematic model is

$$\dot{x} = v\cos\theta, \qquad \dot{y} = v\sin\theta, \qquad \dot{\theta} = \frac{v}{L}\tan\delta.$$

On intervals where the vehicle does not stop, the same Cartesian outputs reconstruct its state and inputs as

$$\begin{aligned}
\theta &= \operatorname{atan2}(\dot{y}, \dot{x}) \quad \textit{(modulo } \pi\textit{)},\\
v &= \pm\sqrt{\dot{x}^2 + \dot{y}^2},\\
\delta &= \arctan\left(L\,\frac{\operatorname{sgn}(v)(\dot{x}\ddot{y} - \dot{y}\ddot{x})}{(\dot{x}^2 + \dot{y}^2)^{3/2}}\right).
\end{aligned}$$

The sign of v selects forward or backward motion, with the corresponding heading branch.

Derivation for the unicycle. From $x' = \cos\theta$ and $y' = \sin\theta$ we get $(x')^2 + (y')^2 = ^2$, hence $= \pm\sqrt{(x')^2 + (y')^2}$, and $y'/x' = \tan\theta$, hence $\theta = \arctan(y'/x')$ up to a multiple of π. Differentiating $\tan\theta = y'/x'$:

$$\frac{\theta'}{\cos^2\theta} = \frac{y''x' - x''y'}{(x')^2} \implies = \theta' = \frac{y''x' - x''y'}{(x')^2}\cos^2\theta = \frac{y''x' - x''y'}{(x')^2 + (y')^2},$$

since $\cos^2\theta = (x')^2/\big((x')^2 + (y')^2\big)$. □

Derivation for the bicycle. The first two bicycle equations have the same form as the unicycle equations, so they give v and θ directly from (x, y) and their first derivatives, as above. Differentiating the heading yields

$$\dot{\theta} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2 + \dot{y}^2}.$$

Using $\dot{\theta} = (v/L)\tan\delta$ and $|v| = \sqrt{\dot{x}^2 + \dot{y}^2}$ (with the heading branch chosen consistently with the sign of v), we obtain

$$\tan\delta = L\frac{\operatorname{sgn}(v)(\dot{x}\ddot{y} - \dot{y}\ddot{x})}{(\dot{x}^2 + \dot{y}^2)^{3/2}},$$

which gives the steering reconstruction above. Thus state and inputs are algebraic functions of the flat outputs and their derivatives, proving flatness wherever $v \neq 0$. □

Intuition

is the "speed along the curve" and is times the *curvature* of the Cartesian path: the formula for is the familiar planar curvature formula $\kappa = (y''x' - x''y')/\big((x')^2 + (y')^2\big)^{3/2}$ multiplied by . The sign choice only decides whether the robot runs the curve nose-first or tail-first.

46.2 Flatness and the chained form *(slides 11–12)*

Theorem 46.3 (Flatness, chained form, linearization)**.** *For driftless dynamic systems, differential flatness is a necessary and sufficient condition for transformability into chained form; hence all WMRs with dimension up to 4 are flat. For general dynamic systems, differential flatness is a necessary and sufficient condition for linearizability via dynamic feedback.*

Singular points of the reconstruction.

- If $x'(s) = y'(s) = 0$ at an isolated point (as at cusps or motion inversions), then $(s) = 0$ there; at these points $\theta(s)$ and (s) can be computed by continuity arguments.
- The reconstruction of θ and $$ *fails* if the Cartesian path degenerates to a point ($x'(s) = y'(s) = 0$ identically). In fact, the unicycle may then have any orientation at the contact point, and may even be rotating on the spot.

Geometrically: at an isolated cusp the Cartesian tangent vanishes only at the tip, so $= 0$ there, the robot stops, inverts its motion and leaves along the other branch; if instead $x' \equiv y' \equiv 0$ over an interval, the contact point does not move and θ, cannot be recovered from (x, y) (they are free in the geometric model $\theta' =$).

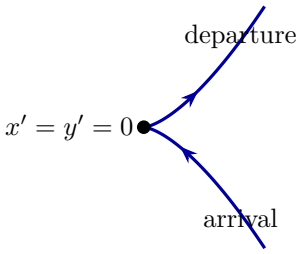


Figure 64: A cusp of the Cartesian path: the tangent vanishes only at the tip, where the unicycle inverts its motion.

Proposition 46.4 (Flatness of the $(2, n)$ chained form)**.** *The $(2, n)$ chained form is differentially flat, with flat outputs z_1 and z_n; all the other state variables and the associated control inputs can be computed from them.*

Example 46.5 (The $(2, 3)$ chained form)**.** Consider

$$\begin{aligned} z_1' &= {}_1, \\ z_2' &= {}_2, \\ z_3' &= z_2\, {}_1. \end{aligned} \tag{14}$$

The reconstruction formulas are

$$z_2 = \frac{z_3'}{z_1'} \quad \text{(state)}, \qquad {}_1 = z_1', \qquad {}_2 = \frac{z_1' z_3'' - z_1'' z_3'}{(z_1')^2} \quad \text{(input)}. \tag{15}$$

Derivation. The first equation gives ${}_1 = z_1'$. Substituting in the third, $z_3' = z_2 z_1'$, so $z_2 = z_3'/z_1'$ (when $z_1' \neq 0$). The second equation then gives ${}_2 = z_2' = \big(z_3'' z_1' - z_3' z_1''\big)/(z_1')^2$ by the quotient rule. □

Intuition

In chained form the first variable z_1 acts as "clock": it advances at the rate $_1$. Dividing the third equation by the first, $z_3/z_1 = z_2$: the slope of z_3 against z_1 *is* z_2. Choose the curve $z_3(z_1)$ and z_2 is forced; its derivative then gives the remaining input. This is exactly the same "choose the curve, read off the rest" mechanism as for (x, y) in the unicycle.

46.3 An algorithm for planning with flat outputs *(slide 13)*

Reconstruction formulas associate admissible paths (trajectories) to the evolution of the flat outputs, which can be chosen *arbitrarily*. This leads to the following algorithm.

Algorithm 46.6 (Path planning via flat outputs)**.**
1. Compute $\boldsymbol{y}_i$ and $\boldsymbol{y}_f$ corresponding to and .
2. Find $\boldsymbol{y}(s)$ by interpolating $\boldsymbol{y}_i$ and $\boldsymbol{y}_f$ with the appropriate boundary conditions for .
3. Use the reconstruction formulas to find (s) and (s).

The resulting path is *automatically* admissible, i.e., it satisfies the nonholonomic constraints at all points. The same algorithm can be used for *trajectory planning* by interpolating over t rather than over s.

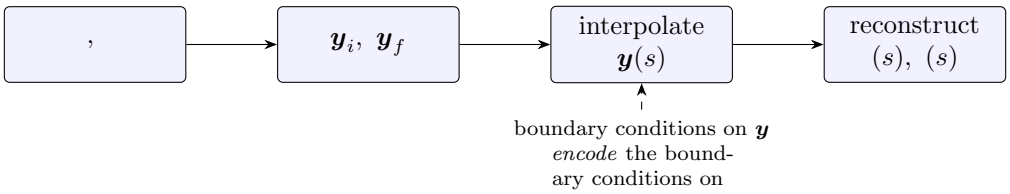


Figure 65: Planning with flat outputs: admissibility is built into the reconstruction step.

Intuition

Normally planning for a nonholonomic robot means solving a constrained problem. Flatness turns it into an *unconstrained* one: the constraints are already "absorbed" by the reconstruction formulas. The only subtlety is that orientation (and higher-order variables) are *not* flat outputs, so their boundary values must be translated into *derivative* conditions on $\boldsymbol{y}$; this is the role of the "appropriate boundary conditions" in step 2.

47 Unicycle path planning via Cartesian polynomials *(slides 14–15)*

We apply Algorithm 46.6 to the unicycle, whose flat outputs are x and y. Given $= (x_i, y_i, \theta_i)$ and $= (x_f, y_f, \theta_f)$, find an admissible path (s), $s \in [s_i, s_f]$, with $(s_i) =$ and $(s_f) =$.

Let $s_i = 0$, $s_f = 1$ and use *pre-fitted cubic polynomials*

$$x(s) = s^3 x_f - (s-1)^3 x_i + \alpha_x s^2(s-1) + \beta_x s(s-1)^2, \tag{16}$$

$$y(s) = s^3 y_f - (s-1)^3 y_i + \alpha_y s^2(s-1) + \beta_y s(s-1)^2, \tag{17}$$

which already satisfy the boundary conditions on x, y (indeed $x(0) = x_i$, $x(1) = x_f$). Since θ_i and θ_f are given, we must also impose

$$\arctan \frac{y'(0)}{x'(0)} = \theta_i, \qquad \arctan \frac{y'(1)}{x'(1)} = \theta_f. \tag{18}$$

Since the interpolation is done independently in x and y, these are separated as

$$\begin{aligned} x'(0) &= k_i \cos\theta_i, \quad x'(1) = k_f \cos\theta_f, \\ y'(0) &= k_i \sin\theta_i, \quad y'(1) = k_f \sin\theta_f, \end{aligned} \tag{19}$$

where $k_i \neq 0$, $k_f \neq 0$ are free parameters. In other words, the flat outputs of the boundary configurations are (x_i, y_i) and (x_f, y_f), while the orientations θ_i, θ_f translate into conditions on the first derivatives. Each of $x(s)$ and $y(s)$ thus carries four boundary conditions (value and derivative at both ends), which matches the four coefficients of a cubic.

Proposition 47.1 (Cartesian cubic planner)**.** *Under the simplifying assumption $k_i = k_f = k$, the additional boundary conditions give*

$$\begin{aligned} \alpha_x &= k\cos\theta_f - 3x_f, \quad \beta_x = k\cos\theta_i + 3x_i, \\ \alpha_y &= k\sin\theta_f - 3y_f, \quad \beta_y = k\sin\theta_i + 3y_i. \end{aligned} \tag{20}$$

Moreover $(0) = (1) = k$; $k > 0$ gives forward motion, $k < 0$ backward motion.

Proof. Differentiating, $x'(s) = 3s^2 x_f - 3(s-1)^2 x_i + \alpha_x(3s^2 - 2s) + \beta_x\big((s-1)^2 + 2s(s-1)\big)$. Then $x'(0) = -3x_i + \beta_x = k\cos\theta_i$ and $x'(1) = 3x_f + \alpha_x = k\cos\theta_f$, which give β_x, α_x. The computation for y is identical. At the extremes $= \pm\sqrt{(x')^2 + (y')^2} = \pm|k|$, with the sign of k for the chosen direction of motion. □

From $x(s)$ and $y(s)$ one obtains $\theta(s)$, (s) and (s) via the reconstruction formulas of Proposition 46.2.

Intuition

This is cubic *Hermite interpolation*: (x_i, y_i) and (x_f, y_f) are the endpoints, and the vectors $k(\cos\theta_i, \sin\theta_i)$ and $k(\cos\theta_f, \sin\theta_f)$ are the end tangents. The heading is forced to be tangent to the path (so the path is admissible), and k is a "tangent length": a large k makes the robot leave and arrive at *straight and for a long stretch* before bending in the middle, a small k gives a more direct, tighter curve.

Example 47.2 (Worked numbers: forward parking)**.** Take $= (0, 0, 0)$ and $= (4, -3, -\pi/2)$. Then $\alpha_x = -12$, $\beta_x = k$, $\alpha_y = 9 - k$, $\beta_y = 0$, so

$$x(s) = (k-8)s^3 + (12-2k)s^2 + ks, \qquad y(s) = (6-k)s^3 - (9-k)s^2.$$

One checks $x'(0) = k$, $y'(0) = 0$ (heading 0, speed k) and $x'(1) = 0$, $y'(1) = -k$ (heading $-\pi/2$, speed k). At the start, $(0) = y''(0)/k = 2(k-9)/k$: for $k = 5$ this is -1.6 (the robot immediately steers right), whereas for $k = 20$ it is $+1.1$ (the robot first swings slightly *left*, up, and then comes down).

Intuition

The sign change of (0) at $k = 9$ is the numerical face of "long tangents overshoot": with a very large k the cubic is stretched along the initial heading, so it first bulges forward and upward before turning down to the target.

48 Unicycle path planning via the chained form *(slides 16–17)*

As an alternative one may transform the unicycle into the $(2, 3)$ chained form, whose flat outputs are z_1 and z_3. (The standard unicycle transformation is $z_1 = \theta$, $z_2 = x\cos\theta + y\sin\theta$, $z_3 =$

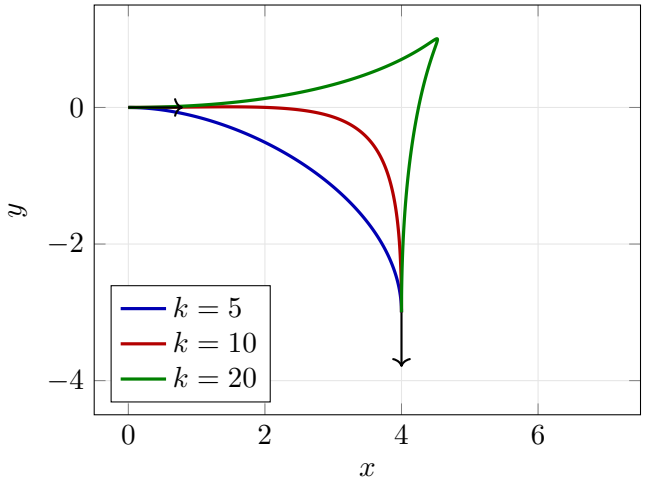


Figure 66: Cartesian cubic planner for the forward-parking problem of Example 47.2, for increasing k (own computation from the formulas above). Larger k produces longer take-off and landing phases.

$x \sin\theta - y\cos\theta$, with inputs $w_1 = \omega$, $w_2 = v - z_3\omega$.)

First compute the initial and final values of z_1, z_2, z_3 corresponding to and . Then, assuming $z_{1,i} \neq z_{1,f}$, define z_1 and z_3 over $[0,1]$ as

$$z_1(s) = z_{1,f}\, s - (s-1)\, z_{1,i}, \tag{21}$$

$$z_3(s) = s^3 z_{3,f} - (s-1)^3 z_{3,i} + \alpha_3 s^2(s-1) + \beta_3 s(s-1)^2, \tag{22}$$

which already satisfy the boundary conditions on z_1, z_3. Since $z_{2,i}$ and $z_{2,f}$ are also given, we must impose

$$\frac{z_3'(0)}{z_1'(0)} = z_{2,i}, \qquad \frac{z_3'(1)}{z_1'(1)} = z_{2,f}. \tag{23}$$

Proposition 48.1 (Chained-form cubic planner)**.** *Imposing these conditions gives*

$$\alpha_3 = z_{2,f}(z_{1,f} - z_{1,i}) - 3z_{3,f}, \qquad \beta_3 = z_{2,i}(z_{1,f} - z_{1,i}) + 3z_{3,i}. \tag{24}$$

Proof. $z_1'(s) = z_{1,f} - z_{1,i} =: \Delta$ is constant. As in Proposition 47.1, $z_3'(0) = -3z_{3,i} + \beta_3$ and $z_3'(1) = 3z_{3,f} + \alpha_3$; setting them equal to $z_{2,i}\Delta$ and $z_{2,f}\Delta$ respectively gives the result. □

From $z_1(s)$ and $z_3(s)$ one obtains $z_2(s)$ and $_1(s)$, $_2(s)$ via the reconstruction formulas of Example 46.5. Finally (s), (s) and (s) are computed by inverting the chained form coordinate and input transformations.

The singular case. If $z_{1,i} = z_{1,f}$ (i.e., $\theta_i = \theta_f$) the construction breaks down (z_1 would be constant). One can introduce a *via point* $_v = (x_v, y_v, \theta_v)$ with $\theta_v \neq \theta_i$ and split the original problem in two; alternatively, set $z_{1,f} = z_{1,i} + 2\pi$ (a full turn).

Direct trajectory planning. Both previous schemes (Cartesian polynomials and chained form) can be used for direct trajectory planning: just let $s = t$ and use the kinematic model.

Intuition

Here $z_1 = \theta$ is used as the interpolation "clock" and it moves *linearly* between θ_i and θ_f: the heading changes monotonically. That is why $\theta_i = \theta_f$ is a problem: the clock does not move at all and the curve $z_3(z_1)$ has nothing to be a function of. Adding a via point or a full 2π turn restores a clock that ticks.

Remark on the numerical results. For the unicycle chained form, the position enters z_2, z_3 through the current (x, y). Placing the origin of z_2, z_3 at (so that $z_{2,i} = z_{3,i} = 0$) changes the plan: with the global origin a reorientation generates a Cartesian displacement, whereas with the origin at pure rotations on the spot are representable (in the numerical check for $= (1, 1, 0) \to = (1, 1, \pi/2)$ with origin at the integration gives $c_0 = c_1 = 0$, i.e. $x \equiv y \equiv$ const).

49 Path planning via parametrized inputs *(slides 18–21)*

49.1 Chained form with parametrized inputs *(slides 18–20)*

A different approach, not directly based on flatness, consists in writing the inputs in *parametrized form* and then imposing that the robot goes from to . Assume the robot can be put in chained form,

$$\begin{aligned} z_1' &= {}_1, \\ z_2' &= {}_2, \\ z_3' &= z_{21}, \\ &\vdots \\ z_n' &= z_{n-11}, \end{aligned} \tag{25}$$

which becomes *linear* (hence easily integrable) if ${}_1(s)$ is constant.

Let $\Delta = z_{1,f} - z_{1,i}$ and choose the geometric inputs for $s \in [s_i, s_f] = [0, |\Delta|]$ as

$$ {}_1 = (\Delta), \tag{26}$$

$$ {}_2 = c_0 + c_1 s + \cdots + c_{n-2} s^{n-2}. \tag{27}$$

Integrating the chained-form equations from $\boldsymbol{z}(0) = \boldsymbol{z}_i$ and imposing $\boldsymbol{z}(|\Delta|) = \boldsymbol{z}_f$ leads to a *linear system*

$$D(\Delta) \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-2} \end{pmatrix} = \boldsymbol{d}(\boldsymbol{z}_i, \boldsymbol{z}_f, \Delta), \tag{28}$$

with $D(\Delta)$ invertible if $\Delta \neq 0$.

Proposition 49.1 (The $(2, 3)$ case)**.** *For the* $(2, 3)$ *chained form,*

$$D = \begin{pmatrix} |\Delta| & \dfrac{\Delta^2}{2} \\ (\Delta)\dfrac{\Delta^2}{2} & \dfrac{\Delta^3}{6} \end{pmatrix}, \qquad \boldsymbol{d} = \begin{pmatrix} z_{2,f} - z_{2,i} \\ z_{3,f} - z_{3,i} - z_{2,i}\Delta \end{pmatrix}, \tag{29}$$

and $\det D = -(\Delta)\,\Delta^4/12 \neq 0$ *whenever* $\Delta \neq 0$.

Proof. With $_1 = (\Delta)$, $_2 = c_0 + c_1 s$: $z_1(s) = z_{1,i} + (\Delta)s$, which reaches $z_{1,f}$ at $s = |\Delta|$; $z_2(s) = z_{2,i} + c_0 s + c_1 s^2/2$; and $z_3' = (\Delta) z_2$ gives $z_3(s) = z_{3,i} + (\Delta)\big(z_{2,i} s + c_0 s^2/2 + c_1 s^3/6\big)$. Imposing $z_2(|\Delta|) = z_{2,f}$ gives $c_0|\Delta| + c_1\Delta^2/2 = z_{2,f} - z_{2,i}$. Imposing $z_3(|\Delta|) = z_{3,f}$ and using $(\Delta)|\Delta| = \Delta$, $(\Delta)|\Delta|^3 = \Delta^3$ gives $(\Delta)\frac{\Delta^2}{2} c_0 + \frac{\Delta^3}{6} c_1 = z_{3,f} - z_{3,i} - z_{2,i}\Delta$. Finally $\det D = |\Delta|\Delta^3/6 - (\Delta)\Delta^4/4 = -(\Delta)\Delta^4/12$. □

If $\Delta = 0$ ($z_{1,i} = z_{1,f}$) a singular case is met, which can be handled as before (e.g., with a via point). Once planning on the chained form is complete, (s) and (s) for the original model are computed via the inverse coordinate and input transformations. Although this method does not make explicit use of flat outputs, it still relies on transformability in chained form, which is strongly related to flatness (Theorem 46.3).

Intuition

With $_1$ constant, s simply counts how far the "clock" variable z_1 has advanced, and the remaining equations are a *chain of integrators* driven by the polynomial $_2$. The $n-1$ unknown coefficients are exactly as many as the $n-1$ boundary conditions on $z_2, \ldots, z_n$ (those on z_1 are met by the choice of the interval $[0, |\Delta|]$), so a square linear system results. For $n = 3$ there are two coefficients (c_0, c_1) for two conditions.

49.2 Other parametrizations and trajectory planning *(slide 21)*

Other classes of parametrized inputs can be used, such as sinusoidal and piecewise-constant functions. Note that in *all* the presented path planning methods the path parameter s *does not represent the arc length.* This scheme too can be used for direct trajectory planning by replacing s with t and using the kinematic model in place of the geometric model.

50 Parking a unicycle: numerical results *(slides 22–27)*

50.1 Planner based on Cartesian polynomials *(slides 22–24)*

- **Forward parking** ($k = 5 > 0$): the motion is forward; a negative k would produce a backward motion. There are no motion inversions (a general result for this planner).
- **Effect of k:** increasing k (e.g., $k = 10$ and $k = 20$) leads to elongated take-off and landing phases; still no motion inversions.
- **Parallel parking** and **reorientation** (again $k = 10$ and $k = 20$): since the planner is based on flat outputs, a Cartesian path of *finite length* is needed for *any* kind of reconfiguration.

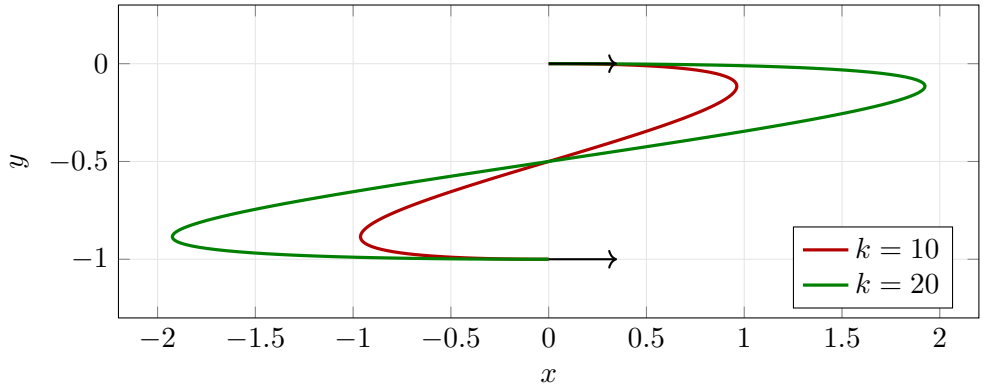


Figure 67: Parallel parking with the Cartesian planner, $= (0, 0, 0) \rightarrow = (0, -1, 0)$ (own computation). A larger k widens the forward and backward excursions needed to shift sideways.

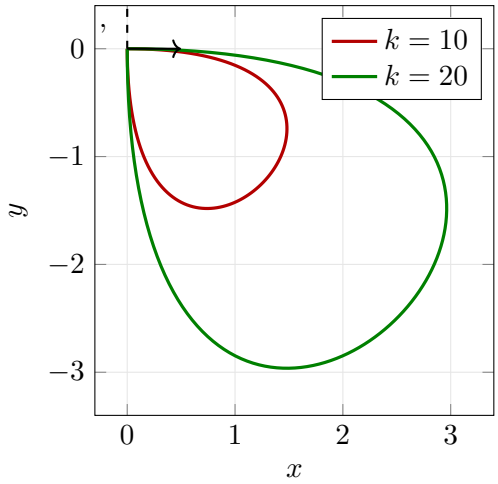


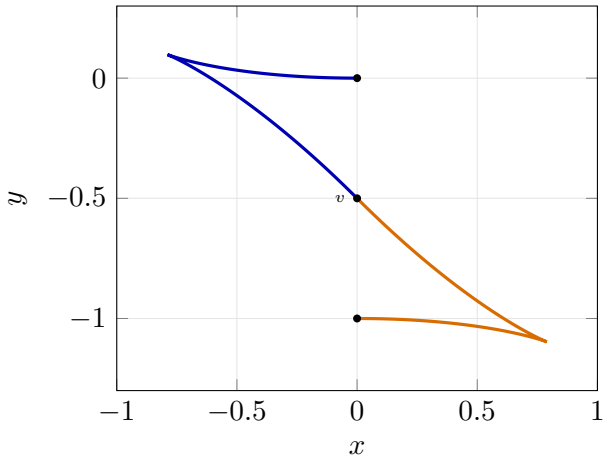
Figure 68: Reorientation $= (0, 0, 0) \rightarrow = (0, 0, \pi/2)$ with the Cartesian planner (own computation): the robot leaves heading east (solid arrow), draws a finite loop and returns heading north (dashed arrow).

Intuition

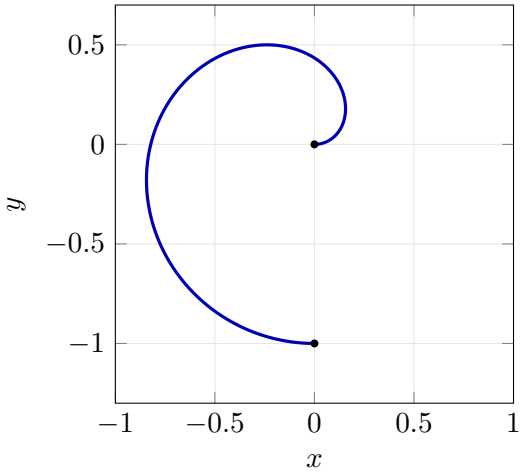
Reorienting a unicycle in place is easy for the physical robot (rotate on the spot), but the Cartesian planner has no way to say "stay at a point". The reconstruction formulas fail when the Cartesian path degenerates to a point (Section 4), so the planner must draw a small loop in the plane to change orientation.

50.2 Planner based on parametrized inputs *(slides 25–27)*

- **Forward parking:** the first maneuver is similar to that of the other planner. In the second maneuver a *cusp* appears: this planner can only generate paths along which the orientation varies monotonically.
- **Parallel parking:** the problem is solved with a via point, or by imposing a full rotation.
- **Reorientation:** using the classical chained form transformation, a Cartesian displacement is generated; placing the origin of z_2, z_3 at avoids it.

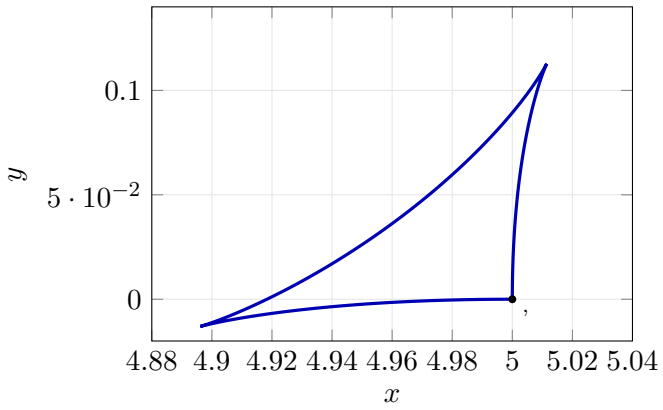


with a via point

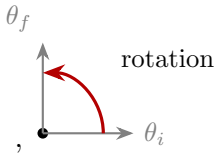


imposing a full rotation ($z_{1,f} = z_{1,i} + 2\pi$)

Figure 69: Parallel parking $= (0, 0, 0) \rightarrow = (0, -1, 0)$ with the parametrized-input planner. Since $\theta_i = \theta_f$ ($\Delta = 0$), the problem is either split at a via point (here $_v = (0, -0.5, -\pi/4)$) or solved with a full rotation of the orientation (own computation).



classical transformation (origin of z_2, z_3 elsewhere)



origin of z_2, z_3 placed at

Figure 70: Reorientation $= (x, y, 0) \rightarrow = (x, y, \pi/2)$ with the parametrized-input planner (own computation). With the classical transformation a small Cartesian displacement is generated (note the scale of the axes); placing the origin at gives $c_0 = c_1 = 0$, i.e., a pure rotation on the spot.

51 Choice of timing law *(outline item)*

This item appears in the outline of the lecture; the slides provided end with the parking examples, so only the basic idea is given here (supplementary material, not taken from the slides).

Given a geometric path (s) with inputs (s), any $s = s(t)$ with $\dot{s} > 0$ yields an admissible trajectory with inputs $v(t) = (s)\dot{s}$, $\omega(t) = (s)\dot{s}$. Velocity bounds $|v| \leq v_{\max}$, $|\omega| \leq \omega_{\max}$ become

$$\dot{s}(t) \leq \min \left\{ \frac{v_{\max}}{|(s)|}, \frac{\omega_{\max}}{|(s)|} \right\},$$

so the timing law can be chosen, for instance, as the fastest one compatible with these bounds, or as a smooth profile (e.g., cubic) satisfying boundary conditions on s and $\dot{s}$.

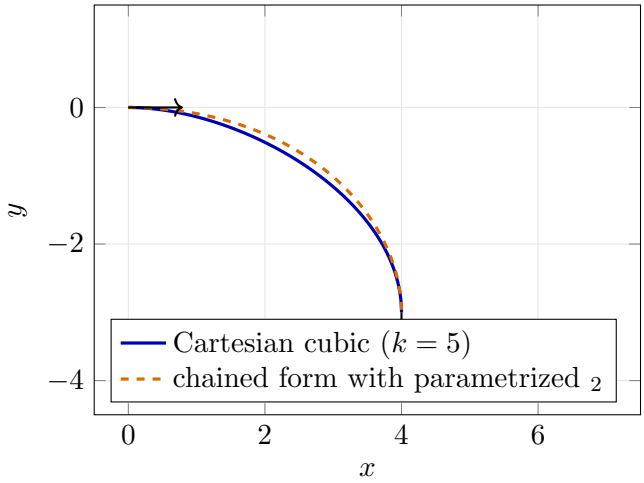


Figure 71: The two planners on the same forward-parking problem $= (0, 0, 0) \rightarrow = (4, -3, -\pi/2)$ (own computation). The parametrized-input planner turns monotonically from 0 to $-\pi/2$ with $= \pm 1$ per unit of s, so s is the angle swept, not the arc length.

Intuition

Slowing down scales *both* inputs by the same factor, so the shape of the path never changes; it is the sharp parts of the path (large $||$) that force the robot to slow down.

52 Summary

	Cartesian polynomials	**Chained form (cubic z_3)**	**Parametrized inputs**
Flat outputs used	$x,\ y$	$z_1,\ z_3$	not directly (needs chained form)
What is chosen	cubic $x(s), y(s)$; k free	z_1 linear, z_3 cubic	$_1 = \Delta$, $_2$ polynomial
Unknowns	$\alpha_x, \beta_x, \alpha_y, \beta_y$ (closed form), k	α_3, β_3 (closed form)	$c_0, \ldots, c_{n-2}$ (linear system $D\boldsymbol{c} = \boldsymbol{d}$)
Singular case	—	$\theta_i = \theta_f$	$\Delta = 0$
Singular case remedy	—	via point or $+2\pi$	via point or $+2\pi$
Orientation	follows the Cartesian curve	monotonic	monotonic
Cusps	none	may appear	may appear
s is arc length?	no	no	no
Direct trajectory planning	$s = t$	$s = t$	$s = t$

Table 7: Comparison of the three planners for the unicycle (all assume obstacle-free motion).

Key take-aways

- A path is admissible for a nonholonomic WMR iff $A()' = 0$; this is independent of the timing law, which only converts a path into a trajectory ($\boldsymbol{u} = \dot{s}$).

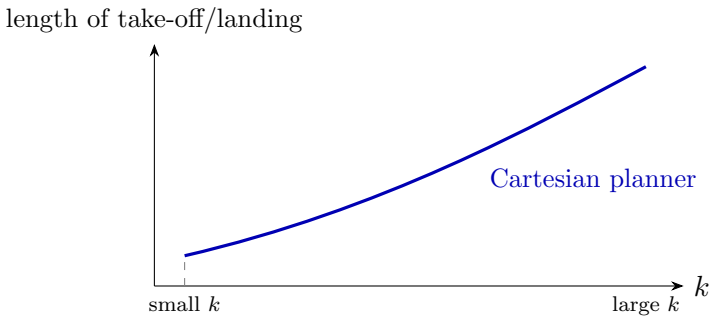


Figure 72: Qualitative trade-off in the Cartesian planner: larger k gives longer straight take-off and landing phases and a possible initial overshoot (qualitative sketch, not measured data).

- Differentially flat systems (unicycle, bicycle, all WMRs up to dimension 4) can be planned by interpolating the flat outputs and reconstructing the rest: admissibility comes for free.
- For the unicycle, a cubic Cartesian Hermite curve with tangent length k gives a closed-form planner; the sign of k selects forward or backward motion and a large k elongates take-off and landing.
- In chained form, either interpolate z_1, z_3 or parametrize $_2$ and solve a linear system; both fail when $\theta_i = \theta_f$ and need a via point or a 2π turn, and both produce monotonically varying orientation.
- In none of the methods does s represent the arc length, and every scheme becomes a direct trajectory planner by setting $s = t$.